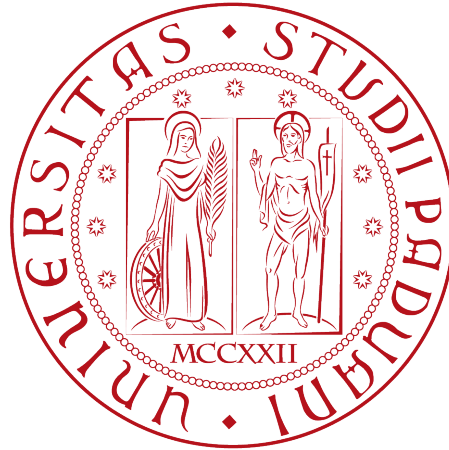


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Unsupervised ensemble pruning technique for Isolation Forest

Master's Thesis

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Abstract

This study focuses on the field of Anomaly Detection (AD) which aims to identify unusual behaviors relative to a specific definition of normality in a specific context. Anomalies are rare events that can have significant implications, in terms of safety, security and operational efficiency, in various application domains, such as medicine, finance, and industry.

A widely used algorithm in AD, adopted in this study, is the *Isolation Forest* (IF), a standard choice for this task due to its lightweight nature and unsupervised formulation: anomalies, which are few and structurally different, are easier to isolate than normal points within the recursive partitioning structure of trees. The strength of this model lies in its randomized recursive partitioning process. By randomly and recursively selecting both features and split values, each tree generates a distinct isolation structure. When aggregated into an ensemble, this diversity makes the model not only efficient but also robust. However, this randomness may also introduce redundancy and noise: some trees may be weakly informative, carry an insignificant structure, or even degrade the overall anomaly score.

This observation motivates the scope of this study: ensemble pruning of the Isolation Forest. The central hypothesis is that harmful or non-informative trees can be identified and removed, thereby reducing the size and computational complexity of the model and potentially improving the quality of the anomaly scores. The challenge is that in unsupervised anomaly detection, true labels are most often unavailable, making traditional pruning strategies — most of which rely on supervised evaluation — inapplicable.

Furthermore, many of the Artificial Intelligence (AI) models used for Anomaly Detection, including the Isolation Forest, are complex and cannot be easily interpretable, which is essential in safety-critical applications. In fact, the lack of interpretability can lead to a lack of trust in the system and, in some cases, to catastrophic consequences. Therefore, it is important to provide explanations for the decisions made by AI systems. For these reasons, the *eXplainable Artificial Intelligence* (XAI) or *Interpretable Machine Learning* field is a recent research area that aims to provide tools and techniques to make AI systems more human-understandable, allowing them to gain insight into the decision-making process.

Then, to address such challenge, experiments were conducted in order to develop a novel unsupervised pruning strategy for the Isolation Forest by exploiting two main research directions:

1. Investigating the potential of interpretability techniques like DIFFI as explanatory tools and also as mechanisms for model optimization. The basic idea is to leverage the feature importance scores provided by DIFFI to identify

and prune trees that contribute to overall performance of the ensemble.

2. Investigating the intrinsic properties of the Isolation Forest, focusing in particular on the depth distribution to identify trees that may not be informative about ensemble performance.

DIFFI, an explainable technique specifically tailored for Isolation Forest, is an algorithm that returns both global and local feature importance scores, which are not only useful for providing explanations of the model decisions to end-users, but also allow developers to better understand the global functioning of the ensemble. The experiments carried out span various synthetic and real-world datasets, allowing us to assess the effectiveness of our proposed pruning strategies under different data distributions and anomaly patterns.

The results demonstrate that the DIFFI-based strategy has the potential to improve Isolation Forest performance; however, its effectiveness is highly dataset-dependent and not universally guaranteed. On the other hand, the Depth-based strategy shows more consistent improvements across different datasets, indicating that leveraging the depth property of the Isolation Forest, i.e., the anomalies are likely to be isolated at shallower levels while the normal instances are likely to be isolated at deeper levels, it is a more reliable approach for ensemble pruning in unsupervised anomaly detection tasks. Indeed, the depth-based *Separation Ratio* method, where trees are evaluated based on the ratio of the average depth of inliers to that of outliers, demonstrates significant improvements in both F1-Score and Average Precision across a wide range of datasets.

Then further analysis is conducted to assess the impact of the number of trees in the original Isolation Forest on the effectiveness of pruning. The results suggest that identifying an optimal number of trees is crucial to maximizing the benefits of pruning, in order to further improve the performance and efficiency of the Isolation Forest model.

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Chapter 1

Introduction

*This chapter introduces the main concepts used in this thesis: Section 1.1 introduces the general topic of **Anomaly Detection**, highlighting its importance in various application domains. Section 1.2 discusses the importance of interpretability in AI models, focusing on the field of **eXplainable Artificial Intelligence (XAI)** and its contribution.*

1.1 Anomaly Detection

Anomaly Detection is a challenge that addresses the problem of discriminating between two main categories of data:

- **Normal data:** the majority of the data that conform to a defined notion of normal behavior within a specific context and represent the standard operating conditions.
- **Anomalous data:** the minority of data that occurs very rarely and whose features are significantly different from the normal data. These points can be caused by several factors, such as system failures, cyber-attacks, or human errors.

It is a crucial task in a wide range of application areas - including healthcare, finance, manufacturing, cybersecurity, and many others - where the ability to identify unusual behaviors can be considered more important than normal data, helping to ensure safety, security, and operational efficiency. Anomalies can represent useful and valuable information that requires immediate attention; otherwise, they could lead to system failure.

There are several approaches to anomaly detection, ranging from using models of the nominal behavior of the monitored system to Statistical Process Control (Oakland, 2007). However, with the advancement of Machine Learning technologies, sophisticated methods have been developed to detect outliers by exploiting large amount of historical data. Unfortunately, since anomalies are rare and cannot be freely generated, manually building labeled datasets is costly and prone to errors. For this reason, most anomaly detection ML models are designed to operate in an unsupervised manner.

In tabular domain, established unsupervised models include *Isolation Forest* (IF), *Local Outlier Factor* (LOF), *k-Nearest Neighbors* (KNN), and *Principal Component*

Analysis (PCA) among others (Han et al., 2022). Deep Learning methods are the state-of-the-art with visual and sequential data, but they are rarely the primary choice for tabular inputs due to lower accuracy, high computational costs, and reduced interpretability. Indeed, model quality is no longer assessed by anomaly detection accuracy alone: efficiency and explainability have also become essential. Efficiency is crucial in industrial or edge scenarios, where computational resources are limited and a low latency is required by the process, in terms of inference time. Explainability is critical when model predictions must be trusted or acted upon by human users, as it enables understanding of why certain points are flagged as anomalous, supporting root-cause analysis and decision-making (Li et al., 2023, Section 1.2). For this reason, in recent years, research has focused on explainability of ML models, because of their complexity, and as a result, their opacity, which makes it difficult to interpret their decisions, especially in safety-critical applications.

In this context, when considering the multi-objective lens of **accuracy**, **computational complexity** and **explainability**, Isolation Forest surpasses other methods. Beyond achieving high performance (Liu et al., 2009), its tree-based structure makes it highly efficient during both training and inference, even on large or high-dimensional datasets. Moreover, while not inherently interpretable, a model-specific explanation technique *DIFFI* (Carletti et al., 2021) has been developed to highlight the most important features that influence each decision.

For these reasons, Isolation Forest will be the focus of this thesis.

1.2 eXplainable Artificial Intelligence

Today, with the growing popularity of Artificial Intelligence (AI) technologies, significant practical progress has been made in this field, providing extremely powerful and sophisticated algorithms. Currently, this advancement in algorithmic capability has been accompanied by a substantial increase in model complexity, underscoring the critical need for researchers to develop methods that provide transparency in these sophisticated decision-making processes (Loyola-González, 2019).

But firstly, we must distinguish between two main categories of AI algorithms:

- **White-box:** Models whose decision-making process is transparent and easily understandable to humans. Their unique characteristic is that they allow for a clear explanation of how inputs are transformed into outputs, revealing the logic behind predictions. Examples of models in this category include *Linear Regression*, *Logistic Regression*, *Rule-based systems*, and *Decision Trees*.
- **Black-box:** Models which, unlike the previous ones, the decision-making process is not transparent, thus preventing a clear explanation of the internal work. Some examples of black-box models include *Neural Networks*, *Support Vector Machines (SVMs)* and *Random Forests*.

In the scientific community, there is a debate about the relevance between these two approaches (see Table 1.1 for a summary of the differences). The first, given its explainability, is preferable for applications in safety-critical fields such as health

Aspect	White-box Models	Black-box Models
<i>Characteristic</i>	Transparent decision-making	Opaque decision-making
<i>Explainability</i>	High	Low
<i>Application Domains</i>	Safety-Critical applications	Complex tasks

Table 1.1: Comparison between White-box and Black-box Models

care, finance, law, and autonomous-drive vehicles, while the second, due to its nature, is more accurate and therefore preferable for solving more complex tasks (Loyola-González, 2019).

However, the goal is to achieve the best of both approaches: **understandable** and **accurate** models. For this reason, in recent years, a new field of research called "*eXplainable Artificial Intelligence*" (XAI) or "*Interpretable Machine Learning*" has emerged, aiming to develop techniques that enable interpretability in black-box models.

Some might wonder why interpretability is considered such an important factor in the context of AI. The answer lies in two aspects: *philosophical* and *practical*.

In high-risk contexts, it is no longer sufficient to know **what** was predicted; it is also necessary to know **why**: this can help us to understand a problem more deeply and suggest appropriate interventions when system failures occur (Molnar, 2025). Considering the first of the two aspects, when an unexpected event occurs, human beings, by nature, require an update of their mental model. This update can be accomplished by seeking an explanation for this unexpected event. All comes down to the desire to seek meaning in the world (Molnar, 2025).

However, the second aspect concerns the more technical part: explainability is also a useful tool for debugging, which would otherwise be impossible without knowing the true cause of an error (Molnar, 2025).

There exist several categorizations in the field of XAI (Molnar, 2025). The first is called **intrinsicity**, which refers to the inherent explainability of a model. In this case, we can distinguish between:

- **by-design**: includes all models that are inherently explainable and do not require additional techniques (also known as *white-box* models).
- **post-hoc**: includes all models that require the application of additional explainability techniques after the model has been trained (also known as *black-box* models).

Among post-hoc techniques, we can make a further distinction based on **agnosticity**:

- **model-specific**: explainability techniques that are designed for specific models, and therefore they are not applicable to any other model since they are tailored to the specific architecture and characteristics of the model.
- **model-agnostic**: explainability techniques that can be applied to any model. In fact, everything that happens within the model is ignored, but how the output changes with a specific modification to the features in input is analyzed. It is based on the **SIPA** principle: **S**ample from the data, perform an

Intervention, get the **Prediction** for manipulated data and **Aggregate** the results.

Finally, for agnostic models, there are still two types of technique, based on the **scope** of the explanation:

- **local**: techniques of this type are concerned with explaining predictions individually, i.e., they provide explanations for each single prediction made by the model and, therefore, they are useful for performing edge-case analysis.
- **global**: techniques that explain the overall behavior of the model with the aim of increasing its accuracy.

Chapter 2

Background

*This chapter provides the necessary background to understand the concepts and techniques used in this thesis. Section 2.1 introduces the **Isolation Forest** algorithm, explaining its structure, functioning, and advantages in Anomaly Detection (AD) tasks. Section 2.2 discusses the **DIFFI** technique, an explainability method designed for Isolation Forest, detailing how it computes feature importance scores both globally and locally.*

Then in Section 2.3, we present an overview of the related works that have influenced this research, while Section 2.4 outlines the main contributions of this thesis.

2.1 Isolation Forest

The Isolation Forest (IF) (Liu et al., 2009) differs from other models designed for Anomaly Detection due to its different approach, based on the concept of isolation. Usually, the majority of models aim to identify normal data, and those that do not meet this standard are classified as anomaly, since they are dependent on distance or density measures to identify outliers (like KNN and LOF). However, these approaches have disadvantages: the anomaly detector is optimized to identify normal data, so it runs the risk of detecting many false anomalies or few true anomalies. Furthermore, their computational cost increases quickly with the number of data samples and features, making them suitable only for low-dimensional and small-scale data.

There are some important quantitative properties that characterize outliers:

- Anomalies are rare events, then they are present in small quantities within the dataset;
- Anomalies have attribute values that differ significantly from those of normal instances.

By leveraging these two properties, IF is able to isolate anomalies in a more effective and efficient way, instead of profiling normal data.

Thus, unlike models that adopt the non-isolation approach, the accuracy is generally higher (Han et al., 2022) and since IF has a **linear complexity** and a low memory requirement, it is capable of scaling with high-dimensional and large data.

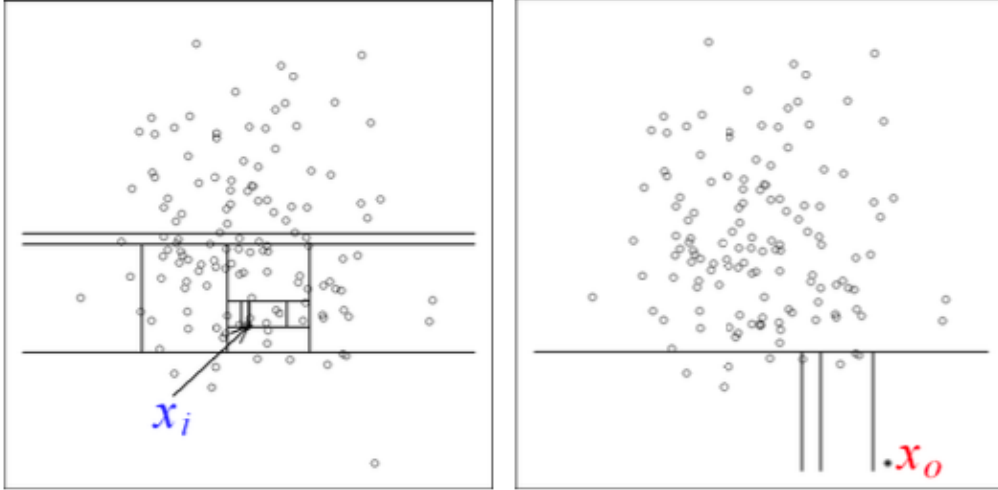


Figure 2.1: Comparison of the number of partitions required to isolate a *normal* point (on the left) and an *anomalous* point (on the right) in an Isolation Tree (Liu et al., 2009).

2.1.1 Isolation Tree

The Isolation Forest is an ensemble of Isolation Trees (iTrees), which are binary trees constructed through *random recursive* partitioning of the dataset. The key idea is that anomalies, being both rare and different from the majority, tend to be isolated by random splits more quickly than normal instances. In other words, when many iTrees are built, anomalous points exhibit a shorter average path length. Although this may not hold for any single trees, it is true in expectation. Figure 2.1 shows an example of this behavior: a normal point (in the left figure), usually requires a greater number of partitions to be isolated, unlike an anomalous one (in the right figure).

Now let us take a closer look at how an Isolation Tree is built, where each of them is assigned a subset $X' = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset X$, where X is the entire dataset and each $\mathbf{x}_i \in X'$ is a p -dimensional vector representing a data point:

1. X' is split recursively by randomly selecting both a feature f and its threshold value t ;
2. the partitioning continues until the stopping criterion is reached, which can be one of the following:
 - the tree reaches its maximum depth;
 - $|X'| = 1$ (i.e. the sample contains only one instance);
 - all data points in X' have the same value and therefore no further splits can be made.

Therefore, an Isolation Tree is a *proper binary tree* in which a node T can be **external** with no children or **internal** with exactly two children (T_l, T_r), obtained by verifying the condition $f < t$ that divides the data points into sets T_l and T_r . Assuming that all instances in X' are distinct, we will obtain a tree with a total

number of nodes corresponding to $2n - 1$, in which the number of external nodes is n and that of internal nodes is $n - 1$.

Let us now explore some characteristics derived from the properties of IF and the AD task.

Two aspects play a key role in the efficacy and efficiency of the forest: the *maximum depth* and *sub-sampling* of the dataset to build each tree. The maximum depth controls the maximum number of splits allowed in each iTree: since we are interested in isolating anomalies and not normal points, continuing to split beyond a pre-defined value (usually the expected isolating depth) does not improve the detection and only increases computational cost. Regarding sub-sampling, instead, it is not necessary to use the entire dataset X since most of the points are normal instances: using all data could be counterproductive, the ability to isolate anomalies could be reduced due to the potential interference caused by normal instances, as it would increase the number of partitions required to isolate each anomaly. For this reason, it is sufficient to use a small subset X' sampled from the entire dataset X to build an Isolation Tree.

In the field of AD, two problems arise:

- **Swamping:** The misclassification of a normal point, i.e. a false anomaly: this happens when a normal point is too close to an anomalous point;
- **Masking:** A phenomenon that can occur in two circumstances: the existence of too many anomalies causes them to be obscured, or they are too close, forming regions of high density and therefore they are not easy to be isolated; in both cases, detection is made more difficult.

These two problems can occur when using a large dataset, then sampling a small subset of data is sufficient to alleviate them.

It is clear that IF is perfectly suited to these circumstances, unlike other AD algorithms that require large datasets to perform well.

2.1.2 Anomaly degree ranking

After constructing the Isolation Trees, the goal is to assign to each point an anomaly score that reflects their degree of outlierness. After that, scores above a certain threshold are considered anomalies. Given a point $\mathbf{x} \in X'$, where $|X'| = n$, the anomaly score is defined as follows:

$$s(\mathbf{x}, n) = 2^{-\frac{E(h(\mathbf{x}))}{c(n)}} \quad (2.1)$$

where $E(h(\mathbf{x}))$ is the average of $h(\mathbf{x})$ in all trees in the forest. The length of the path $h(x)$ is the distance, in terms of the number of edges traversing the tree, between the root node and the external node $\mathbf{x}$.

While $c(n)$ is the normalization constant defined as:

$$c(n) = 2H(n - 1) - \left(\frac{2(n - 1)}{n}\right) \quad (2.2)$$

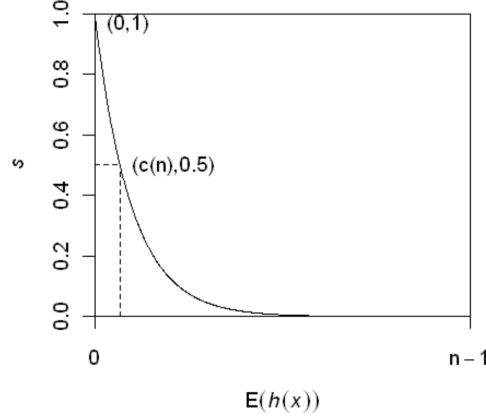


Figure 2.2: The relationship between the expected path length $E(h(x))$ and the anomaly score $s(x, n)$ (Liu et al., 2009).

where $H(n - 1)$ is the $n - 1$ -th harmonic number (of internal nodes), defined as:

$$H(n) = \sum_{i=1}^n \frac{1}{i} \quad (2.3)$$

The anomaly score $s(\mathbf{x}, n)$ is a value within the interval $[0, 1]$ that exhibits an inverse relationship with the length of the path $h(\mathbf{x})$, which ranges from 0 to $n - 1$ (see Figure 2.2). More precisely, a shorter path length corresponds to a higher score, indicating a higher likelihood of being an anomaly.

The classification process either flags points with anomaly score above a certain threshold as anomalies, or it ranks all data points according to their anomaly scores and selects the top-k points with the highest anomaly score as anomalies.

In the training phase, the theoretical computational complexity is $O(tn \log n)$, where $n = |X'|$ and t is the number of iTrees. The inference complexity is instead $O(Nt \log n)$, where N is the number of points to be evaluated. Hence, the Isolation Forest is an efficient method as it can easily handle large and high-dimensional datasets.

However, due to its randomness in the partitioning process, redundancy and noise trees may be present in the ensemble, which can negatively impact the quality of the anomaly scores. Furthermore, this has a negative impact on the explainability of the model, as it becomes more difficult to understand the decision-making process of the ensemble.

Thus, over the years, several improved versions of the IF have been proposed and techniques that provide explainability to this model have been developed.

Barbariol et al., 2022 provides a comprehensive overview of these different versions, highlighting their strengths and weaknesses. In section 4.1 of Barbariol et al., 2022, there is a comparison in which no algorithm emerges that shows significantly better performance than others. Furthermore, it should be noted that improved versions experience an increase in computational complexity. Therefore, it can be concluded that the basic IF is excellent as a baseline.

2.2 Depth-based feature importance of Isolation Forest (DIFFI)

The DIFFI technique (Carletti et al., 2021) is a *model-specific post-hoc* method to explain Isolation Forest. DIFFI falls into the *feature-based* category (Li et al., 2023): a relevance score is calculated for each feature that reflects its importance in isolating outliers.

This method is also proposed in two versions:

- **Global DIFFI:** provides *Global Feature Importance* (GFI), which quantifies the overall influence of each feature on the global model behavior during the training phase.
- **Local DIFFI:** provides *Local Feature Importance* (LFI), which measures how much each feature contributed to a specific individual prediction, during the evaluation phase.

DIFFI computes the importance of a feature by observing the internal nodes across the trees in which it is used. The underlying idea is based on two assumptions:

- (A1) A feature is important if, when used in a split, it induces the isolation of an outlier at a small depth and the isolation of normal points at high depths;
- (A2) A split and its associated feature are good and so important, if they produce a high imbalance on anomalous data points, to isolate them as soon as possible. This is an important factor because it affects the final calculation of the importance of the overall feature, defined in Equation 2.7.

The entire DIFFI process is illustrated in Figure 2.3, which shows the steps required to calculate the GFI values. Calculating GFI requires first determining the *Cumulative Feature Importance* (CFI), computed for both inliers and outliers. These values are then updated through an incremental process that uses normal and anomalous samples, respectively.

The CFI update procedure is based on two additional elements:

- the depth of the leaf node of a given point (A1);
- the *Induced Imbalance Coefficient* (IIC), which is a measure, associated with a specific internal node, of the imbalance in the number of anomalous and normal points in the leaf node, which is used to determine how much a feature contributes to the isolation of anomalies (A2).

2.2.1 Induced Imbalance Coefficient (IIC)

Being a post-hoc method, we exploit the output of the Isolation Forest, after it is trained. In particular, given the subset X'_t associated with each tree $t, t \in$

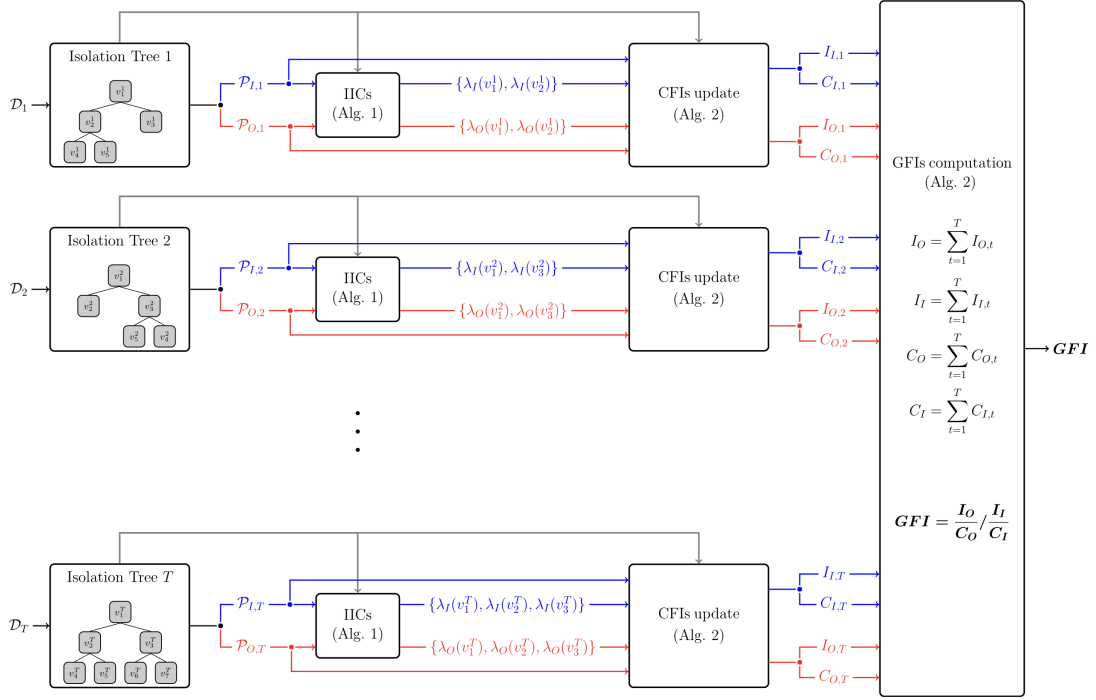


Figure 2.3: Overview of the DIFFI process for computing Global Feature Importance (GFI, Carletti et al., 2021).

$\{0, \dots, T\}$, we can divide it based on two disjoint subsets, depending on the associated prediction.

$$P_{I,t} = \{\mathbf{x}_i \in X'_t | \hat{y}_{i,t} = 0\} \quad (\text{inliers})$$

$$P_{O,t} = \{\mathbf{x}_i \in X'_t | \hat{y}_{i,t} = 1\} \quad (\text{outliers})$$

where $\hat{y}_{i,t}$ is the prediction of the point $\mathbf{x}_i$ by the tree t , and the properties:

$$P_{I,t} \cup P_{O,t} = X'_t$$

$$P_{I,t} \cap P_{O,t} = \emptyset$$

are satisfied.

Then, for each internal node v in the tree t , the IIC $\lambda(v)$ is defined as:

$$\lambda(v) = \begin{cases} 0 & \text{if } n_l(v) = 0 \text{ or } n_r(v) = 0 \\ \lambda(\tilde{v}) & \text{otherwise} \end{cases} \quad (2.4)$$

where $n_l(v)$ and $n_r(v)$, respectively, represent the number of data points on the left and right child of node v , then $n(v)$ is the total number of data points in node v . Although $\lambda(\tilde{v})$ is defined as:

$$\lambda(\tilde{v}) = g\left(\frac{\max(n_l(v), n_r(v))}{n(v)}\right) \quad (2.5)$$

where g is a scaling function that maps the input value to the range $[0.5, 1]$. Then we can interpret the values of $\lambda(v)$ as follows:

- $\lambda(v) = 0$: represents a **useless split** in which all data points are sent to one of the two children, meaning that there is no partitioning of the data points;
- $\lambda(v) = 1$: node v is perfectly imbalanced, i.e. a **perfect split**, meaning that one data point is sent to one of the two children and all the others to the other child, obtaining then an actual isolation;
- $0.5 \leq \lambda(v) < 1$: in this case **intermediate splits** are included, where there is no isolation but they are still useful for achieving it, with $\lambda(v) = 0.5$ we have a perfectly balanced split, i.e. the worst split possible.

The ICC is computed for both inliers ($\lambda_I(v)$) and outliers ($\lambda_O(v)$), taking advantage of, respectively, the subsets $P_{I,t}$ and $P_{O,t}$, and then the ideal situation is when $\lambda_I(v) = 0$ and $\lambda_O(v) = 1$, which means that the feature used to split the node v is significant for isolating anomalies, while it is not for normal points.

2.2.2 Cumulative Feature Importance (CFI)

The CFI is calculated for both inliers (I_I) and outliers (I_O). I_I and I_O are both p -dimensional vectors, where p is the number of features that represent the data points, and each j -th component represents the CFI value for the corresponding feature.

To demonstrate the calculation procedure, let us consider the case of inliers, which is the same as for outliers. However, two more elements are required:

- $path(\mathbf{x}, t)$, all internal nodes present in the path from the root node to the leaf node containing the point $\mathbf{x}$ in the tree t ;
- $h(\mathbf{x})$, the depth of the leaf node that contains the point $\mathbf{x}$ in the tree t .

The CFI calculation, for inliers, is performed as follows:

1. Initialize the CFI vector I_I to zero, i.e. $I_I = \mathbf{0}_p$;
2. For each tree t in the forest, iterate over all inliers in $P_{I,t}$:
 - (a) Choose a data point $\mathbf{x}_I \in P_{I,t}$;
 - (b) For each internal node $v \in path(\mathbf{x}_I, t)$
 - Each time a feature f_j is reached, i.e. the splitting feature associated with the node v , the j -th component of the CFI vector $I_{I,t}$ is updated by adding the quantity:

$$\Delta = \frac{1}{h_t(\mathbf{x}_I)} \cdot \lambda_I(v) \quad (2.6)$$

In other words, the CFI vector is updated by adding the contribution of the ICC of each internal node in the path from the root to the leaf node containing the point $\mathbf{x}_I$, weighted by the inverse of the depth of the leaf node $h_t(\mathbf{x}_I)$. This ensures that shallow nodes contribute more to the CFI than deeper nodes, reflecting the importance of features used in earlier splits, reflecting the typical characteristic of

the anomalies.

Finally, after iterating over all trees, the CFI vector I_I is then obtained by aggregating the contributions from all trees by summation, i.e., $I_I = \sum_{t=1}^T I_{I,t}$. The CFI for outliers is calculated in the same way, but using the subset $P_{O,t}$ and the corresponding ICC $\lambda_O(v)$, resulting in the vector I_O .

2.2.3 Global Feature Importance (GFI)

Before the final GFI calculation, it is important to note a bias in the CFI values: Since in the IF the choice of the splitting feature is random, some features may be selected more frequently than others, leading to an overestimation of their importance by increasing the CFI values, but this does not necessarily mean that they are more significant for isolating anomalies. To mitigate this bias, two counters are introduced, one for inliers (C_I) and one for outliers (C_O), which are p -dimensional vectors that count the number of times j -th feature is selected while updating the CFI vectors. In the same way as for the CFI, these counters are first computed for each tree and then aggregated by summation.

Finally, the GFI is calculated by normalizing the CFI values with the corresponding counters for each feature f_j , thus obtaining a more accurate measure of feature importance:

$$GFI(f_j) = \frac{\frac{I_O}{C_O}}{\frac{I_I}{C_I}} \quad (2.7)$$

The denominator $\frac{I_I}{C_I}$ tells us how important a given feature is to isolate an outlier: if this value is high, it means that the feature is important for isolating normal points, which is not desirable for Anomaly Detection.

This is consistent with assumption (A1), where a higher importance of features for inliers implies a lower overall importance of features.

2.2.4 Local Feature Importance (LFI)

The local version of DIFFI, i.e. LFI, allows you to provide an explanation for a single prediction. The LFI calculation process is similar to that of the GFI, but with some differences.

First of all, it is not possible to calculate the IICs because more than one data point is required, then since the LFI is focused on anomalies, all quantities related to predicted inliers $P_{I,t}$ are not considered, but only those related to outliers $P_{O,t}$. Therefore, given a predicted anomalous point $\mathbf{x}_O$, the corresponding LFI value is defined as follows:

$$LFI(\mathbf{x}_O) = \frac{I_O^{loc}}{C_O^{loc}} \quad (2.8)$$

where C_O^{loc} is the feature counter, which counts the number of times the j -th feature is selected while updating the CFI vector I_O^{loc} , which is computed similarly to the GFI: only considering the path $Path(\mathbf{x}_O, t)$ from the root node to the leaf node that contains the point $\mathbf{x}_O$ in the tree t .

Both values are updated in the same way as the GFI, that is, incrementally. A

further difference is the formula for updating these quantities:

$$\Delta_{loc} = \frac{1}{h_t(\mathbf{x}_O)} - \frac{1}{h_{max}} \quad (2.9)$$

in which no longer contains the IIC as weight, but is replaced by the correction term $-\frac{1}{h_{max}}$. Its purpose is to prevent the ICC from being updated when a leaf node associated with the expected outlier $\mathbf{x}_O$ is at the maximum depth of the tree, i.e., when that data point is not isolated and therefore to eliminate the contributions of useless splits.

2.3 Related works

Ensemble methods are widely adopted in machine learning. In a supervised setting, a combination of multiple base learners helps to achieve superior predictive performance. There are different paradigms, such as *Bayesian optimal classifiers*, *Boosting*, *Bagging*, and others.

Random Forest (RF) (Breiman, 2001) is a well-known application of the Bagging-like method. More precisely, it is an extension: in addition to the bootstrap sampling of the training data and aggregating the predictions, it also introduces the random feature selection at each split during the construction of the decision trees. This technique has proven effectiveness in reducing the correlation between individual trees, leading to improved performance.

Despite their effectiveness, ensemble methods can be highly computationally complex, especially when the number of base learners is large and there is redundancy: some of them may not contribute significantly to the overall performance (Banfield et al., 2005).

For this reason, over the years, various techniques have been studied to make these ensembles more efficient, particularly by removing weak learners that do not contribute to the improvement of performance; this process is known as **ensemble pruning**.

This is particularly relevant in the field of Anomaly Detection, as computational resources are often limited. In this regard, empirical evidence suggests an interesting fact: "The size of the ensemble can be reduced by up to 80% without significantly affecting the performance of the model" (Khalifa et al., 2024).

Ensemble pruning techniques can be taxonomically organized into three main categories (Tsoumakas et al., 2008):

- **Search-based pruning:** the pruning problem is formulated as an optimization problem, where the goal is to find a subset of base learners that maximizes or minimizes a certain objective function;
- **Clustering-based pruning:** this approach aims to identify the most representative learners in order to ensure *diversity* in the pruned ensemble. Classifiers that provide similar predictions are grouped into clusters, and the most representative one is selected from each cluster;
- **Ranking-based pruning:** establish a ranking of the base learners according to specific criteria and retain the highest-performing learners.

The pruning methods, included in the clustering-based category, depend on the important concept of *diversity*: a desirable property in which weak learners are expected to make uncorrelated errors. This is because their combination cancels out individual errors, leading to a more accurate and robust model (Banfield et al., 2005). Furthermore, a correlation has been shown between the increased accuracy of an ensemble and its diversity value. However, models involved in mentioned study were based on supervised learning, such as *Bagging* and *Random Forest* (Banfield et al., 2005).

In Khalifa et al., 2024, an approach to reducing model complexity involves first performing a pruning phase and then constructing the corresponding logical conjunctions from the pruned Random Forest. These are then aggregated into a tree-like structure that allows for a more efficient representation of the model (Khalifa et al., 2024).

Another interesting study has proposed an innovative technique in which an additional step is added to the pruning phase, namely *leaf-refinement*, which consists in a procedure to refine the probability estimates of the leaf nodes in a tree with the aim of improving the performance of the Random Forest model (Buschjäger and Morik, 2023).

Empirical experiments conducted by research have shown that there is a certain threshold beyond which adding more trees to the forest does not lead to a significant improvement in performance, but rather increases the computational complexity of the model (Bernard et al., 2009). Based on this observation, selection criteria were proposed to retain the most informative estimators in the ensemble, rather than relying on independent trees selection, which may not ensure effective cooperation. The intersection of explainability and pruning can represent a promising avenue for improving the performance of the ensemble methods. One such approach is to leverage *Shapley values* (Rozemberczki et al., 2022), which measure the contribution of each feature to the model predictions, helping to identify and remove weak learners from the ensemble that do not improve performance. This method is a game in which for each individual learner, it is quantified how much it contributes to the overall performance of the ensemble by exploiting the Shapley values. But this evaluation is unfortunately performed within a supervised learning context because it requires an oracle that provides the true labels of the data points.

Unfortunately, in all these proposed methods is not possible to directly apply them to IFs, since most are applied to Random Forests or true labels are required due to the supervised learning paradigm.

In this regard, a study has proposed an interesting approach, which is based not only on the concept of pruning to make the model more effective and efficient, but also on its construction (Ding and Xing, 2020b). Indeed, taking into account how an IF is constructed (Liu et al., 2009), that is, by randomly selecting features and their threshold values, this study instead proposes selecting the such value for a selected feature, also randomly, considering the distribution of the threshold values. For each feature, a histogram is created with all possible values, in order to analyze its distribution and select then low-frequency values, usually at the edges of the histogram, which are more likely to effectively isolate anomalies, following the logic that "they are rare and diverse."

However, as mentioned above, the work presented in this thesis is based on the

basic version of the IF (Section 2.1.2), so this alternative version is not taken into account.

Moving on to the proposed pruning method (Ding and Xing, 2020b), it is based on the concept that a good Isolation Tree should easily separate anomalies and hardly separate normal points. Then two corresponding lists are created, one for outliers and one for inliers, each representing the average isolation depth. They are then sorted in ascending order and descending order, respectively, following the aforementioned concept, but this is a limitation, because creating these two lists requires knowledge of the data point labels, which is impossible in real-world scenarios, since AD is an unsupervised task.

2.4 Contribution

As anticipated in Section 2.3, several pruning techniques have been proposed to make ensemble methods more efficient, but most of them are applied to Random Forest; therefore, it is not possible to directly apply the same techniques to Isolation Forests, given the differences between these two models:

- **Learning Paradigm:** RF is a *supervised* learning algorithm, while IF is an *unsupervised* learning algorithm;
- **Objective:** RF is designed to solve the classification and regression task, while IF aims to isolate anomalies.

Other researchers, instead, have proposed a pruning method for IFs, but they require the knowledge of the labels of the data points, which is not possible in real-world scenarios, as AD is an unsupervised task. Or they are based on alternative versions of the IF, which is not the case in this thesis.

Therefore, the literature thus reveals a significant research gap: although numerous pruning techniques exist for supervised ensemble methods, their application to unsupervised AD frameworks, like IF, remains underexplored. This thesis fills this gap by developing unsupervised pruning strategies specifically tailored to Isolation Forest.

Recall that the goal of pruning is to reduce the number of base learners in order to make the model more efficient, without significantly compromising its performance. Furthermore, the basic idea of pruning the Isolation Forest is to remove potentially redundant and noisy trees — an inevitable consequence of the random partitioning process — which may lead to a decrease in performance.

We adopt two complementary research directions in order to achieve this goal:

- **DIFFI-based pruning** (Section 3.2): We take advantage of the explainability provided by the DIFFI technique (Section 2.2) to identify and remove weak learners. Given the GFI scores, we can identify the most meaningful features to isolate anomalies (Section 3.2.2), then by analyzing the usage of these features in each tree, we can remove those that do not meet a certain threshold, which we can set in different ways (Section 3.2.5). The goal is therefore to remove trees in which the usage of important features is low and then they are less likely to be useful for detecting anomalies.

- **Depth-based pruning** (Section 3.3): We take advantage of the intrinsic properties of the IF (Section 2.1), focusing on the depth distribution. This approach is based on the fundamental premise that outliers should have path lengths significantly shorter than those at normal points. Trees that fail to maintain this separation — where anomalies and normal points exhibit similar path lengths — contribute little to effective detection and are candidates for removal.

There are several motivations for choosing DIFFI over other explainability techniques, such as SHAP and LIME. Firstly, DIFFI is a model-specific technique designed specifically for IFs, which allows it to capture the unique characteristics of this model more effectively, whereas the others two, in contrast, are model-agnostic techniques that may not fully exploit the specific properties of the model. Additionally, DIFFI demonstrated superior performance in terms of execution time (Carletti et al., 2021), making it a more efficient choice for our pruning strategy.

Our research makes several key contributions to both AD and XAI:

- Development of unsupervised pruning techniques, designed for IFs, addresses a significant gap in the existing literature. The strategies proposed are:
 - **DIFFI-based pruning** (Section 3.2)
 - **Depth-based pruning** (Section 3.3)
- Investigation of the relationship between explainability technique DIFFI and model optimization (i.e., ensemble pruning);
- Empirical evaluation of pruning effectiveness across various synthetic and real-world datasets;
- A comprehensive analysis of the impact of the number of estimators on pruning effectiveness, shedding light on how ensemble size influences the balance between efficiency and performance.

Chapter 3

Methodology

This chapter presents the methodological approaches developed to optimize the Isolation Forest algorithm through strategic pruning techniques. Section 3.1 defines the concepts of redundant and uninformative trees within the context of Isolation Forest.

The research explores two distinct pruning strategies: Section 3.2 illustrates a technique based on feature importance scores derived from DIFFI, while Section 3.3 describes a method that leverages the intrinsic depth properties of Isolation Trees.

In this chapter, the methodological decisions are explained and justified, laying the foundation for the experimental evaluation presented in Chapter 4. Each proposed technique aims to maintain or improve the anomaly detection performance while significantly reducing the computational complexity of the model.

3.1 Redundant and non-informative trees

In the context of Isolation Forest, *non-informative trees* refer to those individual decision trees within the ensemble that do not effectively contribute to the ability of the model in anomaly isolation. These trees may fail to distinguish the two classes of data points, resulting in suboptimal anomaly detection performance.

The presence of non-informative trees can be caused by several factors, including:

- **Poor Feature Selection:** The inherent randomness in the tree-building process can lead to the selection of features that do not provide meaningful splits for isolating anomalies;
- **Swamping Effects:** When anomalies are located near normal instances, it is difficult for some trees to effectively separate them. Although Isolation Forest mitigates this effect through sub-sampling, some trees may still be affected, resulting in uninformative behavior.

Figure 3.1 shows the tendency of this kind of tree to isolate outliers at depths similar to those of inliers, leading to an increase in the number of splits required to isolate anomalies, which is inconsistent with the fundamental principle of Isolation Forest (Section 2.1).

On the other hand, *redundant trees* are those that produce duplicate ones or very similar structures within the ensemble. They can be either non-informative or informative, but their presence does not add new information to the ensemble.

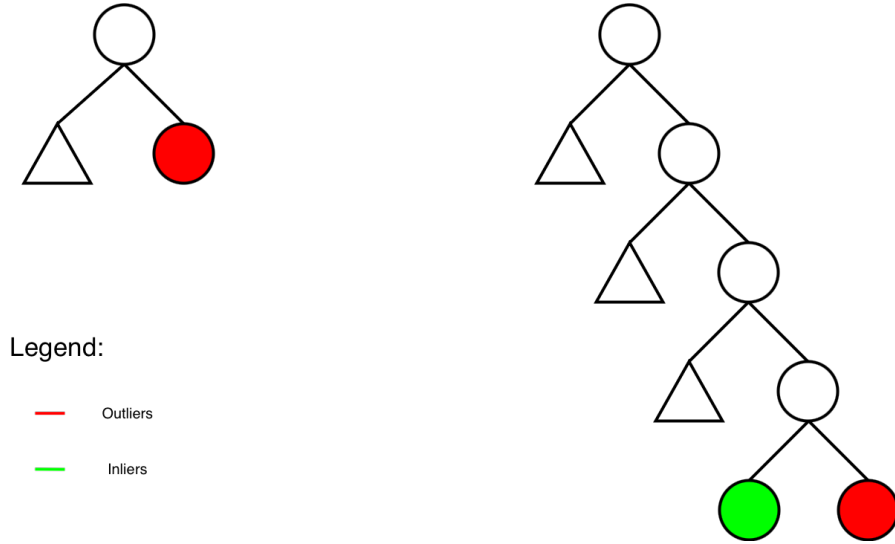


Figure 3.1: Example of an informative (left) and a non-informative tree (right) within an Isolation Forest. In the informative tree, anomalies (red points) are isolated at shallower depths compared to normal points (green points), while in the non-informative one, both anomalies and normal points are isolated at similar depths, indicating a lack of effective separation.

Redundancy can be caused by the random nature of the tree construction process. Since features and split values are randomly selected, multiple trees can be very similar or even identical, especially in datasets with limited variability or when the number of trees is large relative to the dataset size. However, random sub-sampling of data points can also contribute to the creation of redundant trees, as different trees can end up being trained on very similar subsets of data.

Both non-informative and redundant trees can negatively impact the Isolation Forest performance, increasing computational complexity without providing additional value in terms of anomaly detection. Therefore, the primary goal of this study is to identify and remove non-informative trees using pruning techniques, with the aim of improving the efficiency and effectiveness of the Isolation Forest model. Regarding redundant trees, rather than explicitly identifying and removing them, which requires a specific analysis, the pruning strategies proposed in this thesis intrinsically address redundancy, not directly, by focusing on the overall informativeness of each tree.

To this end, two distinct pruning strategies are proposed in the following sections.

3.2 Pruning based on DIFFI feature importance

This approach relies on the Global Feature Importance (GFI) scores provided by DIFFI (Section 2.2.3) to identify and remove the least significant trees from the Isolation Forest. The underlying assumption is that trees that mostly use less important features contribute minimally to the anomaly detection capabilities of the model. By pruning these trees, we aim to simplify the IF, reducing computational overhead, and preserving, or even improving, detection performance. More precisely,

the GFI takes into account the importance of each feature in anomaly isolation: A feature with higher GFI values means it likely plays a crucial role in distinguishing anomalies from normal data points. In contrast, features with lower GFI values are less influential in the isolation process. Therefore, trees that predominantly use features with high GFI scores are considered more effective for AD and are ideal candidates for retention.

The DIFFI feature importance-based pruning process involves three key steps, as described below:

1. **Feature Selection:** A subset of significant features is selected based on their GFI scores.
2. **Usage Score:** For each tree in the Isolation Forest, a usage score is computed to determine the frequency with which the selected features are used in a tree.
3. **Threshold Computation:** A threshold is determined to decide which trees to prune based on their usage scores.

For completeness, two additional essential methodological steps are included in the development of our *feature selection* and *thresholding* approach. These steps essentially consist of an analysis of the distribution of feature importance scores before pruning (Sections 3.2.1), which helped us better understand the dependence of the model on different features.

3.2.1 Pre-pruning analysis of feature importance distribution

To identify the most significant features for anomaly detection, a comprehensive analysis of the GFI score distribution is performed. The trained and unpruned Isolation Forest model serves as a starting point from which DIFFI generates feature importance metrics (as formally described in Section 2.2.3). Specifically, it provides two critical sets of importance vectors:

- the GFI vector for the entire forest, which provides an importance score for each feature across all trees;
- the FI vectors for *outliers* and *inliers* for each individual tree, respectively, $\mathcal{I}_O$ and $\mathcal{I}_I$ in DIFFI paper (Carletti et al., 2021), which indicate how important each feature is in isolating anomalies and normal points in that specific tree.

These scores serve as the foundation for interpretability of the Isolation Forest decision-making process. By examining these importance metrics, we can identify which features exhibit the greatest discriminative power in the AD task, distinguishing between the respective contributions to isolation of both inliers and outliers. This analysis is crucial to understanding the underlying mechanisms through which the model distinguishes between normal and anomalous data points.

For this purpose, some visual representations are used to better understand the distribution of these scores. The GFI vector is visualized using a bar chart, which shows the average importance scores for each feature across all trees in the Isolation Forest. For instance, Figure 3.2 highlights the differential contribution of features

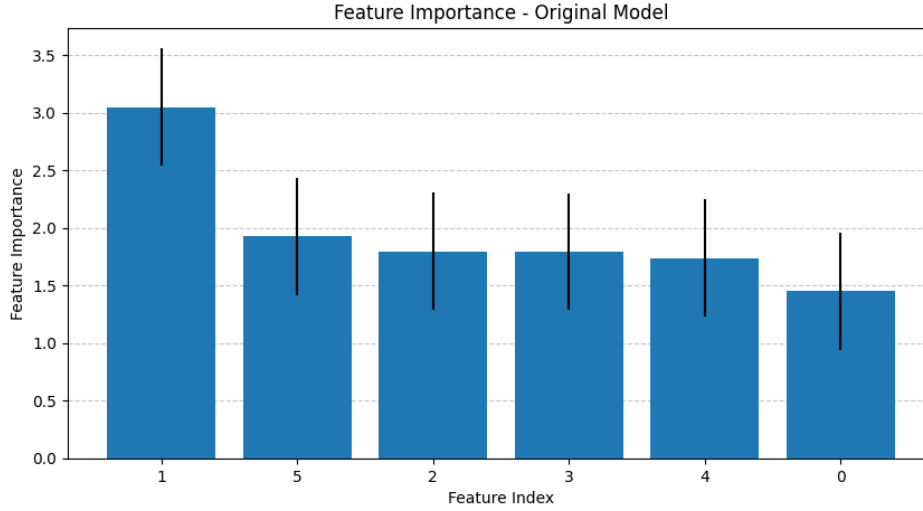


Figure 3.2: Bar plot of the average GFI scores for each feature across all trees in 10 Isolation Forests. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

to the anomaly detection ability of models. In this example, a 10-Isolation Forest setup is adopted based on experimental considerations detailed in Section 4.2.2. The bar chart reveals a pronounced importance for **feature 1**, which exhibits a higher GFI score compared to the others, indicating that it plays a crucial role in isolating anomalies. Furthermore, since the GFI is calculated as the average of 10 Isolation Forests, the bar plot provides a more statistically robust estimate of feature importance by aggregating information across multiple models (i.e., Isolation Forests). This suggests that the importance of this feature mitigates the effects of random partitioning inherent in individual Isolation Tree. Moreover, it indicates that this feature is likely to be more relevant for effective anomaly detection and positions it as an optimal candidate for inclusion in the subset during the *meaningful features selection* stage (more details in Section 3.2.2) in the subsequent pruning process, as it will guide the identification of trees that predominantly use these significant features. By focusing on trees that exploit the most important features, we can effectively prune the Isolation Forest while preserving its anomaly detection capabilities.

To better understand the relationship between the importance of the features and their utilization in the model, Figure 3.3 shows a heat map in which the FI scores are plotted for both inliers and outliers in all individual trees of a single Isolation Forest. For instance, **feature 1**, which has the highest GFI score, as shown in Figure 3.2, was later found to be the most important feature among the trees in the forest for isolating outliers and, to a lesser extent, for isolating inliers. This correspondence between high GFI scores and FIs, for both outliers and inliers, indicates that the model effectively leverages the most significant features for anomaly detection. Overall, this analysis of feature importance distribution provides valuable insights into the model’s dependence on different features and guides the subsequent steps in

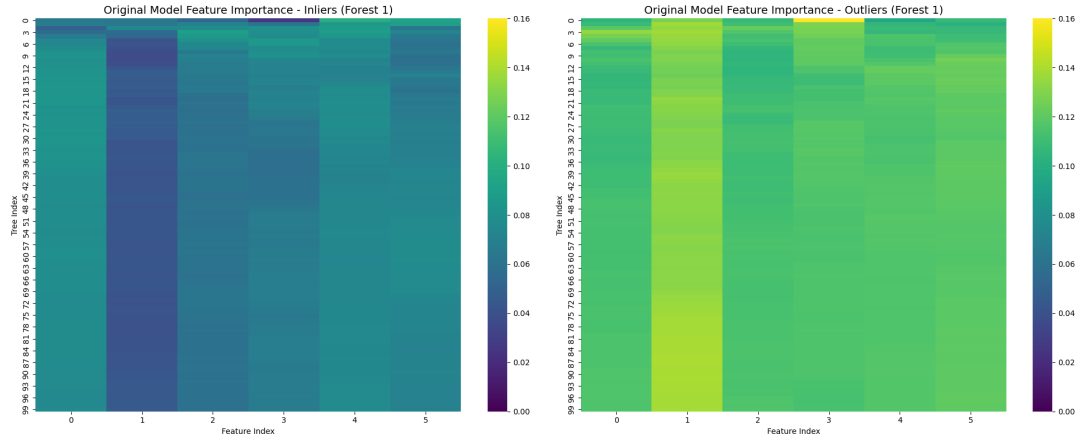


Figure 3.3: Heatmap of the FI scores for inliers (left) and outliers (right) across all individual trees in the first Isolation Forest. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

the pruning process, particularly in selecting meaningful features and determining which trees to prune based on their feature usage.

3.2.2 Meaningful features selection

This stage focuses on identifying a subset of significant features based on their GFI scores, which are used to guide tree pruning in the Isolation Forest. It arises from the question of which and how many actual features contribute most to the anomaly detection capabilities of the model.

A preliminary analysis of the distribution of GFI values (Section 3.2.1) reveals that some features have significantly higher importance scores compared to others. Looking at the frequency with which these important features are used in the trees in the forest, it becomes clear that the trees that exploit these significant features are likely more effective in isolating anomalies.

Visualizing the GFI distribution, the bar chart presented in Figure 3.2 suggests a way to identify and select the most meaningful features. In particular, a pronounced drop in importance scores is observed after the first feature. By sorting then the features by their GFI values in descending order and calculating the differences (gaps) between consecutive scores, we can determine a cut-off point where the feature’s importance drops significantly, i.e. where there is the largest gap between two consecutive GFI scores. This observation motivates the adoption of the **elbow method** for feature selection, the procedure of which is formally described in Algorithm 1.

First, we need to check whether at least two features are available for selection. Otherwise, this process cannot proceed. Therefore, the only available feature is returned, with the gap threshold and the gap sequence set to zero.

Another limited case occurs when there are exactly two features. In this situation, we have two possible scenarios, by determining the magnitude of the difference between their GFI scores:

- If the gap between the two features is significant: the importance of the second feature is considerably lower than that of the first feature (i.e., 50% or less of the first feature's GFI score), then only the first feature is selected as meaningful for the pruning process.
- Otherwise, if the gap is not significant, both features are retained as meaningful.

If there are at least two features, instead, we sort the GFIs in descending order for each Isolation Forest, first obtaining the sorting indices and then applying them as mask to the original GFI vector in order to obtain a sorted version (lines 7-8 of Algorithm 1).

Next, we compute the gap sequence by calculating the differences between consecutive GFIs for an Isolation Forest, as given in Equation 3.1:

$$\begin{aligned} \mathcal{G}[i] &= gfi^{sorted}[i] - gfi^{sorted}[i + 1] \\ \text{where } i &= 1, \dots, n_{features} - 1 \end{aligned} \quad (3.1)$$

where $\mathcal{G}[i]$ represents the gap between the i -th and $(i + 1)$ -th sorted GFI scores and $n_{features}$ is the total number of features. We therefore have a vector of gap sequences between each pair of consecutive features, in which it is possible to identify the maximum gap value, indicating the most significant drop in feature importance and therefore an ideal cut-off point for feature selection. Then we can obtain the index of the feature in which its GFI score is greater than the maximum calculated gap, as given in Equation 3.2:

$$\begin{aligned} g^{max} &= \arg \max_i (\mathcal{G}[i] + \varepsilon \cdot U(0, 1)) \\ \text{where } i &= 1, \dots, n_{features} - 1 \end{aligned} \quad (3.2)$$

where ε is a small perturbation added to the gap values to break the bonds in the case of multiple maximum gaps, and $U(0, 1)$ is a uniform random variable between 0 and 1.

Finally, with the largest feature index calculated for an Isolation Forest, we can use it as a mask to obtain the set of indices of selected features, as given in Equation 3.3:

$$\mathcal{S} = \mathbf{idx}[1 : g^{max}], \text{ where } \mathcal{S} \in \mathbb{R}^s \text{ and } s \leq n_{features} \quad (3.3)$$

where $\mathcal{S}$ represents the set of s selected feature indices for an Isolation Forest obtained from the sorting indices $\mathbf{idx}$, with features up to the index of the maximum gap included.

The output, for a single Isolation Forest, of this algorithm consists of three components:

- the set of selected feature indices $\mathcal{S}$;
- the gap threshold τ corresponding to the maximum gaps identified;
- the gap sequences $\mathcal{G}$.

These components are then used in the subsequent stage for calculating the usage score (Section 3.2.4) to determine the score for each tree, in order to perform pruning based on its dependence on the selected meaningful features.

As an example, Figure 3.4 illustrates the result of applying the elbow method to select meaningful features based on GFIs. Consistent with the pre-pruning analysis presented in Section 3.2.1, the elbow method identifies **feature 1** as the only meaningful feature to select for the pruning process, given its significantly higher GFI score compared to the others, which exhibit a substantial drop in importance score, i.e., the maximum gap (referring to Figure 3.2).

Algorithm 1 Elbow-based Feature Selection

Require: GFI vector $\mathbf{gfi} \in \mathbb{R}^{n_{features}}$

Ensure: Selected feature indices $\mathcal{S} \in \mathbb{R}^s$ where $s \leq n_{features}$, gap threshold τ and gap sequences $\mathcal{G} \in \mathbb{R}^{n_{features}-1}$

```

1: Initialize empty lists:  $\mathcal{S} \leftarrow [], \mathcal{G} \leftarrow []$ 
2: Initialize  $\tau \leftarrow 0$ 
3: if  $n_{features} < 2$  then                                 $\triangleright$  Handle edge case with insufficient features
4:    $\mathcal{S} \leftarrow \arg(\mathbf{gfi})$                                  $\triangleright$  Only one feature available
5: else if  $n_{features} == 2$  then                             $\triangleright$  Handle edge case with exactly two features
6:   if  $\mathbf{gfi}[1] - \mathbf{gfi}[2] \geq 0.5 \cdot \mathbf{gfi}[1]$  then
7:      $\mathcal{S} \leftarrow [\arg(\mathbf{gfi}[1])]$                          $\triangleright$  Select only the first feature
8:   else
9:      $\mathcal{S} \leftarrow [\arg(\mathbf{gfi}[1]), \arg(\mathbf{gfi}[2])]$          $\triangleright$  Select both features
10:  end if
11: else
12:   Initialize  $\mathbf{gfi}^{sorted} \leftarrow [], \mathbf{idx} \leftarrow []$ 
13:    $\mathbf{idx} \leftarrow \text{argsort}(\mathbf{gfi})$  in descending order
14:    $\mathbf{gfi}^{sorted} \leftarrow \mathbf{gfi}[\mathbf{idx}]$                          $\triangleright$  Sorted GFI scores for an Isolation Forest
15:   Compute gap sequences  $\mathcal{G}$  using Equation 3.1,
16:   Compute maximum gap indices  $g^{max}$  using Equation 3.2
17:   Compute selected feature indices  $\mathcal{S}$  using Equation 3.3
18:    $\tau \leftarrow \mathcal{G}[g^{max}]$ 
19: end if
20: return  $\mathcal{S}, \tau, \mathcal{G}$ 

```

3.2.3 Soft features selection

Another approach we have adopted is not selecting features based on highest GFI scores but rather to consider all of them as meaningful for the pruning process. This method, referred to as **soft selection**, assumes that all features contribute to some extent to the anomaly detection capabilities of the model, then the returned value is $\mathcal{S} \in \mathbb{R}^{n_{features}}$. Therefore, rather than filtering out features based on their importance scores, we include all of them by weighting their contributions with corresponding GFI values, as defined in Equation 3.4. This approach allows us to retain the entire feature set while still leveraging the GFIs to inform the pruning

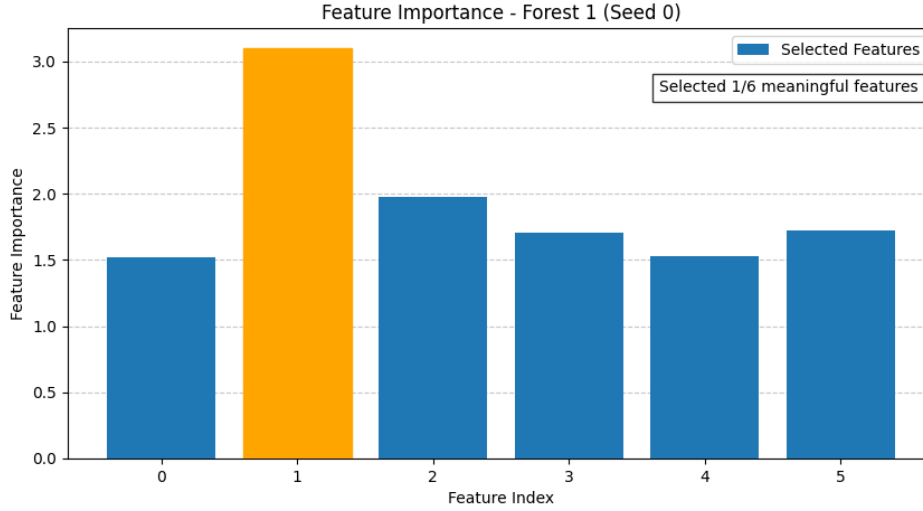


Figure 3.4: Elbow method for meaningful features selection based on GFI scores. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

process. The idea is that considering all features, we ensure that no potentially valuable information is discarded, which could be particularly beneficial in cases where some anomalies are isolated by a specific feature that may not have a high GFI score overall.

In fact, as illustrated in Figure 3.5, all features are selected for the pruning process, but their contributions are weighted according to their GFI scores. This means that while all features are considered, those with greater importance will have a more significant influence on the score calculation used for thresholding in the pruning process (detailed in Section 3.2.4).

3.2.4 Usage Score

Before computing the threshold to decide which trees to remove, it is required first to quantify how much each tree depends on identified meaningful features selected (Section 3.2.2) or on all the features (Section 3.2.3). This quantification is achieved through the computation of a **usage score** for each tree in the Isolation Forest, that reflects the frequency with which the selected features are used in the splits of a tree.

The detailed procedure to compute the usage score involves the following steps:

- **Calculating Feature Usage Score:** Formally described in Algorithm 2.
- **Normalization of the GFI:** The GFI vector of the selected features for a single Isolation Forest is normalized by dividing each GFI score by the sum of all GFIs of the selected features.

For this purpose, we first extract the **feature usage** across the entire model, i.e, a list of features' occurrences for each tree of an Isolation Forest. Each feature used

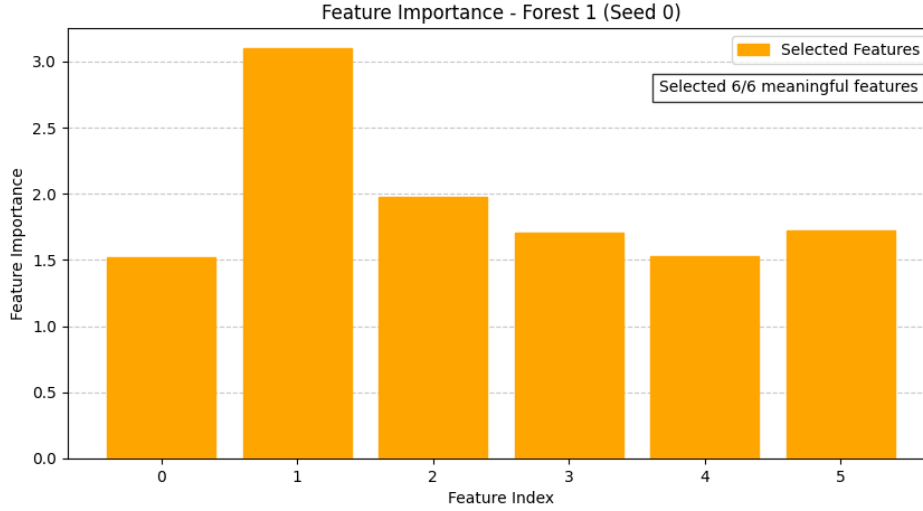


Figure 3.5: Soft features selection based on GFI scores. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

in at least one split of the tree is considered to be used by that tree. Then, we iterate through such an occurrence vector $\mathbf{f}$ and a counter vector $\mathbf{u}$ keeping track of how many times each $\mathbf{f}[i]$ -th selected feature is used in the splits of a tree. Then it is normalized by dividing it by the total number of splits in the tree to get a usage score for each selected feature.

Both usage and GFI scores are normalized to ensure that they are on a comparable scale. Normalizing feature usage scores based on the total number of tree splits allows us to account for differences in tree size and structure, ensuring that trees with more splits do not inherently have higher usage scores. Similarly, normalizing GFI scores ensures that the importance of each feature is considered relative to the others, allowing for a fair comparison when weighting feature usage scores.

Algorithm 2 Feature Usage Quantification

Require: Occurrence vector $\mathbf{f} \in \mathbb{R}^d$ for a tree, where d represents the total number of occurrences of all features used in splits of the tree; Selected feature indices $\mathcal{S}$

Ensure: Normalized features usage score vector $\mathbf{u} \in \mathbb{R}^s$ for a tree

```

1: Initialize  $\mathbf{u} \leftarrow \mathbf{0}_s$ 
2: for  $i = 1$  to  $d$  do
3:   if  $\mathbf{f}[i] \in \mathcal{S}$  then
4:      $\mathbf{u}[\mathbf{f}[i]] \leftarrow \mathbf{u}[\mathbf{f}[i]] + 1$ 
5:   end if
6:   Initialize  $total\_splits \leftarrow 0$ 
7:   if feature index  $i$  is used to split an internal node then
8:      $total\_splits \leftarrow total\_splits + 1$ 
9:   end if
10: end for
11: Normalize  $\mathbf{u}$  by dividing each element by  $total\_splits$ 
12: return  $\mathbf{u}$ 

```

3.2.5 Threshold computation

At this point, we can proceed with the computation of the threshold to decide which trees to prune from the Isolation Forest, which is calculated based on usage score. Three different approaches are proposed: **mean**, **percentile**, and **majority vote**. Moreover, it is possible to combine different feature selection methods (elbow method or soft selection) with any of the threshold methods, with the purpose of evaluating their effectiveness in the pruning process and determining the best combination for optimal performance.

However, an additional step must be performed, which differs slightly between the mean and percentile methods and the majority vote approach. Specifically, it is necessary to calculate the weighted sum for each tree within the Isolation Forest, as defined in Equation 3.4:

$$ws_i = \sum_{j \in \mathcal{S}} u[j] \cdot gfi_norm[j] \quad (3.4)$$

where $gfi_norm \in \mathbb{R}^s$ is the normalized GFI vector for the selected features $\mathcal{S}$ for an Isolation Forest. The weighted sum ws_i represents the general reliance of the i -th tree on the selected j -th meaningful features, taking into account both the frequency with which those features are used in the tree and their importance, as indicated by GFIs. Then, similar to feature usage scores, the weighted sums are normalized by dividing each weighted sum ws_i by the sum of all normalized GFIs of the selected features for the tree i -th, as defined in Equation 3.5:

$$ws_i^{norm} = \frac{ws_i}{\sum_{j \in \mathcal{S}} gfi_norm[j]} \quad (3.5)$$

where ws_i^{norm} is the normalized weighted sum for the tree i -th. This normalization ensures that the weighted sums are on a comparable scale, facilitating the threshold

computation in subsequent steps.

Finally, for the **mean** and **percentile** methods, we compute the threshold θ based on the normalized weighted sum of each tree in the Isolation Forest. The **mean** threshold is simply the average of all normalized weighted sums, as given in Equation 3.6:

$$\theta_{mean} = \frac{1}{n_{trees}} \sum_{i=1}^{n_{trees}} ws_i^{norm} \quad (3.6)$$

where n_{trees} is the total number of trees in the Isolation Forest.

Although the **percentile** threshold is determined by selecting a specific percentile of the normalized weighted sum distribution, as given in Equation 3.7:

$$\theta_{percentile} = P_p(\{ws_1^{norm}, ws_2^{norm}, \dots, ws_{n_{trees}}^{norm}\}) \quad (3.7)$$

where $P_p = x_{(\lceil p \cdot \frac{n_{trees}+1}{100} \rceil)}$

where P_p denotes the p -th percentile function. The choice of percentile (e.g., 80th, 70th, 60th, 50th) allows for flexibility in determining the aggressiveness of pruning trees based on their dependence on the selected features.

When deciding which trees to be removed, for the mean and percentile methods, a comparison is made between the normalized weighted sum ws_i^{norm} of the i -th tree and the calculated threshold θ . Naturally, trees with a normalized weighted sum below the threshold are pruned from the Isolation Forest, as they are considered less dependent on the selected meaningful features and therefore less effective for anomaly detection.

3.2.5.1 Majority vote mechanism

The final step for the **majority vote**, which differs from the previous two methods, involves a different approach to quantifying the reliance of each tree on the selected meaningful features. Instead of calculating a weighted sum as in Equation 3.4, a vote mechanism is employed, the complete procedure of which is formally described in Algorithm 3. In this approach, each tree receives a vote based on whether its feature usage score, for a given selected feature, is greater than or equal to the average score of that feature across all trees in the Isolation Forest. This average score is computed as given in Equation 3.8:

$$a_j = \frac{1}{n_{trees}} \sum_{i=1}^{n_{trees}} U[i][j] \quad \text{for } j \in \mathcal{S} \quad (3.8)$$

where a_j represents the average feature usage score for the j -th selected in the entire Isolation Forest, $U[i][j]$ is the normalized usage score for the j -th selected feature in the tree i -th, and n_{trees} is the total number of trees in the Isolation Forest.

Then, for each tree i and a given selected feature j , if the feature usage score $U[i][j]$ is greater than or equal to the average score a_j , the tree votes for that feature (i.e., it gets a vote of 1); otherwise, it does not vote for that feature (i.e., it gets a vote of 0). Once all votes are collected for each selected feature, the total vote score for each tree is computed, as defined in Equation 3.9:

$$v_i = \sum_{j \in \mathcal{S}} B[i][j] \cdot gfi_norm[j] \quad (3.9)$$

where v_i represents the total vote score for the tree i -th, $B[i][j]$ is the binary vote (1 or 0) for the selected feature j -th in the tree i -th and $gfi_norm[j]$ is the normalized GFI value for the selected feature j -th. This vote score reflects how many selected features a tree uses more than average, weighted by their importance as indicated by the GFI scores.

Equation 3.9 clearly shows a difference from the weighted sum calculation in Equation 3.4, where actual usage scores are considered instead of the vote score. This difference allows the majority vote mechanism to focus on whether a tree uses selected features more than average, rather than on how much it uses them.

To remove trees using the majority vote mechanism, a comparison is made between vote score of each tree v_i and a threshold θ_{vote} , the value of which is a parameter to be set within the range $[0, 1]$. Again, trees with a vote score lower than the threshold are removed from the Isolation Forest, as they are considered less effective for anomaly detection.

Algorithm 3 Majority vote Feature Usage Quantification

Require: Feature usage score matrix $\mathbf{U} \in \mathbb{R}^{n_{trees} \times s}$, where each row $\mathbf{u} \in \mathbb{R}^s$ represents the normalized feature usage scores for every tree; Normalized GFI vector $\mathbf{gfi_norm} \in \mathbb{R}^s$

Ensure: vote score vector $\mathbf{v} \in \mathbb{R}^{n_{trees}}$

```

1: Initialize  $\mathbf{v} \leftarrow \mathbf{0}_{n_{trees}}$ 
2: Initialize  $\mathcal{B} \leftarrow []$ ,  $\mathcal{B} \in \mathbb{R}^{n_{trees} \times s}$ , matrix of binary votes
3: Initialize  $\mathbf{a} \leftarrow \mathbf{0}_s$ , vector of average feature usage scores across all trees in the
   Isolation Forest
4: Compute  $\mathbf{a}$  using Equation 3.8
5: for  $i = 1$  to  $n_{trees}$  do
6:   for  $j = 1$  to  $s$  do
7:     if  $U[i][j] \geq \mathbf{a}[j]$  then
8:        $B[i][j] \leftarrow 1$  ▷ Tree  $i$  votes for feature  $j$ 
9:     else
10:       $B[i][j] \leftarrow 0$  ▷ Tree  $i$  does not vote for feature  $j$ 
11:    end if
12:  end for
13:  Compute  $\mathbf{v}[i]$  using Equation 3.9
14: end for
15: return  $\mathbf{v}$ 

```

3.3 Pruning based on depth

Another pruning strategy investigated in this thesis works leverages the intrinsic structure of the Isolation Forest, specifically focusing on the **depth** at which data points (both normal and anomalous) are isolated within each tree. The rationale behind this approach is that trees isolating normal points too shallowly or anomalies too deeply may be misaligned with the expected behavior of the algorithm. Therefore, by examining depth-based statistics, we can identify and remove trees that provide limited or even harmful contributions to the anomaly detection performance

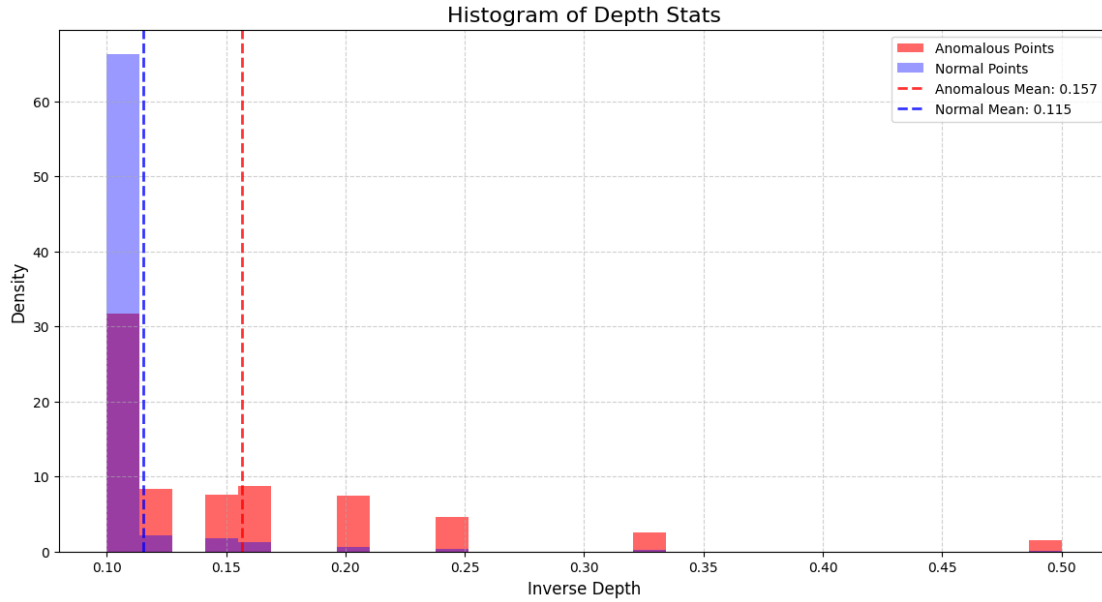


Figure 3.6: Density distributions of inverse depth scores for normal points (blue) and anomalous points (red) in a single Isolation Forest. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

of the model.

Regarding DIFFI, in Section 3.2.1, we start with a trained and unpruned Isolation Forest model to conduct a preliminary analysis of the depth distribution corresponding to the predicted normal and anomalous points (Section 3.3.1). The comparison of these distributions is designed with the purpose of validating the core principle of Isolation Forest, mentioned in Section 2.1: *anomalies are expected to be isolated at shallower depths than at normal points*.

3.3.1 Depth distribution analysis

Once the model was trained, the isolation depth is extracted for each data point in each tree in the forest. This produces two separate distributions: one for predicted *inliers* and one for predicted *outliers*. Using them, we look for the difference in isolation depths between the two classes of data points.

The histogram in Figure 3.6 illustrates the density distributions of the inverse depth scores for both classes of data points within a single Isolation Forest. The *x-axis* shows the inverse depth scores, calculated as the reciprocal of the isolation depth (i.e., $1/\text{depth}$). This transformation is used to better visualize the distribution, since shallower depths (indicating easier isolation) correspond to higher inverse depth scores. The *y-axis* represents the density of data points observed at different score levels, indicating how frequently instances fall within each score interval.

Examining the histogram, we can observe that the center of the depth distribution for anomalies is shifted towards the right side, towards shallower depths, compared to that of normal points. This indicates that the outliers tend to be isolated earlier in the tree than inliers, i.e., the mean value of the inverse depth scores

for anomalous points is higher than that for normal points. This observation is consistent with the fundamental principle of Isolation Forest, according to which anomalies are usually isolated at shallower depths than at inliers.

However, it is also evident that there is a greater degree of overlap between the two distributions, revealing that many individual trees do not effectively separate normal from anomalous points. This overlap can be caused by two main factors:

- Unfortunate random splits in the tree construction process, i.e., the random selection of features and threshold values, are not optimal for isolating anomalies quickly.
- Swamping effect, i.e., some outliers are located close to the normal data points in the feature space, making it difficult for the Isolation Forest to isolate them at shallower depths.

Looking closely at Figure 3.6 in the overlap region, in the inverse depth score range from approximately 0.10 to 0.25, we can see that both normal and anomalous points are isolated at the same depth levels. This suggests that some trees in the Isolation Forest may not be effectively contributing to anomaly detection capabilities, as they fail to isolate normal points at significantly lower depths.

It is therefore clear that within an Isolation Forest ensemble, there are non-informative trees, and that this *overlap* effect can be exploited to identify and remove such trees from the model.

Based on this preliminary analysis, a natural question arises: *What are the criteria for identifying and removing these non-discriminating trees from the Isolation Forest?* To address this question, we exploit the result obtained from the histogram in Figure 3.6: since the overlap between the two distributions indicates that some trees do not effectively separate inliers from outliers, we can use this information to define pruning criteria. More in detail, we can calculate the ratio between the average isolation depth for the anomalous and normal points in each tree, called the **separation ratio**, as defined in Equation 3.10:

$$SR_i = \frac{D_{outliers}^i}{D_{inliers}^i} \quad (3.10)$$

where SR_i is the separation ratio for the tree i -th, $D_{outliers}^i$ is the average isolation depth of the outliers, and $D_{inliers}^i$ is the average isolation depth of the inliers.

This ratio provides a quantitative measure of how well each tree isolates the two classes of data points based on their isolation depth.

Figure 3.7 illustrates the separation ratio scores for each tree in a single Isolation Forest. The x -axis represents the tree indices, while the y -axis shows the corresponding separation ratio scores. A higher value means that a tree effectively isolates anomalous points at depths lower than normal points, thus positively contributing to the anomaly isolation capabilities of the model. In contrast, trees with lower separation ratios are considered "non-informative", as they do not effectively separate the two classes.

By analyzing these scores, we can identify the least effective trees to isolate anomalies and consider them to prune the isolation forest.

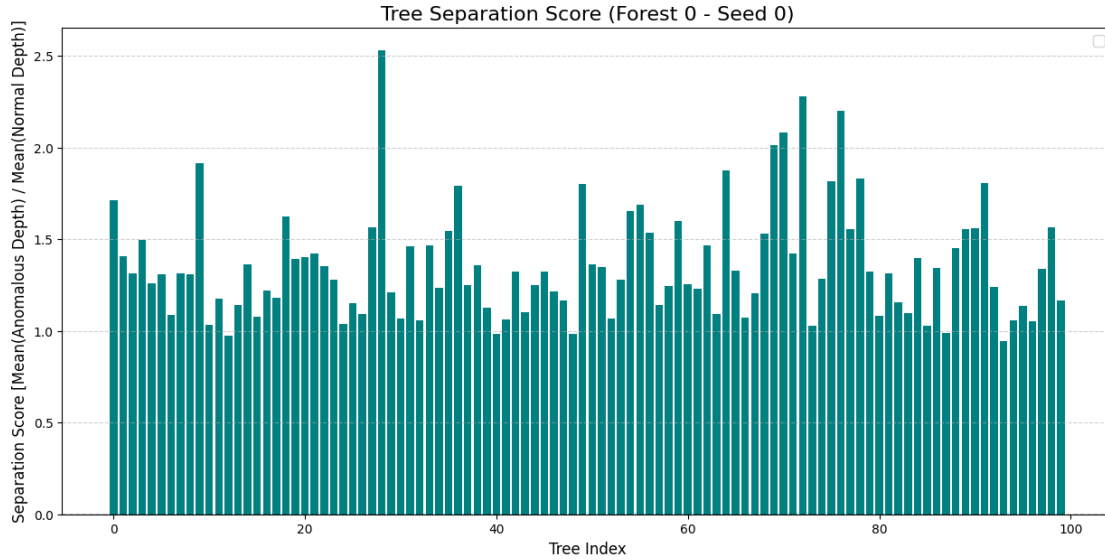


Figure 3.7: Separation ratio scores for each tree in a single Isolation Forest. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

In addition, it is also interesting to group and count the number of trees based on their separation ratio scores. This grouping allows us to see how many trees fall into different separation efficiency ranges, providing insight into the overall quality of the trees in the Isolation Forest.

Figure 3.8 shows the distribution of trees based on their separation ratio scores in a single Isolation Forest. From the plot, we can observe that a significant number of trees have low scores. This insight justifies the need to eliminate such trees from the model to improve its effectiveness. Furthermore, we can note that very few trees have high separation ratio scores, indicating that only a small subset of trees is highly effective at the isolation of anomalies, and therefore the set size can be reduced by up to 80% without losing significant anomaly detection capabilities, as mentioned in Section 2.3.

Another possible criterion for pruning trees is to focus exclusively on the average isolation depth of anomalous points ($D_{outliers}^i$) within each tree i -th, referred to as **anomaly depth**. Trees with higher isolation depths for anomalies may be less useful for anomaly detection, as they do not isolate outliers earlier in the tree.

This latter criterion is a weaker strategy than the separation ratio, at theoretical level, since it does not take into account the isolation depth of normal points. However, it is still adopted to evaluate its effectiveness in the pruning process and to validate the eventual superiority of the separation ratio over it.

3.3.2 Threshold computation

Based on the analysis of depth distribution presented in Section 3.3.1, we have defined two criteria for the pruning of trees from the Isolation Forest: **separation ratio** and **anomaly depth**.

Similarly to the threshold computation methods described in Section 3.2.5, we

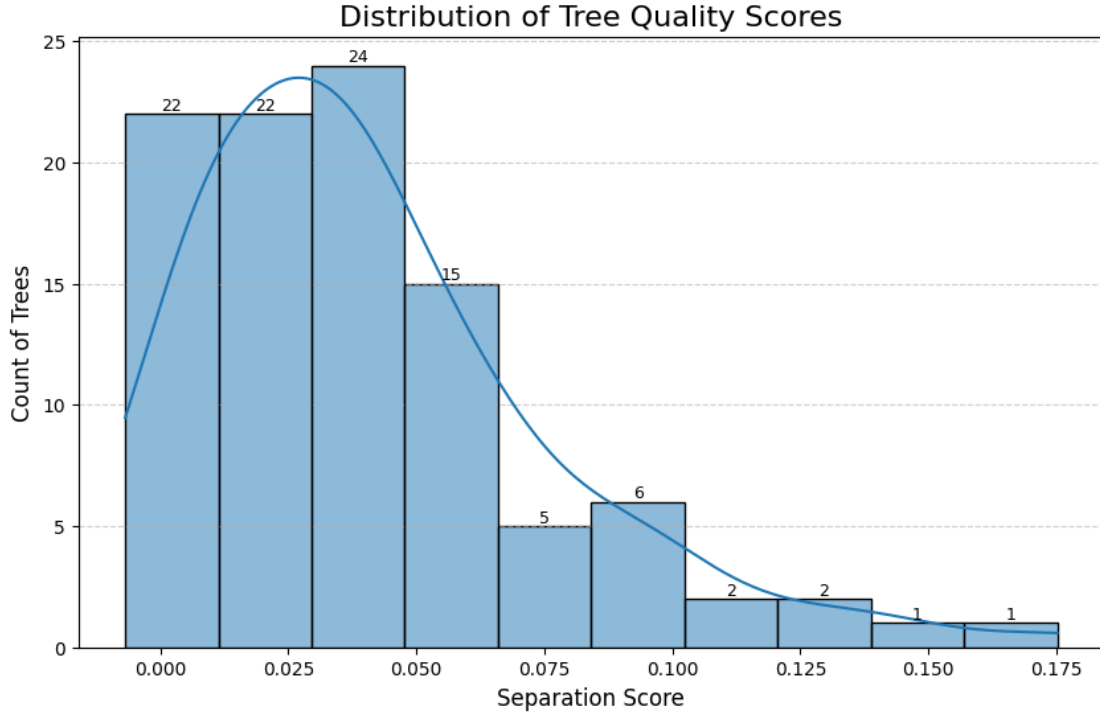


Figure 3.8: Distribution of trees based on their separation ratio scores in a single Isolation Forest. This example comes from the experiments conducted on the *Thyroid* dataset (Section 4.1.2.1). For further details on the experimental setup, refer to Section 4.2.2.

can again apply both the **mean** and **percentile** methods to calculate the pruning thresholds based on these two criteria.

The **mean** threshold is calculated as the average *separation ratio* or the average *anomaly depth* in all trees in the Isolation Forest, as defined in Equations 3.11 and 3.12, respectively:

$$\theta_{SR_mean} = \frac{1}{n_{trees}} \sum_{i=1}^{n_{trees}} SR_i \quad (3.11)$$

$$\theta_{AD_mean} = \frac{1}{n_{trees}} \sum_{i=1}^{n_{trees}} D_{outliers}^i \quad (3.12)$$

where θ_{SR_mean} is the mean threshold for the separation ratio, θ_{AD_mean} is the mean threshold for the depth of anomaly, and n_{trees} is the total number of trees in the Isolation Forest.

The **percentile** threshold is determined by selecting a specific percentile of the *separation ratio* or *anomaly depth* distribution, as defined in Equations 3.13 and 3.14, respectively:

$$\theta_{SR_percentile} = P_p(\{SR_1, SR_2, \dots, SR_{n_{trees}}\}) \quad (3.13)$$

where $P_p = x_{(\lceil p \cdot \frac{n_{trees}+1}{100} \rceil)}$

$$\theta_{AD_percentile} = P_p(\{D_{outliers}^1, D_{outliers}^2, \dots, D_{outliers}^{n_{trees}}\}) \quad (3.14)$$

where $P_p = x_{(\lceil p \cdot \frac{n_{trees}+1}{100} \rceil)}$

where $\theta_{SR_percentile}$ is the percentile threshold for the separation ratio, $\theta_{AD_percentile}$ is the percentile threshold for the anomaly depth, and P_p denotes the function of p -th percentile.

When deciding which trees to remove, for the mean and percentile methods, a comparison is made again between the separation ratio SR_i or the anomaly depth $D_{anomalies}^i$ of the i -th tree and the calculated threshold θ .

Chapter 4

Experiments

This chapter describes the experiments conducted to evaluate the proposed pruning strategies for the Isolation Forest. The experiments are designed to assess the effectiveness of both the DIFFI- and Depth-based pruning methods on various datasets, including synthetic and real-world scenarios. The chapter is structured as follows: Section 4.1 details the datasets used in the experiments, Section 4.2 outlines the experimental setup and evaluation metrics, Section 4.3 presents the results obtained from the experiments and Section 4.4 discusses the impact of the number of estimators on the performance of the pruned models.

4.1 Datasets

In the context of Anomaly Detection, there are a wide variety of datasets that can be used to evaluate the performance of different models and techniques. Several categories of datasets are available, including:

- **Tabular datasets:** Structured data organized into rows and columns, where each row represents an instance and each column represents a feature;
- **Time-series datasets:** Contain sequential data points collected over time, often used in applications such as sensor data analysis and network traffic monitoring;
- **Image datasets:** They consist of images, often used in computer vision tasks to detect anomalies in visual data, such as identifying manufacturing defects or unusual patterns in medical imaging.

Since the model used in this study, Isolation Forest, is mainly designed for tabular data, experiments are conducted using this kind of dataset. We selected a combination of synthetic and real-world datasets to comprehensively evaluate the effectiveness of the proposed pruning strategies for Isolation Forest. The synthetic dataset allows controlled experiments where specific features can be manipulated, while the real-world datasets provide practical scenarios that reflect the complexities encountered in actual applications.

For the experiments, we assume that the ground truth labels are unknown during the training stage, as is typical in unsupervised anomaly detection tasks. For

evaluation purposes, the ground truth labels are used to evaluate the performance of the model after pruning.

4.1.1 Synthetic dataset

The synthetic dataset used in the experiments is generated to create a controlled environment in which to evaluate the pruning strategies. The dataset includes a mix of normal and anomalous instances, strategically placed to test the ability of the model to effectively identify them.

The generation process involved defining specific distributions for the normal data points and introducing anomalies that significantly deviate from those distributions. This setup allows for a clear assessment of the effectiveness of the pruning methods in maintaining or improving the model’s anomaly detection performance while reducing the ensemble size.

The configuration parameters for generating the synthetic dataset are identical to those used in the DIFFI paper (Carletti et al., 2021), ensuring consistency with established research in the field. Each point in the dataset is generated as shown in Equation 4.1:

$$\mathbf{x}_i = [\rho \cos(\theta), \rho \sin(\theta), n_1, \dots, n_{p-2}]^T \quad (4.1)$$

where $n_j \sim \mathcal{N}(0, 1)$ for $j = 1, \dots, p - 2$, are white noise samples. The parameters ρ and θ are used to generate data points and for regular data we have:

$$\theta \sim \mathcal{U}(0, 2\pi), \quad \rho \sim \mathcal{U}(0, 3); \quad (4.2)$$

while for anomalies:

$$\theta \sim \mathcal{U}(0, 2\pi), \quad \rho \sim \mathcal{U}(4, 30). \quad (4.3)$$

The number of dimensions p is set to 6, resulting in a six-dimensional dataset: Two informative features (the first two dimensions) and four noise features. The dataset consists of a total of 1000 data points, with 100 anomalies, representing a contamination rate of 10%.

Figure 4.1 provides a visual representation of an instance of the generated synthetic dataset, projected onto the first two informative dimensions.

4.1.2 Real-world datasets

The real-world datasets, reflecting some practical domains, vary in terms of dimensionality, size, and nature of the anomalies present, allowing for a comprehensive evaluation of the proposed pruning strategies. The datasets used in this study are described in the following section.

4.1.2.1 ODDS dataset

The **Outlier Detection DataSets** (ODDS, Rayana, 2016) is a popular collection of datasets specifically curated to evaluate anomaly detection algorithms. It includes a variety of datasets from different domains, each with its own characteristics and challenges. For the experiments on this study, several datasets from the ODDS repository were selected based on their relevance to the Isolation Forest model and

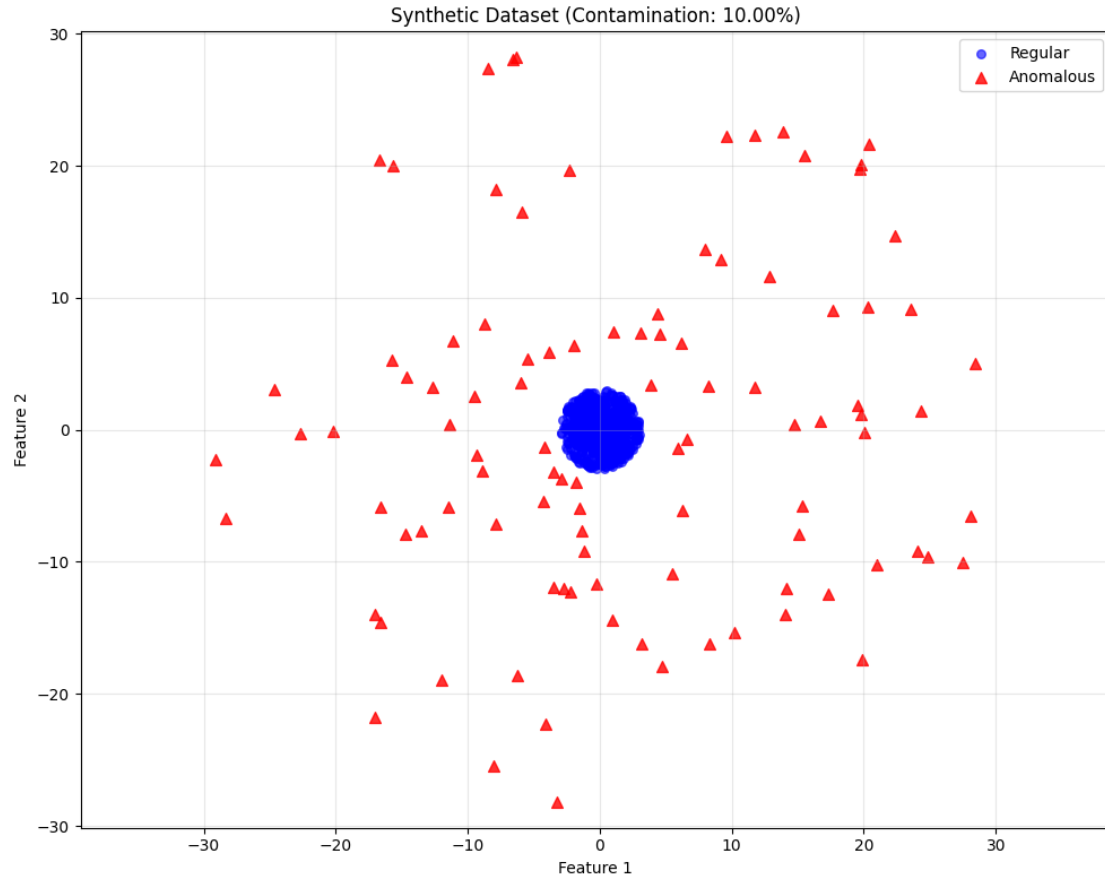


Figure 4.1: Visualization of a synthetic dataset instance used in the experiments, projected onto the first two informative dimensions. Normal instances are represented in blue, while anomalies are shown in red.

Table 4.1: A summary of characteristics of the datasets used from ODDS Rayana, 2016.

Dataset	#points	#dim.	#outliers (%)
Vowels	1456	12	50 (3.4%)
Cardio	1831	21	176 (9.6%)
Thyroid	3772	6	93 (2.5%)
Satimage-2	5803	36	71 (1.2%)
Letter Recognition	1600	32	100 (6.25%)
Pima	768	8	268 (35%)
Satellite	6435	36	2036 (32%)
BreastW	683	9	239 (35%)
Ionosphere	351	33	126 (36%)
Mnist	7603	100	700 (9.2%)
Optdigits	5216	64	150 (3%)
Anthyroid	7200	6	534 (7.42%)
Pendigits	6870	16	156 (2.27%)
Wine	129	13	10 (7.7%)
Vertebral	240	6	30 (12.5%)

the presence of labeled anomalies for evaluation purposes. Table 4.1 provides a summary of the characteristics of the selected datasets, including size, dimensionality (i.e., the number of features) and contamination rates (the amount of outliers). These datasets were chosen to cover a range of scenarios, from low-dimensional to high-dimensional spaces, and different levels of contamination, enabling a reliable evaluation of pruning strategies.

4.2 Experimental setup

The experimental setup for evaluating the proposed Isolation Forest pruning strategies involved several key components to ensure a systematic and reproducible approach:

- **Model configurations:** Definition of parameters for Isolation Forest model, as detailed in Section 4.2.1;
- **Logging procedures:** Establishing protocols for recording relevant data before and after the pruning process to simplify analysis;
- **Pruning processes:** Implementation of DIFFI- and depth-based pruning strategies, as outlined in Sections 4.2.2 and 4.2.3;
- **Evaluation metrics:** Select appropriate metrics to assess the performance of the pruned models, as explained in Section 4.2.4.

In this section, we describe in detail the experimental setup used in the study. Figure 4.2 illustrates the overall workflow adopted for the experiments, highlighting the steps involved in the evaluation process:

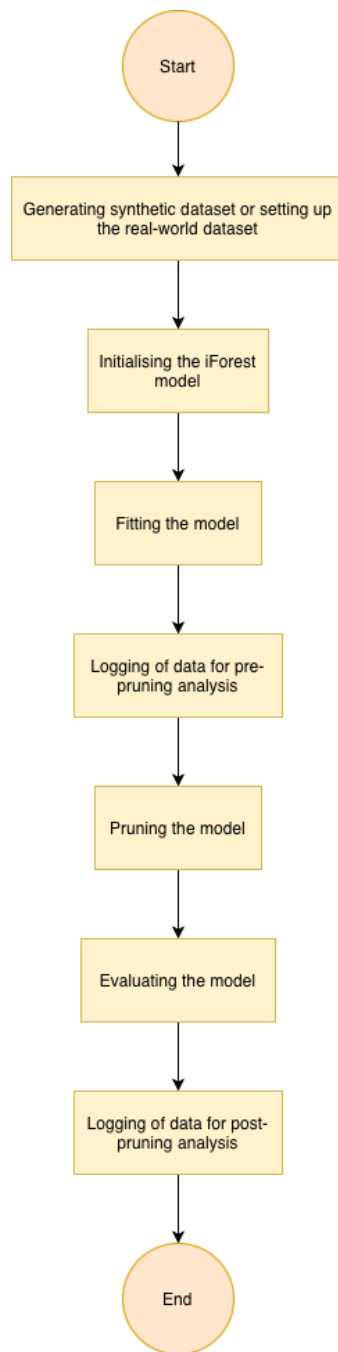


Figure 4.2: Workflow adopted for the experiments conducted to evaluate the pruning strategies for the Isolation Forest.

1. **Setting up the dataset:** In this initial phase, for synthetic dataset only, the data generation process is performed based on predefined configurations. Subsequently, for both synthetic and real-world datasets, the contamination rate is calculated to appropriately initialize the Isolation Forest model, ensuring it is aware of the expected proportion of anomalies in the data.
2. **Initialization of the Isolation Forest model:** It is configured with specific parameters, such as the amount of estimators and the maximum number of samples, as detailed in Section 4.2.1. This step ensures that the model is set up consistently across all experiments.
3. **Training the Isolation Forest:** The model is trained on the selected dataset.
4. **Logging pre-pruning data:** Relevant data, such as *GFI* and *depth* distributions, are recorded before the pruning process to facilitate subsequent analysis, as described in Sections 3.2.1 and 3.3.1.
5. **Applying pruning strategies:** DIFFI- and depth-based pruning methods are applied to the trained Isolation Forest, explained in detail in Sections 3.2 and 3.3.
6. **Evaluating pruned models:** The performance of pruned models is assessed using selected evaluation metrics, as described in Section 4.2.4.
7. **Logging post-pruning data:** Recording data obtained after the pruning process in order to analyze its impact on model performance and compare the results of the different pruning strategies. More specifically, the same data logged before pruning is recorded again afterwards, along with the number of estimators remaining in the pruned model and the values of the evaluation metrics.

To maintain controlled randomness in experiments, seed values are set to offer several benefits:

- **Reproducibility:** Setting a seed allows for exact replication of results, which is crucial to validate the findings and compare different approaches.
- **Controlled variability:** The inherent randomness of the Isolation Forest algorithm can be controlled, allowing for a more consistent evaluation of pruning strategies across runs.
- **Fair comparisons:** When comparing different pruning methods, using the same seed ensures that any observed differences in performance are due to the methods themselves rather than random variations in the data or model initialization.
- **Debugging and analysis:** Seeds simplify debugging and in-depth analysis of the model behavior, as the same random sequences can be reproduced for investigative purposes.

- **Statistical significance:** Using multiple seeds allows us to assess the statistical significance of our results, providing insights on the robustness of our pruning strategies under different random conditions.
- **Consistent data splits:** In cases where the dataset is split into training and testing sets, using a seed ensures that the same split is used across different experiments, allowing for a fair comparison of model performance.

In the Isolation Forest implementation, its `random_state` parameter is also set using seed values to control pseudo-randomness in features selection and their split values during the tree construction process.

4.2.1 Model configurations

The Isolation Forest model is configured with specific parameters to ensure consistent and fair evaluation across all experiments. Key configurations included:

- **Number of estimators (`n_estimators`):** The size of the ensemble is set to a fixed value to evaluate all experiments. However, a discussion about the impact of varying this parameter is provided in Section 4.4.
- **Maximum samples (`max_samples`):** This parameter controls the number of samples drawn to train each base estimator in the ensemble. As mentioned in Section 2.1.1, it is useful in reducing *swamping* and *masking* effects.
- **Number of forests (`n_forests`):** Sets the number of independent Isolation Forests to train for each experiment.
- **Test size (`test_size`):** This parameter defines the proportion of the dataset used to testing the performance of the model after pruning.

For some of the above parameters, specific values are chosen based on preliminary experiments and established practices in the literature. For instance, the values chosen for `n_estimators` and `max_samples` parameters are standard choices, according to the original Isolation Forest paper (Liu et al., 2009). A summary of the model configurations is provided in Table 4.2.

Table 4.2: Summary of Isolation Forest model configurations used in the experiments.

Parameter	Value
<code>n_estimators</code>	100
<code>max_samples</code>	256
<code>n_forests</code>	10
<code>test_size</code>	[0, 0.2]

To ensure robustness and account for variability of results due to the random nature of the Isolation Forest, each experiment is repeated across multiple independent forests. The final performance metrics are averaged across these multiple runs to provide a more reliable assessment of the pruning strategies.

The test size is initially set to 0 (i.e., no hold-out set), then the entire dataset is used

to train the model. Because the datasets used are static — fixed and invariant over time — and even though a hold-out test is created, the two smaller subsets come from the same distribution as the entire original dataset. Therefore, for explanatory purposes, a separate test set is not necessary, as the primary goal is to evaluate the pruning strategies rather than generalizing to unseen data. Even without train-test split, no information leakage occurs, as no ground-truth labels are used during training, given the unsupervised nature of the task.

However, for predictive scenarios, in which the model is expected to generalize to previously unseen data, a test size of 0.2 is also considered (i.e., 20% of the dataset is held out for testing). This allows us to evaluate the performance of pruned models on new data, providing insights into their generalization capabilities.

The number of seeds used for the experiments depends on the test size configuration. When the `test_size` is set to 0, 5 different seeds are used to ensure sufficient variability in the results, then obtaining a total of $n_forests \cdot seed$ independent runs. On the other hand, when the `test_size` = 0.2, a larger number of seeds, precisely 30, is used to provide a more comprehensive evaluation of the performance of the model on unseen data. This setting allows for a thorough assessment of the pruning strategies under different conditions, ensuring that the results are reliable and generalizable.

4.2.2 DIFFI-based

For DIFFI-based pruning strategy, the theoretical details of which are provided in Section 3.2, the experimental setup involved several specific configurations:

- **Feature selection method:** The method used to select the meaningful features from the DIFFI explanations, as described in Section 3.2.2;
- **Thresholding approach:** The approach used to determine the pruning threshold based on *usage scores*, as described in Section 3.2.5.

A summary of DIFFI-based pruning configurations is provided in Table 4.3 For

Table 4.3: Summary of DIFFI-based pruning configurations used in the experiments.

Parameter	Values
Feature selection method	[<i>elbow</i> , <i>soft</i>]
Thresholding approach	[<i>mean</i> , <i>majority vote</i> , <i>percentile</i>]

majority vote approach, a threshold value of 0.5 is chosen, which means that the trees must vote for at least half of the selected features to be retained in the model. However, other threshold values can be experimented with to observe their impact on pruning effectiveness.

For the *percentile* approach, several values - [50%, 60%, 70%, 80%] - are tested to evaluate the impact of varying pruning aggressiveness on model performance. The choice of these percentile values is based on empirical observations from preliminary experiments, in which it was observed that pruning above 80% often leads to significant performance degradation, while lower percentiles may not provide

sufficient pruning to observe a performance increase.

The experiments are conducted by systematically varying the *feature selection* and *thresholding* methods to evaluate their impact on the performance of the pruned Isolation Forest model and to determine the most effective configuration for anomaly detection tasks.

4.2.3 Depth-based

For the depth-based pruning strategy, described in detail in Section 3.3, the experimental setup included specific configurations:

- **Depth metrics:** These are used to create a classification, in decreasing order, of trees in the Isolation Forest, as described in Section 3.3.2;
- **Thresholding approach:** The approach used to determine the pruning threshold based on depth metrics.

Table 4.4 provides a summary of the depth-based pruning configurations. For the

Table 4.4: Summary of depth-based pruning configurations used in the experiments.

Parameter	Values
Depth metrics	[<i>separation ratio</i> , <i>anomaly depth</i>]
Thresholding approach	[<i>mean</i> , <i>percentile</i>]

percentile approach, identical to the DIFFI-based method, the same percentile values are tested.

For this pruning strategy, the *majority vote* approach is not considered as it has proven to be one of the least effective methods in the DIFFI-based pruning experiments (Section 4.3.2) and is therefore not included in the depth-based pruning ones.

4.2.4 Evaluation metrics

To assess the performance of pruned Isolation Forest models, two main evaluation metrics are employed:

- **F1-Score:** is useful for a classification model on an imbalanced dataset where one class (normal instances) is more frequent than the other class (anomalies). It is calculated as the harmonic mean of *precision* and *recall* (Buitinck et al., 2013).

Precision measures the percentage of the anomaly predictions that are actually correct - the amount of actual anomalies out of all those predicted. Recall, on the other hand, measures the percentage of actual anomalies correctly identified by the model.

The impact of the precision metric is particularly significant in anomaly detection tasks, where false positives (normal instances incorrectly classified as anomalies) can lead to unnecessary investigations. Therefore, high precision indicates that the model is effective in minimizing false alarms, which is

crucial in practical applications, but it can also lead to some actual anomalies being missed (i.e., low recall).

Higher recall is equally important, as it ensures that most of the actual anomalies are detected, reducing the risk of overlooking critical issues. However, focusing solely on recall can lead to an increase in false positives, which can be costly and inefficient.

Therefore, F1-score provides a balanced measure that considers both precision and recall, making it a suitable metric to evaluate anomaly detection models. Using this metric, a model with a high F1-score indicates that both precision and recall are high too, thus being effective at identifying anomalies, avoiding missing real anomalies, and minimizing false alarms.

However, because the Isolation Forest produces continuous anomaly scores rather than binary classifications, it is important to note that the F1-score is calculated based on a specific threshold for classifying instances as anomalies or normal (i.e., *threshold-dependent*). Therefore, it provides a snapshot of the performance at that particular threshold, which may not capture the full spectrum of the capabilities of the model at different thresholds. For example, two models might have the same F1-score at a given threshold, but one might perform better at other thresholds. Hence, while F1-score is a valuable metric, it should be interpreted in the context of the chosen threshold.

- **Average Precision (AP):** A single score that summarizes the performance of a model by calculating the area under its *precision-recall curve* (Buitinck et al., 2013). The core idea behind AP is to rank all instances of a model based on their anomaly scores, in descending order, and then calculate the precision at various threshold levels. All these precision values are then averaged, giving greater weight to thresholds that result in greater recall increases (i.e., *threshold-independent*).

It is particularly useful as a complement to the F1-score, as the choice of threshold can significantly impact the performance of the model. By considering the area under the precision-recall curve, AP provides a more comprehensive evaluation of the effectiveness of the model in identifying anomalies.

This characteristic makes AP a valuable metric for assessing the overall quality of the anomaly detection model, providing insights into its robustness and reliability in different operating conditions.

These evaluation metrics are calculated both before and after the pruning process to assess the impact of the proposed strategies on performance of the model. This comparative analysis allows for a comprehensive understanding of how pruning affects the ability to detect anomalies Isolation Forest.

For the purposes of this study, the **number of estimators** after pruning is also recorded to analyze the trade-off between model complexity and performance. Specifically, we examine the relationship between the number of trees retained in the ensemble and the corresponding evaluation metrics. The effectiveness of pruning strategies can be assessed in terms of both *efficiency* (i.e., reduction in model size) and *effectiveness* (i.e., maintenance or improvement in anomaly detection performance).

4.3 Results analysis

In this Section, we report and analyze the results obtained from experiments conducted to evaluate the proposed pruning strategies for the Isolation Forest, on both synthetic and real-world datasets.

The presentation of results is structured around the three pruning strategies examined in this work. First, Section 4.3.1 introduces the baseline scenario, where trees are removed from the Isolation Forest at random. Section 4.3.2 then details the outcomes of the DIFFI-based pruning method, highlighting its behaviour across the evaluated datasets. Finally, Section 4.3.3 discusses the results associated with the Depth-based approach.

The following sections are organized to make it easier to clearly understand the performance of each pruning strategy and to allow for a comprehensive comparison between them.

First, let us analyze the results obtained within each pruning strategy, discussing each configuration used and the performance achieved to determine the best-performing setups for each method.

Section 4.3.4 then provides a comparative analysis of the best results obtained by the three strategies. This analysis highlights the strategy that yields the most effective pruning of the Isolation Forest in terms of both model size reduction and anomaly detection performance.

For each strategy, the results presented include only those obtained with `test_size` = 0.2, since their trends are similar to those obtained with `test_size` = 0. Appendix A provides the complete set of results without test set for each configuration of pruning strategies. These results generally demonstrate and confirm the same considerations made for the results with test set, with slight differences in terms of performance values and trends.

4.3.1 Baseline

In order to establish a baseline for evaluating the effectiveness of the proposed pruning strategies, a random removal of trees from the Isolation Forest is performed. This method serves as a control experiment, allowing us to compare the performance of the DIFFI- and depth-based pruning methods against a simple, non-informative-based approach. Another rationale for using random removal is that if the proposed methods do not produce significant improvements over a naive approach, it would be preferable to use a simpler one that does not require additional computations.

The process involves a random selection of a specified percentage of trees from the ensemble to be pruned. The percentages of trees removed are set to values from 50% to 80%, providing a range of degrees of aggressiveness pruning for analysis. This means that, for example, at pruning degree of 50%, half of the trees in the Isolation Forest are randomly removed, and so on for the other levels.

For each percentage value, 30 independent runs are conducted within a seed, obtaining a comprehensive set of partial results that capture the variability inherent in the random pruning process. These are then averaged to obtain a more reliable estimate of the performance after random pruning.

These final results are used as benchmark to evaluate the effectiveness of the DIFFI-

and Depth-based pruning strategies, allowing us to assess whether these informed methods provide significant improvements over a random approach.

Table 4.5: Comparison of performance ($mean \pm std$) for Original vs. Pruned Models at different random pruning levels (50% to 80%). Results are obtained from the *Synthetic* dataset with `test_size` = 0.2 by averaging the results over seeds.

(a) F1-Score			
Pruning	Original	Pruned	Variation (%)
50%		0.3922 ± 0.0551	-4.67%
60%	0.4114	0.3821 ± 0.0515	-7.12%
70%	$\pm$ 0.0649	0.3695 ± 0.0467	-10.18%
80%		0.3466 ± 0.0419	-15.75%

(b) Average Precision			
Pruning	Original	Pruned	Variation (%)
50%		0.4048 ± 0.0694	-6.94%
60%	0.4350	0.3906 ± 0.0652	-10.21%
70%	$\pm$ 0.0821	0.3733 ± 0.0590	-14.18%
80%		0.3435 ± 0.0519	-21.03%

Table 4.5 presents the results obtained from random removal in the synthetic dataset, and are averaged over 30 different seeds. The results are summarized for both F1-Score (Table 4.5a) and AP (Table 4.5b) metrics, comparing the performance of the original model (i.e., non-pruned model) with that of models pruned at different degrees.

It is clear that all random pruning configurations result in performance degradation. More in detail, we can make the following observations:

- **Consistent performance degradation** is observed as the percentage of trees removed increases. As the pruning degree increases from 50% to 80%, both the F1-Score and AP metrics show a noticeable decline. This trend indicates that randomly removing trees negatively impacts the ability of the model to accurately detect anomalies. Therefore, there are no cases where random pruning results in performance improvements over the original model.
- **Performance degradation is gradual** as the number of pruned trees increases. For instance, at 50% pruning, the F1 Score drops by a smaller margin for the original model than with 80% pruning. This suggests that while some degree of random pruning may be tolerable if the performance penalty is limited, excessive pruning significantly compromises the effectiveness of the model.
- The **original model significantly outperforms the pruned versions** in all runs and at all levels of pruning. This means that the entire set of estimators provides the most effective performance in detecting anomalies.

These results demonstrate that, for this synthetic dataset, random removal causes a performance penalty. This behavior is predictable, as random pruning ignores any information that could guide the selection of trees to remove, resulting in the loss of potentially important trees that significantly contribute to the ability to detect anomalies of the model. Therefore, it is not surprising that the original model consistently outperforms the randomly pruned versions and that a more intelligent pruning strategy is needed to maintain or improve performance while reducing model complexity.

Table 4.6 summarizes the results obtained by randomly removing trees from various *ODDS* datasets.

Note that *ODDS* experiments with test sets are conducted only for the pruning degrees of 50% and 80% due to computational constraints, as each configuration is averaged over 30 different seeds.

Looking at these results, we can observe different behaviors across the datasets, for which some considerations are provided below:

- A subset of datasets - *breastw*, *cardio*, *pima*, *satellite*, *satimage-2*, *thyroid*, *vowels*, *pendigits* - exhibit a **monotonic performance degradation** for both metrics as the pruning degree increases. This trend, similar to that of the synthetic dataset, indicates that randomly removing trees negatively impacts the ability of the model to detect anomalies on these datasets. Even in these cases, the original model consistently outperforms the pruned versions, demonstrating that the full ensemble of trees achieves better performance. This behavior can be attributed to the fact that each tree in the ensemble contributes valuable information for anomaly detection, and random pruning likely removes some of these important trees, resulting in a performance penalty.
- A different behavior is observed in other datasets - *letter*, *optdigits*, *vertebral*, *wine* - where there is a **performance improvement**, in terms of F1-Score, at every pruning degree compared to the original model. This suggests that, in these cases, the original ensemble may contain a significant number of non-informative trees that lead to suboptimal performance. Random pruning, by chance, removes some of these less useful trees, resulting in a more effective model. While the AP for the *letter*, *optdigits* and *wine* datasets exhibits the behavior described above, indicating that, while random pruning may enhance the performance of the model at specific thresholds (as reflected in F1-Score), it does not significantly impact the overall ranking of instances across all thresholds (as reflected in AP). This discrepancy suggests that pruned models may be more effective at identifying anomalies at certain decision points, but they do not necessarily improve the overall detection capability as thresholds vary. For *vertebral* dataset, a slight improvement in AP is also observed, indicating that random pruning may have a more consistent positive effect on the capability of the model to rank instances effectively in this specific case.
- Subset of datasets - *annthyroid*, *ionosphere* - instead show a more stable behavior across different pruning degrees, with small fluctuations in perfor-

Table 4.6: Comparison of performance ($mean \pm std$) for Original and randomly Pruned Models. Pruning levels indicate the percentage of trees removed. Results are obtained from various *ODDS* datasets with `test_size` = 0.2 by averaging the results over seeds.

(a) F1-Score					
Dataset	Original	80% Pruning		50% Pruning	
		Pruned	Var. (%)	Pruned	Var. (%)
annthyroid	0.3151 $\pm$ 0.0390	0.3160 $\pm$ 0.0351	+0.29%	0.3153 $\pm$ 0.0373	+0.06%
breastw	0.9367 $\pm$ 0.0209	0.9173 $\pm$ 0.0201	-2.07%	0.9317 $\pm$ 0.0204	-0.53%
cardio	0.5066 $\pm$ 0.0672	0.4869 $\pm$ 0.0500	-3.89%	0.5026 $\pm$ 0.0616	-0.79%
ionosphere	0.6734 $\pm$ 0.0622	0.6722 $\pm$ 0.0563	-0.18%	0.6729 $\pm$ 0.0598	-0.07%
letter	0.0753 $\pm$ 0.0414	0.1051 $\pm$ 0.0318	+39.58%	0.0872 $\pm$ 0.0387	+15.80%
mnist	0.3023 $\pm$ 0.0292	0.2933 $\pm$ 0.0212	-2.98%	0.2996 $\pm$ 0.0265	-0.89%
optdigits	0.0225 $\pm$ 0.0148	0.0447 $\pm$ 0.0118	+98.67%	0.0300 $\pm$ 0.0132	+33.33%
pendigits	0.3358 $\pm$ 0.0592	0.2905 $\pm$ 0.0402	-13.49%	0.3183 $\pm$ 0.0522	-5.21%
pima	0.5153 $\pm$ 0.0388	0.5025 $\pm$ 0.0340	-2.48%	0.5113 $\pm$ 0.0362	-0.78%
satellite	0.5793 $\pm$ 0.0215	0.5619 $\pm$ 0.0177	-3.00%	0.5747 $\pm$ 0.0199	-0.79%
satimage-2	0.8481 $\pm$ 0.0417	0.7758 $\pm$ 0.0369	-8.52%	0.8307 $\pm$ 0.0377	-2.05%
thyroid	0.5623 $\pm$ 0.0663	0.5287 $\pm$ 0.0604	-5.98%	0.5497 $\pm$ 0.0668	-2.24%
vertebral	0.0262 $\pm$ 0.0507	0.0335 $\pm$ 0.0469	+27.86%	0.0297 $\pm$ 0.0507	+13.36%
vowels	0.1698 $\pm$ 0.0693	0.1502 $\pm$ 0.0465	-11.54%	0.1656 $\pm$ 0.0637	-2.47%
wine	0.1856 $\pm$ 0.1799	0.2001 $\pm$ 0.1287	+7.81%	0.1928 $\pm$ 0.1503	+3.88%

(b) Average Precision					
Dataset	Original	80% Pruning		50% Pruning	
		Pruned	Var. (%)	Pruned	Var. (%)
annthyroid	0.3051 $\pm$ 0.0343	0.2985 $\pm$ 0.0331	-2.16%	0.3029 $\pm$ 0.0339	-0.72%
breastw	0.9714 $\pm$ 0.0162	0.9616 $\pm$ 0.0179	-1.01%	0.9690 $\pm$ 0.0167	-0.25%
cardio	0.5685 $\pm$ 0.0700	0.5191 $\pm$ 0.0622	-8.69%	0.5550 $\pm$ 0.0686	-2.37%
ionosphere	0.8235 $\pm$ 0.0522	0.8143 $\pm$ 0.0510	-1.12%	0.8218 $\pm$ 0.0518	-0.21%
letter	0.0988 $\pm$ 0.0198	0.0987 $\pm$ 0.0164	-0.10%	0.0985 $\pm$ 0.0186	-0.30%
mnist	0.2777 $\pm$ 0.0262	0.2579 $\pm$ 0.0207	-7.13%	0.2724 $\pm$ 0.0248	-1.91%
optdigits	0.0540 $\pm$ 0.0063	0.0534 $\pm$ 0.0058	-1.11%	0.0539 $\pm$ 0.0062	-0.19%
pendigits	0.2795 $\pm$ 0.0508	0.2498 $\pm$ 0.0407	-10.63%	0.2716 $\pm$ 0.0473	-2.83%
pima	0.5055 $\pm$ 0.0420	0.4966 $\pm$ 0.0399	-1.76%	0.5030 $\pm$ 0.0413	-0.49%
satellite	0.6729 $\pm$ 0.0218	0.6543 $\pm$ 0.0216	-2.76%	0.6689 $\pm$ 0.0215	-0.59%
satimage-2	0.9089 $\pm$ 0.0427	0.8720 $\pm$ 0.0432	-4.06%	0.9013 $\pm$ 0.0421	-0.84%
thyroid	0.5763 $\pm$ 0.0677	0.5486 $\pm$ 0.0599	-4.81%	0.5685 $\pm$ 0.0658	-1.35%
vertebral	0.1129 $\pm$ 0.0194	0.1170 $\pm$ 0.0203	+3.63%	0.1139 $\pm$ 0.0191	+0.89%
vowels	0.1712 $\pm$ 0.0623	0.1421 $\pm$ 0.0397	-17.00%	0.1596 $\pm$ 0.0522	-6.78%
wine	0.3228 $\pm$ 0.1750	0.3079 $\pm$ 0.1383	-4.62%	0.3191 $\pm$ 0.1617	-1.15%

mance (i.e., **performance is flat**) for both metrics. In these cases, random removal of trees does not lead to significant changes in the model effectiveness, indicating that the original set is robust and that redundant trees are therefore

present, which is why performance does not significantly deteriorate with pruning.

Indeed, for these datasets, it is reasonable, hypothetically speaking, to choose the 80% pruned model as the best model, as it achieves a performance comparable to the original model while significantly reducing the number of trees in the ensemble, thus improving efficiency without sacrificing effectiveness.

We conclude that random tree removal generally leads to a degradation in model performance across different datasets, with some exceptions where performance improvements are observed. However, these improvements are not consistent across both evaluation metrics, indicating that random pruning is not a reliable strategy for optimizing Isolation Forest models: Even if an improvement in F1-Score occurs as AP decreases, this suggests that the small improvements observed may be due to a fluke rather than a systematic improvement in model performance. Therefore, we cannot consider random pruning as a reliable strategy for optimizing Isolation Forest models.

4.3.2 DIFFI-based

In this Section, we present the results obtained with DIFFI-based pruning strategy, as detailed in Section 3.2, following the experimental setup described in Section 4.2.2.

Since we have different combinations of *feature selection* and *thresholding* approaches, we first analyze the performance of each configuration to identify the most effective one, then compare the results obtained with the best *feature selections* in Section 4.3.2.3.

4.3.2.1 Feature selection: Elbow

As discussed in Section 3.2.2 regarding the *Elbow* feature selection method, it focuses on identifying a point in the importance values of ranked features where the marginal gain in importance decreases significantly, indicating that subsequent features contribute less to the performance of the model. All features whose importance is higher than this identified "elbow" point are then selected.

We expect that after identifying the most informative features and calculating the usage scores—the frequency with which these features are used in each tree—the trees that rely more heavily on these important features will be retained, resulting in improved performance of the pruned model.

As we can see in Table 4.7, the trend of the results, for *Synthetic* dataset, is opposite compared to the random pruning presented in Table 4.5: Each thresholding approach leads to a performance improvement. With random removal, we observed a performance degradation, indicating that the best choice was to keep the original model; in this case, however, we can see that pruning based on usage score is effective in enhancing the ability of the model to detect anomalies. This validates the underlying hypothesis: an ensemble can contain a significant number of non-informative trees that compromise its performance, and the DIFFI-elbow successfully identifies a criterion to eliminate them.

Table 4.7: Comparison of performance (*mean $\pm$ std*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Elbow* method. Results are obtained from the *Synthetic* dataset with `test_size = 0.2` by averaging the results over seeds.

(a) F1-Score			
Method	Original	Pruned	Variation (%)
Majority vote		0.4739 ± 0.0622	+15.19%
Mean		0.4854 ± 0.0574	+17.99%
Percentile 50%	0.4114	0.4802 ± 0.0583	+16.72%
Percentile 60%	± 0.0649	0.4889 ± 0.0590	+18.84%
Percentile 70%		0.4966 ± 0.0557	+20.71%
Percentile 80%		0.4916 ± 0.0483	+19.49%

(b) Average Precision			
Method	Original	Pruned	Variation (%)
Majority vote		0.5055 ± 0.0826	+16.21%
Mean		0.5289 ± 0.0785	+21.59%
Percentile 50%	0.4350	0.5218 ± 0.0795	+19.95%
Percentile 60%	± 0.0821	0.5360 ± 0.0794	+23.22%
Percentile 70%		0.5435 ± 0.0729	+24.94%
Percentile 80%		0.5371 ± 0.0626	+23.47%

While all thresholding methods yield performance improvements, there are clear differences in their effectiveness:

- **Majority vote** provides the smallest gains in both F1-Score and AP and reduces the ensemble size by approximately 50% of the trees on average;
- **Mean** thresholding offers better performance improvements than Majority vote, still achieving an average reduction of approximately 50%;
- **Percentile-based** approaches show a consistent trend of increased performance with higher percentile thresholding at the *70th percentile* achieving the largest improvements in both metrics, with 30% of the trees retained.

When comparing the results of different thresholding methods, it is clear that *percentile* approaches provide the most substantial performance enhancements and granular control over the pruning process. Although the *70th percentile* achieve the best result, it may be reasonable to consider choosing the *80th percentile*, as it involves a negligible performance loss but leads to a more efficient model, often a requirement in practical anomaly detection applications.

The standard deviation values indicate that the pruned models maintain consistent performance across runs, suggesting that the pruning strategy is robust and does not introduce significant variability, resulting a more stable model.

We can also observe the effects of different degrees of pruning aggressiveness applied by each thresholding method:

- The **most conservative approaches** are represented by *Majority vote* and *Mean* methods, which remove approximately 50% of the trees on average. The *50th* and *60th percentile* methods also exhibit similar behavior, pruning exactly 50% and 60% of the trees, respectively.
- *70th percentile* and *80th percentile* methods are **more aggressive**. This suggests that these methods are more selective, retaining only the most informative trees that contribute significantly to the performance of the model.

However, it is important to note that, while more aggressive pruning may lead to better performance in this case, it may also risk removing trees that could be beneficial in other contexts or datasets. Therefore, the choice of thresholding method should consider the performance of the initial model: if the original model already performs well, a more conservative approach may be preferable to avoid potential over pruning. In contrast, if the original model has room for improvement, more aggressive pruning could be beneficial.

In conclusion, for these configurations, the DIFFI-Elbow strategy, particularly when combined with higher percentile thresholding, demonstrates significant potential for enhancing the performance of Isolation Forest models on the synthetic dataset by effectively removing less informative trees from the ensemble. This finding suggests that targeted pruning based on the usage of informative features can lead to more robust anomaly detection capabilities compared to random pruning methods. This demonstrates that by intelligently identifying and removing non-informative estimators, it is possible to create not only a smaller and more efficient model, but also substantially more accurate one. Indeed, using the *Elbow* method for feature selection in conjunction with the *70th percentile* thresholding approach yields the best results, meaning that the trees retained after pruning are those that primarily utilize the features identified as significant.

Although the DIFFI-elbow proved effective for the synthetic dataset, the same cannot be said for the results obtained on the *ODDS* datasets, as shown in Table 4.8. Indeed, unlike in the synthetic case, where all thresholding methods led to performance improvements, here we observe a mixed outcome, similar to the results of random pruning, described in Section 4.3.1.

While some datasets exhibit performance improvements with certain thresholding approaches, others experience a performance drop compared to the original model. The observed trend in the *ODDS* datasets indicates that the effectiveness of the DIFFI-elbow is highly dataset-dependent.

Analyzing the results in detail, we can distinguish three distinct groups of datasets based on their response to the DIFFI-elbow:

- **Performance improvements:** The datasets - *annthyroid*, *letter*, *thyroid*, *opdigits* - show consistent performance improvements across most thresholding methods, indicating that the original ensemble for these datasets likely

Table 4.8: Comparison of performance (*mean*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Elbow* method. Results are obtained from various *ODDS* datasets with `test_size` = 0.2 by averaging the results over seeds.

(a) F1-Score

Dataset	Original	MV		Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3151	0.3535	+12.2%	0.3534	+12.1%	0.3518	+11.6%	0.3777	+19.9%
breastw	0.9367	0.9251	-1.2%	0.9251	-1.2%	0.9262	-1.1%	0.9034	-3.6%
cardio	0.5066	0.4962	-2.0%	0.4974	-1.8%	0.5000	-1.3%	0.4826	-4.7%
ionosphere	0.6734	0.6749	+0.2%	0.6732	-0.0%	0.6730	-0.1%	0.6743	+0.1%
letter	0.0753	0.0927	+23.1%	0.0903	+19.9%	0.0913	+21.2%	0.1015	+34.8%
mnist	0.3023	0.3022	-0.0%	0.3028	+0.2%	0.3041	+0.6%	0.2884	-4.6%
optdigits	0.0225	0.0294	+30.7%	0.0241	+7.1%	0.0248	+10.2%	0.0173	-23.1%
pendigits	0.3358	0.3119	-7.1%	0.3106	-7.5%	0.3113	-7.3%	0.2863	-14.7%
pima	0.5153	0.5099	-1.0%	0.5089	-1.2%	0.5099	-1.0%	0.4794	-7.0%
satellite	0.5793	0.5732	-1.0%	0.5755	-0.7%	0.5751	-0.7%	0.5630	-2.8%
satimage-2	0.8481	0.8197	-3.3%	0.8230	-3.0%	0.8252	-2.7%	0.7685	-9.4%
thyroid	0.5623	0.5924	+5.3%	0.5933	+5.5%	0.5918	+5.2%	0.5960	+6.0%
vertebral	0.0262	0.0256	-2.3%	0.0247	-5.7%	0.0240	-8.4%	0.0189	-27.9%
vowels	0.1698	0.1629	-4.1%	0.1650	-2.8%	0.1668	-1.8%	0.1499	-11.7%
wine	0.1856	0.1841	-0.8%	0.1812	-2.4%	0.1726	-7.0%	0.1380	-25.6%

(b) Average Precision

Dataset	Original	MV		Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3051	0.3625	+18.8%	0.3625	+18.8%	0.3603	+18.1%	0.3929	+28.8%
breastw	0.9714	0.9698	-0.2%	0.9698	-0.2%	0.9702	-0.1%	0.9651	-0.6%
cardio	0.5685	0.5600	-1.5%	0.5613	-1.3%	0.5644	-0.7%	0.5302	-6.7%
ionosphere	0.8235	0.8223	-0.1%	0.8223	-0.1%	0.8226	-0.1%	0.8185	-0.6%
letter	0.0988	0.0987	-0.1%	0.0992	+0.4%	0.0997	+0.9%	0.0991	+0.3%
mnist	0.2777	0.2754	-0.8%	0.2779	+0.1%	0.2778	+0.0%	0.2702	-2.7%
optdigits	0.0540	0.0511	-5.4%	0.0500	-7.4%	0.0503	-6.8%	0.0461	-14.6%
pendigits	0.2795	0.2646	-5.3%	0.2619	-6.3%	0.2639	-5.6%	0.2407	-13.9%
pima	0.5055	0.4987	-1.3%	0.4985	-1.4%	0.4992	-1.2%	0.4891	-3.2%
satellite	0.6729	0.6676	-0.8%	0.6687	-0.6%	0.6692	-0.5%	0.6521	-3.1%
satimage-2	0.9089	0.8980	-1.2%	0.8981	-1.2%	0.8994	-1.0%	0.8709	-4.2%
thyroid	0.5763	0.6427	+11.5%	0.6443	+11.8%	0.6427	+11.5%	0.6640	+15.2%
vertebral	0.1129	0.1135	+0.5%	0.1134	+0.4%	0.1132	+0.3%	0.1151	+1.9%
vowels	0.1712	0.1606	-6.2%	0.1628	-4.9%	0.1629	-4.8%	0.1415	-17.3%
wine	0.3228	0.3019	-6.5%	0.3154	-2.3%	0.3148	-2.5%	0.2891	-10.4%

contained a significant number of non-informative trees, the removal of which improved the anomaly detection capabilities of the model.

- **Performance degradation:** This trend is observed in datasets - *breastw*, *satellite*, *satimage-2*, *pendigits*, *vertebral*, *wine* - which show a decline in performance across all thresholding methods. This suggests that for these datasets, the original ensemble was already well-constructed and the pruning process inadvertently removed informative trees, reducing the effectiveness of anomaly detection, even with an informed strategy.
- **Stable or mixed results:** Other datasets, however - *ionosphere*, *mnist* - demonstrate relatively stable performance with minor variations ($< 1\%$), indicating that the pruning had a limited impact on the model's effectiveness, likely due to the presence of redundant trees that neither contributed nor significantly reduced overall performance.

From these results, it is clear that applying DIFFI with *Elbow* method for feature selection has the potential to enhance model performance in certain contexts, but is not universally applicable. In fact, for high-performing models (i.e., dataset that

already achieve good results with the original ensemble), the pruning process may lead to performance degradation, as important trees could be mistakenly removed. In conclusion, the DIFFI-elbow method demonstrates its potential to enhance model performance on certain real-world datasets, while its effectiveness varies significantly across different datasets. This observation indicates that the relationship between the usage frequency of important features and the performance of the model is not straightforward. Specifically, simply counting the frequency with which important features appear within trees does not necessarily guaranty the preservation of the most informative components of the ensemble. In a tree, an important feature can contribute substantially to anomaly identification when used at shallower depths, even if it is rarely used. In another tree, however, it might appear more frequently but only at deeper levels, thus exerting relatively limited influence on the model’s decision-making. Therefore, the DIFFI-elbow method might not consistently identify and retain the most informative trees across all datasets.

4.3.2.2 Feature selection: Soft

An identical experimental setup for the *Elbow* method is applied to the *Soft* feature selection approach, as described in Section 3.2.2.

The *Soft* selection method involves the inclusion of all features, rather than selecting a subset based on a specific criterion. Each feature is assigned a weight corresponding to its importance, allowing the model to utilize all available features, highlighting those deemed most significant. In this case, the usage score for each tree includes all features, then the weight of each feature, i.e., the GFI value, is included in its calculation, rather than the contribution of only the selected features as in the *Elbow* method.

We evaluate the performance of the DIFFI-based pruning strategy using the *Soft* method for feature selection combined with the thresholding approaches mentioned above.

As we can see in Table 4.9, the trend of the results, for synthetic dataset, is similar to the one obtained with the *Elbow* method (in Table 4.7): each thresholding approach leads to an improvement in performance.

Applying the *Soft* feature selection method to the *ODDS* datasets, as shown in Table 4.10, we observe again a mixed outcome, similar to that of the *Elbow* method. Some datasets benefit from pruning, while others suffer a performance penalty.

We can draw similar conclusions to those drawn for the *Elbow* method in Section 4.3.2.1: the effectiveness of the DIFFI-Soft method is highly dataset-dependent, likely due to the complex relationship between the utilization scores of all features, as well as those not specifically selected, and the overall model performance.

However, a more detailed analysis of all results obtained with DIFFI-based pruning methods and a comparison between them is presented in Section 4.3.2.3.

Table 4.9: Comparison of performance (*mean $\pm$ std*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Soft* method. Results are obtained from the *Synthetic* dataset with `test_size = 0.2` by averaging the results over seeds.

(a) F1-Score			
Method	Original	Pruned	Variation (%)
Majority vote		0.4633 ± 0.0595	+12.62%
Mean		0.4952 ± 0.0595	+20.37%
Percentile 50%	0.4114	0.4922 ± 0.0593	+19.64%
Percentile 60%	± 0.0649	0.4992 ± 0.0570	+21.34%
Percentile 70%		0.5020 ± 0.0579	+22.02%
Percentile 80%		0.4976 ± 0.0504	+20.95%

(b) Average Precision			
Method	Original	Pruned	Variation (%)
Majority vote		0.5000 ± 0.0779	+14.94%
Mean		0.5417 ± 0.0821	+24.53%
Percentile 50%	0.4350	0.5373 ± 0.0829	+23.52%
Percentile 60%	± 0.0821	0.5459 ± 0.0799	+25.49%
Percentile 70%		0.5541 ± 0.0759	+27.38%
Percentile 80%		0.5493 ± 0.0618	+26.28%

4.3.2.3 DIFFI-based pruning: results comparison

We present a comparative analysis of the results obtained using the two DIFFI-based feature selection methods: *Elbow* (Section 4.3.2.1) and *Soft* (Section 4.3.2.2). The goal is to identify which combination of feature selection method and thresholding approach provides the best performance on the evaluated datasets.

Starting with the comparison between uninformative and informative pruning strategies, i.e., random pruning (Section 4.3.1) and DIFFI-based pruning (Section 4.3.2), respectively, we observe that the latter approach generally outperforms the former in terms of performance improvements on both datasets. With random pruning, performance degradation was modest but systematic, while DIFFI-based pruning demonstrated the potential for performance improvements in certain datasets, although with a heterogeneous overall result. This indicates that leveraging feature importance information can lead to more informed pruning decisions, which can enhance model performance if applied appropriately.

Comparing the quantitative results obtained through the DIFFI-based pruning experiments, conducted using both the *Elbow* and *Soft* feature selection methods, reveals consistent patterns in how feature-guided pruning impacts model performance on the *ODDS* datasets. The evaluation, performed on F1-Score and AP metrics, enables a detailed comparison of the original models (unpruned) and their

Table 4.10: Comparison of performance (*mean*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Soft* method. Results are obtained from the *ODDS* dataset with `test_size = 0.2` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	MV		Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3151	0.3359	+6.6%	0.3487	+10.7%	0.3475	+10.3%	0.3711	+17.8%
breastw	0.9367	0.9254	-1.2%	0.9250	-1.2%	0.9262	-1.1%	0.9023	-3.7%
cardio	0.5066	0.5063	-0.06%	0.5060	-0.12%	0.5076	+0.2%	0.4920	-2.9%
ionosphere	0.6734	0.6744	+0.1%	0.6752	+0.3%	0.6751	+0.2%	0.6800	+1.0%
letter	0.0753	0.0992	+31.7%	0.0918	+21.9%	0.0904	+20.0%	0.0952	+26.4%
mnist	0.3023	0.3076	+1.8%	0.3074	+1.7%	0.3081	+1.9%	0.3003	-0.7%
optdigits	0.0225	0.0326	+44.9%	0.0152	-32.4%	0.0149	-33.8%	0.0158	-29.8%
pendigits	0.3358	0.3080	-8.3%	0.3114	-7.3%	0.3107	-7.5%	0.2872	-14.5%
pima	0.5153	0.5114	-0.8%	0.5078	-1.5%	0.5098	-1.1%	0.4759	-7.6%
satellite	0.5793	0.5683	-1.9%	0.5771	-0.4%	0.5769	-0.4%	0.5669	-2.1%
satimage-2	0.8481	0.8000	-5.7%	0.8272	-2.5%	0.8277	-2.4%	0.7760	-8.5%
thyroid	0.5623	0.5706	+1.5%	0.5941	+5.7%	0.5909	+5.1%	0.5960	+6.0%
vertebral	0.0262	0.0275	+5.0%	0.0247	-5.7%	0.0265	+1.1%	0.0261	-0.4%
vowels	0.1698	0.1734	+2.1%	0.1622	-2.1%	0.1682	-0.9%	0.1456	-14.2%
wine	0.1856	0.1986	+7.0%	0.1811	-2.4%	0.1848	-0.4%	0.1503	-19.0%

(b) Average Precision

Dataset	Original	MV		Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3051	0.3354	+9.9%	0.3559	+16.6%	0.3535	+15.9%	0.3847	+26.1%
breastw	0.9714	0.9699	-0.2%	0.9700	-0.1%	0.9704	-0.1%	0.9652	-0.6%
cardio	0.5685	0.5668	-0.3%	0.5714	+0.5%	0.5726	+0.7%	0.5468	-3.8%
ionosphere	0.8235	0.8208	-0.3%	0.8232	-0.0%	0.8231	-0.0%	0.8204	-0.4%
letter	0.0988	0.0974	-1.4%	0.0988	+0.0%	0.0988	+0.0%	0.0977	-1.1%
mnist	0.2777	0.2788	+0.4%	0.2850	+2.6%	0.2844	+2.4%	0.2807	+1.1%
optdigits	0.0540	0.0503	-6.8%	0.0487	-9.8%	0.0493	-8.7%	0.0452	-16.3%
pendigits	0.2795	0.2613	-6.5%	0.2658	-4.9%	0.2653	-5.1%	0.2413	-13.7%
pima	0.5055	0.5001	-1.1%	0.4991	-1.3%	0.5004	-1.0%	0.4892	-3.2%
satellite	0.6729	0.6609	-1.8%	0.6704	-0.4%	0.6704	-0.4%	0.6564	-2.4%
satimage-2	0.9089	0.8832	-2.8%	0.9030	-0.6%	0.9033	-0.6%	0.8762	-3.6%
thyroid	0.5763	0.6106	+6.0%	0.6423	+11.4%	0.6387	+10.8%	0.6588	+14.3%
vertebral	0.1129	0.1139	+0.9%	0.1134	+0.4%	0.1125	-0.4%	0.1169	+3.5%
vowels	0.1712	0.1630	-4.8%	0.1641	-4.2%	0.1635	-4.5%	0.1472	-14.0%
wine	0.3228	0.3079	-4.6%	0.3092	-4.2%	0.3098	-4.0%	0.2876	-10.9%

pruned counterparts, across different thresholding approaches: *Majority vote*, *Mean* and *Percentile*.

The Table 4.11 summarizes the performance comparison between the two DIFFI-based pruning methods, highlighting which yields better results for each dataset and metric.

First, we can observe that the *Soft* feature selection method generally outperforms the *Elbow* method, winning on 8 out of 16 datasets for F1-Score (Table 4.11a) and on 4 out of 16 datasets for AP (Table 4.11b). *Elbow* method instead wins in 3 datasets for both metrics.

More importantly, although for F1-Score our DIFFI-based pruning methods achieve improvements in a larger number of datasets compared to the original model, for AP the opposite is true: Only 7 datasets show improvements with DIFFI-based pruning, compared to 9 datasets that have better performance with the original model. This confirms the unreliability of DIFFI-based pruning methods in consistently enhancing model performance across different datasets and metrics.

In conclusion, in the context of DIFFI-based pruning methods, although the *Soft* feature selection method demonstrates a slight edge over the *Elbow* method in terms of performance improvements across different datasets, neither approach

consistently outperforms the original unpruned models.

By analyzing the performance variations across different datasets for both feature selection methods in more detail, we can make further observations. Despite the similarities in the observed trends, the results indicate that neither feature selection method consistently outperforms the other across all datasets. Rather, the effectiveness of each method appears to be dataset-dependent. However, considering the best and worst performances achieved for features selection methods, without distinguishing between thresholding approaches (the analysis of which is already described in Section 4.3.2.1), we can observe that *Elbow* shows the largest improvement, i.e., an increased F1-Score of +34.8% obtained by performing *80th percentile* thresholding on *letter* dataset. Furthermore, since its AP is positively stable, even for the *letter* dataset, this suggests that the pruning helps the model find a better decision boundary without necessarily improving the overall ranking of anomalies. Although the *Soft* method provides the best improvement in the F1-Score (i.e., a +44.9% increase in the F1-Score achieved by performing majority vote thresholding on the *opdigits* dataset), its AP for the same dataset and thresholding approach decreases, indicating that pruning found a "lucky" threshold for classification, but worsens the overall ability to rank anomalies. This trend found with *opdigits* is also confirmed by the results obtained with *Elbow* method, with some differences in terms of percentage variation. Therefore, it seems that *opdigits* represents an interesting case where there is no monotonic relationship between F1-Score and AP after pruning, regardless of the feature selection method used, as also found in the *vertebral* dataset.

On the other hand, the *Elbow* method also achieves the worst performance, i.e., with a -25.6% reduces F1-Score obtained by applying *80th percentile* thresholding in *wine* dataset. Its AP also decreases for the same dataset and thresholding approach, indicating that the pruning is not reliable. These observations exclude *opdigits* and *vertebral* datasets from this consideration because they are examples of uncommon cases, as mentioned above.

By all these considerations, we can conclude that the *Elbow* method appears to be more sensitive to dataset-specific characteristics, leading to more pronounced improvements or declines in performance depending on the dataset. In contrast, the *Soft* method has tendency to yield moderate variations in performance, suggesting a more stable but less impactful pruning effect. Examples of this behavior can be observed in datasets such as *mnist*, *cardio* and *ionosphere*, where *Soft* method provides smaller improvements compared to *Elbow*. In cases of degradation, such as in *satimage-2*, *Soft* method again results in less severe performance drops than *Elbow*, i.e., -8.5% vs. -9.4% , respectively, in F1-Score with *80th percentile* thresholding. This fact can also be proved by how each method selects features: while *Elbow* method focuses on a subset of features deemed most important, potentially leading to more sensitive pruning decisions, *Soft* method, instead, incorporates all features, may offer a more stable basis for pruning decisions.

In summary, between the *Elbow* and *Soft* feature selection methods, despite their small differences, the choice is not crucial, as both can lead to significant performance gains or losses depending on the dataset and therefore do not represent the main factor affecting the effectiveness of DIFFI-based pruning. However, it

Table 4.11: Summary of performance comparison between DIFFI-based pruning methods: *Elbow* vs. *Soft* feature selection, against *original* model. Results are obtained from the *Synthetic* and *ODDS* datasets with `test_size = 0.2`.

(a) F1-Score				
Dataset	Original	Elbow	Soft	Better Method
synthetic	0.4114	0.4866	0.5020	Soft
annthyroid	0.3151	0.3777	0.3711	Elbow
breastw	0.9367	0.9262	0.9262	Original
cardio	0.5066	0.5066	0.5076	Soft
ionosphere	0.6734	0.6749	0.6800	Soft
letter	0.0753	0.1015	0.0952	Elbow
mnist	0.3023	0.3041	0.3081	Soft
optdigits	0.0225	0.0294	0.0326	Soft
pendigits	0.3358	0.3358	0.3358	Original
pima	0.5153	0.5153	0.5153	Original
satellite	0.5793	0.5793	0.5793	Original
satimage-2	0.8481	0.8481	0.8481	Original
thyroid	0.5623	0.5960	0.5960	Tie
vertebral	0.0262	0.0262	0.0275	Soft
vowels	0.1698	0.1698	0.1734	Soft
wine	0.1856	0.1856	0.1986	Soft

(b) Average Precision				
Dataset	Original	Elbow	Soft	Better Method
synthetic	0.4350	0.5435	0.5541	Soft
annthyroid	0.3051	0.3929	0.3847	Elbow
breastw	0.9714	0.9714	0.9714	Original
cardio	0.5685	0.5685	0.5726	Soft
ionosphere	0.8235	0.8235	0.8235	Original
letter	0.0988	0.0997	0.0988	Elbow
mnist	0.2777	0.2779	0.2850	Soft
optdigits	0.0540	0.0540	0.0540	Original
pendigits	0.2795	0.2795	0.2795	Original
pima	0.5055	0.5055	0.5055	Original
satellite	0.6729	0.6729	0.6729	Original
satimage-2	0.9089	0.9089	0.9089	Original
thyroid	0.5763	0.6640	0.6588	Elbow
vertebral	0.1129	0.1151	0.1169	Soft
vowels	0.1712	0.1712	0.1712	Original
wine	0.3228	0.3228	0.3228	Original

seems confirmed that the utilization scores calculated using the GFI provided by DIFFI are not sufficiently effective in consistently identifying and retaining the most informative trees across all datasets, as discussed in Section 4.3.2.1 on *ODDS* datasets, regardless of the feature selection method used.

Future research could further explore the relationship between feature importance, tree usage scores, and model performance to develop more robust pruning strategies that can universally adapt to the specific characteristics of different datasets.

For this reason, in the next section, we will explore an alternative pruning strategy based on a different concept: the depth of all instances in the trees; this approach does not rely on feature importance, but rather on the structural properties of the trees themselves.

4.3.3 Depth-based

In this section, we present the results obtained from the depth-based pruning strategies introduced in Section 3.3. Two different approaches are evaluated: *Anomaly Depth* (Section 4.3.3.2) and *Separation Ratio* (Section 4.3.3.1). The goal is to assess the effectiveness of these methods in enhancing the performance of anomaly detection models by selectively pruning trees based on depth-related criteria.

The reasons for exploring depth-based pruning methods are listed below:

- The results obtained from DIFFI-based pruning methods, discussed in Section 4.3.2, indicate that these strategies do not consistently produce performance improvements across all datasets. This inconsistency suggests that alternative pruning criteria may be needed to better capture the characteristics of informative trees within the ensemble.
- By the conclusion drawn for DIFFI-based pruning in Section 4.3.2.3, combined with the observations made in Section 3.3.1, it is evident that the usage scores derived from feature importance do not consistently identify the most informative trees in all datasets. Specifically, looking at the depth distribution in Figure 3.6, we note an overlap between the depth distributions of the inliers and outliers at specific depth levels. This overlap reveals that there are trees that isolate anomalies at the same depth levels as inliers, indicating that these trees may not be effective for anomaly detection. Therefore, the feature importance-based usage score may not adequately capture this aspect of tree performance because it does not consider the depth of anomaly isolation within trees, but rather focuses on the frequency of feature utilization.

As in previous experiments, we evaluated the performance of depth-based pruning strategies using the same *Synthetic* and *ODDS* datasets, with a `test_size` of 0.2, averaging the results over 30 different seeds.

4.3.3.1 Separation ratio

In this section, we present the results obtained from a depth-based pruning strategy that exploit the concept of *Separation Ratio*, as described in Section 3.3.2. This approach focuses on the depth ratio between outliers and inliers within each tree to

identify and retain the most informative trees for anomaly detection - those with a higher separation ratio, meaning they isolate anomalies at shallower depths than inliers.

Table 4.12: Comparison of performance (*mean $\pm$ std*) for Original vs. Depth-based Pruned Models, exploiting **depth ratio** between outliers and inliers, with different *thresholding approach*. Results are obtained from the *Synthetic* dataset with `test_size` = 0.2 by averaging the results over seeds.

(a) F1-Score			
Method	Original	Pruned	Variation (%)
Mean		0.6182 ± 0.0537	+50.27%
Percentile 50%	0.4114 ± 0.0649	0.5660 ± 0.0626	+37.58%
Percentile 60%		0.5965 ± 0.0602	+44.99%
Percentile 70%		0.6226 ± 0.0559	+51.34%
Percentile 80%		0.6374 ± 0.0500	+54.93%

(b) Average Precision			
Method	Original	Pruned	Variation (%)
Mean		0.7013 ± 0.0634	+61.22%
Percentile 50%	0.4350 ± 0.0821	0.6364 ± 0.0774	+46.30%
Percentile 60%		0.6780 ± 0.0701	+55.86%
Percentile 70%		0.7111 ± 0.0629	+63.47%
Percentile 80%		0.7338 ± 0.0530	+68.69%

As we can see in Table 4.12, the depth-based pruning strategy based on *Separation Ratio* yields significant improvements in performance in the synthetic dataset. The key observations are as follows:

- **Massive performance improvements:** This strategy transforms the original model performance (F1-Score of 0.4114 and AP of 0.4350) into substantially high-performing pruned models (F1-Score up to 0.6374 and AP up to 0.7338).
- **Consistent gains across thresholding methods:** All thresholding approaches produce significant increases in both F1 score and AP compared to the original model. More in detail, as the aggressiveness of pruning increases (from the most conservative *Mean* to the most aggressive *80th percentile*), the performance metrics steadily improve, indicating that maintaining trees with higher separation ratios is beneficial.
- **Reduced variance:** The standard deviations of the pruned models are lower than those of the original one, suggesting that the pruning strategy not only enhances performance, but also stabilizes it across different runs.

These results and observations suggest that leveraging the depth ratio between outliers and inliers is an effective criterion for tree pruning in an Isolation Forest,

as it helps preserve the trees best suited for anomaly isolation. Therefore, the fundamental hypothesis — the presence of non-informative or redundant trees for the detection of anomalies in the original set — is confirmed by these results.

In this case, we can establish that the best combination of pruning and thresholding approaches based on the *Separation Ratio* is the *80th percentile*, which consistently yields the highest performance.

Turning to the results obtained from the *ODDS* datasets, as presented in Table 4.13, we observe a similar trend of performance improvements in all datasets. The *Separation Ratio* depth-based pruning strategy consistently improves both the F1-Score and AP metrics compared to the original models.

However, the magnitude of improvements varies depending on the datasets:

- **Radical improvements:** Datasets such as *optdigits*, *letter*, *wine*, and *vertebral* show the most dramatic improvements, with increases in the F1-Score up to +1412.9% and increases in AP up to +587.0%. This means that for these datasets, the original models struggled significantly, and this strategy is exceptionally good at identifying and retaining the most informative trees and, in combination with *80th percentile* threshold, effectively purifies the ensemble.
- **Significant improvements:** Other datasets, such as *cardio*, *thyroid*, *annthyroid*, and *pendigits* also show substantial improvements, with increases in the F1-Score ranging from +23.4% to +58.4% and AP from +30.6% to +94.2%. These results indicate that the original models, initially characterized by moderate performance, have been consolidated through the pruning process.
- **Moderate improvements:** Datasets such as *ionosphere*, *pima*, and *satellite* show more modest but consistent improvements, with increases in the F1-Score around +2.6% to +8.3% and AP increases from +2.0% to +9.3%. These datasets already had relatively high-performing original models, and the pruning strategy helps to fine-tune their performance further.
- **Stable performance:** The *breastw* and *satimage-2* datasets exhibit minimal performance variations. This suggests that the original models were already highly effective and that the pruning strategy maintains their performance without significant alterations. In particular, in these cases, the most aggressive pruning (*80th percentile*) does not yield the best results, indicating that a more conservative approach is preferable when the original model is already strong.

These results further validate the effectiveness of the *Separation Ratio*-based pruning strategy in identifying and retaining trees that are more proficient at isolating anomalies. Consistent performance enhancements across multiple datasets highlight the robustness and generalizability of this approach to improve anomaly detection models. Once again, the best combination is *Separation Ratio*-based pruning with *80th percentile* threshold, which consistently yields the highest performance (12 out of 15 datasets for both F1-Score and AP).

Table 4.13: Comparison of performance (*mean*) for Original vs. Depth-based Pruned Models, exploiting **depth ratio** between outliers and inliers, with different *thresholding approach*. Results are obtained from the *ODDS* dataset with `test_size = 0.2` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3151	0.3968	+25.9%	0.3820	+21.2%	0.4328	+37.4%
breastw	0.9367	0.9446	+0.8%	0.9436	+0.7%	0.9331	-0.4%
cardio	0.5066	0.6274	+23.8%	0.6159	+21.6%	0.6621	+30.7%
ionosphere	0.6734	0.6910	+2.6%	0.6931	+2.9%	0.7160	+6.3%
letter	0.0753	0.1382	+83.5%	0.1322	+75.6%	0.1657	+120.0%
mnist	0.3023	0.4660	+54.1%	0.4432	+46.6%	0.5110	+69.0%
optdigits	0.0225	0.3139	+1295.1%	0.2313	+928.0%	0.3404	+1412.9%
pendigits	0.3358	0.4986	+48.5%	0.4809	+43.2%	0.4987	+48.5%
pima	0.5153	0.5419	+5.2%	0.5410	+5.0%	0.5444	+5.6%
satellite	0.5793	0.6235	+7.6%	0.6216	+7.3%	0.6275	+8.3%
satimage-2	0.8481	0.8617	+1.6%	0.8640	+1.9%	0.8438	-0.5%
thyroid	0.5623	0.7038	+25.2%	0.6906	+22.8%	0.7198	+28.0%
vertebral	0.0262	0.0805	+207.2%	0.0705	+169.1%	0.1025	+291.2%
vowels	0.1698	0.3084	+81.6%	0.3114	+83.4%	0.2922	+72.1%
wine	0.1856	0.4986	+168.6%	0.4817	+159.5%	0.5341	+187.8%

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3051	0.4437	+45.4%	0.4267	+39.9%	0.4833	+58.4%
breastw	0.9714	0.9795	+0.8%	0.9797	+0.8%	0.9779	+0.7%
cardio	0.5685	0.7166	+26.0%	0.7009	+23.3%	0.7539	+32.6%
ionosphere	0.8235	0.8401	+2.0%	0.8406	+2.1%	0.8466	+2.8%
letter	0.0988	0.1261	+27.6%	0.1208	+22.3%	0.1439	+45.6%
mnist	0.2777	0.4720	+70.0%	0.4435	+59.7%	0.5239	+88.7%
optdigits	0.0540	0.2665	+393.5%	0.1787	+230.9%	0.3710	+587.0%
pendigits	0.2795	0.5108	+82.8%	0.4839	+73.1%	0.5428	+94.2%
pima	0.5055	0.5407	+7.0%	0.5405	+6.9%	0.5523	+9.3%
satellite	0.6729	0.7205	+7.1%	0.7191	+6.9%	0.7216	+7.2%
satimage-2	0.9089	0.9252	+1.8%	0.9253	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8035	+39.4%	0.7888	+36.9%	0.8195	+42.2%
vertebral	0.1129	0.1503	+33.1%	0.1399	+23.9%	0.1678	+48.6%
vowels	0.1712	0.3531	+106.2%	0.3644	+112.8%	0.3324	+94.2%
wine	0.3228	0.6345	+96.6%	0.5854	+81.3%	0.6834	+111.7%

4.3.3.2 Anomaly depth

Now we present the results obtained from the depth-based pruning strategy that exploit the *Anomaly Depth* concept, as described in Section 3.3.2. This approach focuses on the average depth at which outliers are isolated within each tree to identify and retain the most informative trees for anomaly detection, i.e., those that isolate anomalies at shallower depths.

Table 4.14: Comparison of performance (*mean $\pm$ std*) for Original vs. Depth-based Pruned Models, exploiting **depth of outliers**, with different *thresholding approach*. Results are obtained from the *Synthetic* dataset with `test_size` = 0.2 by averaging the results over seeds.

(a) F1-Score			
Method	Original	Pruned	Variation (%)
Mean		0.6076 ± 0.0575	+47.69%
Percentile 50%	0.4114 ± 0.0649	0.5568 ± 0.0619	+35.34%
Percentile 60%		0.5906 ± 0.0591	+43.56%
Percentile 70%		0.6156 ± 0.0540	+49.64%
Percentile 80%		0.6331 ± 0.0457	+53.89%

(b) Average Precision			
Method	Original	Pruned	Variation (%)
Mean		0.6912 ± 0.0675	+58.90%
Percentile 50%	0.4350 ± 0.0821	0.6235 ± 0.0779	+43.33%
Percentile 60%		0.6697 ± 0.0721	+53.95%
Percentile 70%		0.7050 ± 0.0618	+62.07%
Percentile 80%		0.7287 ± 0.0541	+67.52%

As we can see in Table 4.14, the *Anomaly Depth*-based pruning strategy yields significant improvements in performance on the *Synthetic* dataset, following a trend similar to that observed with the *Separation Ratio*-based approach. Therefore, the key observations still hold. However, comparing the two depth-based strategies, we note that the *Separation Ratio*-based pruning marginally outperforms the *Anomaly Depth* one in terms of both F1-Score and AP across all thresholding methods.

Moving to the results obtained from the *ODDS* datasets, as presented in Table 4.15, we again observe a similar trend of performance improvements in all datasets, consistent with the findings of the *Separation Ratio*-based approach. The *Anomaly Depth*-based pruning strategy consistently enhances both F1-Score and AP metrics compared to the original models.

The magnitude of improvements also varies, following the same pattern as that observed with *Separation Ratio*-based pruning. Detailed results comparing the two depth-based pruning strategies are provided in the next Section 4.3.3.3.

Table 4.15: Comparison of performance (*mean*) for Original vs. Depth-based Pruned Models, exploiting **depth of outliers**, with different *thresholding approach*. Results are obtained from the *ODDS* dataset with `test_size = 0.2` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3151	0.3957	+25.6%	0.3805	+20.8%	0.4311	+36.8%
breastw	0.9367	0.9453	+0.9%	0.9458	+1.0%	0.9357	-0.1%
cardio	0.5066	0.6253	+23.4%	0.6126	+20.9%	0.6565	+29.6%
ionosphere	0.6734	0.6888	+2.3%	0.6892	+2.3%	0.7116	+5.7%
letter	0.0753	0.1425	+89.2%	0.1338	+77.7%	0.1664	+121.0%
mnist	0.3023	0.4694	+55.3%	0.4419	+46.2%	0.5061	+67.4%
optdigits	0.0225	0.2974	+1221.8%	0.2237	+894.2%	0.3250	+1344.4%
pendigits	0.3358	0.4965	+47.9%	0.4786	+42.5%	0.4964	+47.8%
pima	0.5153	0.5485	+6.4%	0.5468	+6.1%	0.5473	+6.2%
satellite	0.5793	0.6145	+6.1%	0.6119	+5.6%	0.6200	+7.0%
satimage-2	0.8481	0.8603	+1.4%	0.8640	+1.9%	0.8423	-0.7%
thyroid	0.5623	0.7024	+24.9%	0.6899	+22.7%	0.7174	+27.6%
vertebral	0.0262	0.0833	+217.9%	0.0720	+174.8%	0.1047	+299.6%
vowels	0.1698	0.3072	+80.9%	0.2836	+67.0%	0.3051	+79.7%
wine	0.1856	0.4948	+166.6%	0.4726	+154.6%	0.5355	+188.5%

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3051	0.4419	+44.8%	0.4251	+39.3%	0.4812	+57.7%
breastw	0.9714	0.9787	+0.8%	0.9787	+0.8%	0.9769	+0.6%
cardio	0.5685	0.7130	+25.4%	0.6965	+22.5%	0.7488	+31.7%
ionosphere	0.8235	0.8382	+1.8%	0.8382	+1.8%	0.8439	+2.5%
letter	0.0988	0.1220	+23.5%	0.1174	+18.8%	0.1396	+41.3%
mnist	0.2777	0.4672	+68.2%	0.4351	+56.7%	0.5175	+86.3%
optdigits	0.0540	0.2547	+371.7%	0.1689	+212.8%	0.3544	+556.3%
pendigits	0.2795	0.5123	+83.3%	0.4861	+73.9%	0.5475	+95.9%
pima	0.5055	0.5361	+6.1%	0.5346	+5.8%	0.5430	+7.4%
satellite	0.6729	0.7172	+6.6%	0.7157	+6.4%	0.7199	+7.0%
satimage-2	0.9089	0.9253	+1.8%	0.9250	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8036	+39.4%	0.7888	+36.9%	0.8190	+42.1%
vertebral	0.1129	0.1486	+31.6%	0.1393	+23.4%	0.1649	+46.1%
vowels	0.1712	0.3495	+104.1%	0.3239	+89.2%	0.3644	+112.8%
wine	0.3228	0.6256	+93.8%	0.5856	+81.4%	0.6865	+112.7%

4.3.3.3 Depth-based pruning: results comparison

In this section, we provide a direct comparison between the two depth-based pruning strategies: *Separation Ratio* and *Anomaly Depth*. The goal is to identify which of the two approaches produces better performance on the evaluated datasets.

Table 4.16 summarizes the performance comparison between the two strategies in terms of F1-Score and AP for both the *Synthetic* and *ODDS* datasets. The best results for each dataset and metric are highlighted in bold.

From the presented comparison, we can draw several conclusions regarding the performance of the two depth-based pruning strategies:

- **Overall Performance:** The *Separation Ratio*-based pruning strategy outperforms the *Anomaly Depth* one on most datasets for both performance metrics. Specifically, it achieves better results in 10 out of 16 datasets for F1-Score and 12 out of 16 datasets for AP. This indicates that the *Separation Ratio* is generally more effective at identifying and retaining trees that positively contribute to the anomaly detection performance.
- **Consistency:** The *Separation Ratio*-based method demonstrates more consistent performance across different datasets, suggesting that it is a more robust approach for depth-based pruning in isolation forest models.
- **Close Competition:** Although the *Separation Ratio*-based approach generally provides better results, in some cases the *Anomaly Depth*-based method is competitive in some cases, particularly in datasets like *breastw*, *letter*, *pima*, *vertebral*, and *wine*. This suggests that both strategies have their strengths and can be effective depending on the specific characteristics of the dataset.

In summary, while both depth-based strategies enhance the performance of anomaly detection models, the *Separation Ratio* approach emerges as the most effective method across a wide range of datasets. Its ability to consistently deliver superior results highlights its potential as the preferred strategy for depth-based pruning. In addition to the empirical evidence presented, several theoretical rationales support the superiority of *Separation Ratio* over *Anomaly Depth*: the former considers both the outliers and inliers depth, rewarding trees that not only isolate anomalies quickly (i.e., shallow depth) but also isolate normal instances slowly. This approach directly addresses the primary objective of anomaly detection, which is to maximize the distinction between anomalous and normal data points. In contrast, *Anomaly Depth* focuses exclusively on anomaly depth, potentially neglecting the handling of normal instances. By incorporating both aspects, *Separation Ratio* provides a more holistic assessment of trees effectiveness in isolating anomalies.

Ultimately, the best configuration, provided by *Depth-based* strategies, which generally yields the best performance is **Separation Ratio** combined with **80th percentile** thresholding. We caution that it may be useful to adjust the aggressiveness of the pruning (i.e., thresholding approach) based on the initial performance of the original model, as discussed in Section 4.3.2.1.

Table 4.16: Summary of performance between Depth-base pruning methods: **Separation Ratio** vs **Anomaly Depth**. Results are obtained from the *Synthetic* and *ODDS* datasets with `test_size = 0.2`.

(a) F1-Score

Dataset	Separation Ratio	Anomaly Depth	Better Method
synthetic	0.6374	0.6331	Separation Ratio
annthyroid	0.4328	0.4311	Separation Ratio
breastw	0.9446	0.9458	Anomaly Depth
cardio	0.6621	0.6565	Separation Ratio
ionosphere	0.7160	0.7116	Separation Ratio
letter	0.1657	0.1664	Anomaly Depth
mnist	0.5110	0.5061	Separation Ratio
optdigits	0.3404	0.3250	Separation Ratio
pendigits	0.4987	0.4964	Separation Ratio
pima	0.5444	0.5485	Anomaly Depth
satellite	0.6275	0.6200	Separation Ratio
satimage-2	0.8640	0.8640	Tie
thyroid	0.7198	0.7174	Separation Ratio
vertebral	0.1025	0.1047	Anomaly Depth
vowels	0.3114	0.3072	Separation Ratio
wine	0.5341	0.5355	Anomaly Depth

(b) Average Precision

Dataset	Separation Ratio	Anomaly Depth	Better Method
synthetic	0.7338	0.7287	Separation Ratio
annthyroid	0.4833	0.4812	Separation Ratio
breastw	0.9797	0.9787	Separation Ratio
cardio	0.7539	0.7488	Separation Ratio
ionosphere	0.8466	0.8439	Separation Ratio
letter	0.1439	0.1396	Separation Ratio
mnist	0.5239	0.5175	Separation Ratio
optdigits	0.3710	0.3544	Separation Ratio
pendigits	0.5428	0.5475	Anomaly Depth
pima	0.5523	0.5430	Separation Ratio
satellite	0.7216	0.7199	Separation Ratio
satimage-2	0.9253	0.9253	Tie
thyroid	0.8195	0.8190	Separation Ratio
vertebral	0.1678	0.1649	Separation Ratio
vowels	0.3644	0.3644	Tie
wine	0.6834	0.6865	Anomaly Depth

4.3.4 Results comparison

Finally, we provide a comprehensive comparison of all the pruning strategies evaluated in this study, including DIFFI- and Depth-based methods, which aims to identify the most effective pruning strategy for improving the performance of Isolation Forest models on the evaluated datasets.

Table 4.17: Overall performance comparison of all pruning strategies. The best result for each row is highlighted in **bold**.

(a) F1-Score						
Dataset	Original	Random	DIFFI-based		Depth-based	
			Elbow	Soft	Separation Ratio	Anomaly Depth
synthetic	0.4114	0.3922	0.4866	0.5020	0.6374	0.6331
annthyroid	0.3151	0.3160	0.3777	0.3711	0.4328	0.4311
breastw	0.9367	0.9317	0.9262	0.9262	0.9446	0.9458
cardio	0.5066	0.5026	0.5066	0.5076	0.6621	0.6565
ionosphere	0.6734	0.6729	0.6749	0.6800	0.7160	0.7116
letter	0.0753	0.1051	0.1015	0.0952	0.1657	0.1664
mnist	0.3023	0.2996	0.3041	0.3081	0.5110	0.5061
optdigits	0.0225	0.0447	0.0294	0.0326	0.3404	0.3250
pendigits	0.3358	0.3183	0.3358	0.3358	0.4987	0.4964
pima	0.5153	0.5113	0.5153	0.5153	0.5444	0.5485
satellite	0.5793	0.5747	0.5793	0.5793	0.6275	0.6200
satimage-2	0.8481	0.8307	0.8481	0.8481	0.8640	0.8640
thyroid	0.5623	0.5497	0.5960	0.5960	0.7198	0.7174
vertebral	0.0262	0.0335	0.0262	0.0275	0.1025	0.1047
vowels	0.1698	0.1656	0.1698	0.1734	0.3114	0.3072
wine	0.1856	0.2001	0.1856	0.1986	0.5341	0.5355

(b) Average Precision						
Dataset	Original	Random	DIFFI-based		Depth-based	
			Elbow	Soft	Separation Ratio	Anomaly Depth
synthetic	0.4350	0.4048	0.5435	0.5541	0.7338	0.7287
annthyroid	0.3051	0.3029	0.3929	0.3847	0.4833	0.4812
breastw	0.9714	0.9690	0.9714	0.9714	0.9797	0.9787
cardio	0.5685	0.5550	0.5685	0.5726	0.7539	0.7488
ionosphere	0.8235	0.8218	0.8235	0.8235	0.8466	0.8439
letter	0.0988	0.0987	0.0997	0.0988	0.1439	0.1396
mnist	0.2777	0.2724	0.2779	0.2850	0.5239	0.5175
optdigits	0.0540	0.0539	0.0540	0.0540	0.3710	0.3544
pendigits	0.2795	0.2716	0.2795	0.2795	0.5428	0.5475
pima	0.5055	0.5030	0.5055	0.5055	0.5523	0.5430
satellite	0.6729	0.6689	0.6729	0.6729	0.7216	0.7199
satimage-2	0.9089	0.9013	0.9089	0.9089	0.9253	0.9253
thyroid	0.5763	0.5685	0.6640	0.6588	0.8195	0.8190
vertebral	0.1129	0.1170	0.1151	0.1169	0.1678	0.1649
vowels	0.1712	0.1596	0.1712	0.1712	0.3644	0.3644
wine	0.3228	0.3191	0.3228	0.3228	0.6834	0.6865

Table 4.17 summarizes the performance of all pruning strategies in terms of F1-Score

and AP for the *Synthetic* and *ODDS* datasets. From the comparison, we can draw several conclusions:

- **Dominance of Depth-based Methods:** Depth-based strategies, particularly the *Separation Ratio* approach, consistently outperform DIFFI-based methods on most datasets. This indicates that leveraging depth information is more effective for pruning Isolation Forest models in the context of anomaly detection.
- **Limited Improvement from DIFFI-based Methods:** Although DIFFI-based strategies provide some improvements over the original models, their performance gains are generally modest compared to depth-based methods. This suggests that while feature importance is a valuable metric, it may not capture the tree’s effectiveness in isolating anomalies as well as depth-based metrics.

These results demonstrate that the standard, unpruned Isolation Forest is a suboptimal model for nearly all tested datasets, validating the fundamental hypothesis of this study: it contains a high degree of non-information and redundancy that can be effectively removed through pruning, leading to significant performance improvements. Therefore, an Isolation Forest model should not be considered fully optimized until it has been pruned to eliminate unnecessary trees and improve its detection capabilities.

In conclusion, the findings of this comprehensive comparison underscore the effectiveness of depth-based pruning strategies, particularly the *Separation Ratio*-based approach, usually combined with *80th Percentile*, in enhancing the performance of Isolation Forest models.

To provide further evidence on the effectiveness of the proposed pruning strategies, we report some statistical data about the characteristics of the pruned models. In particular, we analyze the distributions of GFI and depth of the predicted outliers and inliers, before and after pruning. These analyses help illustrate how pruning strategies influence the focus of the model on relevant features and its ability to isolate anomalies effectively. We selected representative datasets from *ODDS* collection, resulting in datasets with low and high initial performance (i.e., *thyroid* and *breastw*, respectively).

In Figures 4.3 and 4.4, we present the GFI and Depth distributions for both datasets. More in detail:

- We first show both distributions for the original Isolation Forest models (Figures 4.3a, 4.3b, 4.4a, and 4.4b).
- Then we present the distributions after applying the DIFFI-based pruning strategy (Figures 4.3c, 4.3d, 4.4c, and 4.4d).
- Finally, we show the distributions after applying the Depth-based pruning strategy (Figures 4.3e, 4.3f, 4.4e, and 4.4f).

Starting with GFI, where attention is paid only to features with higher importance values. On *thyroid* dataset, we observe that:

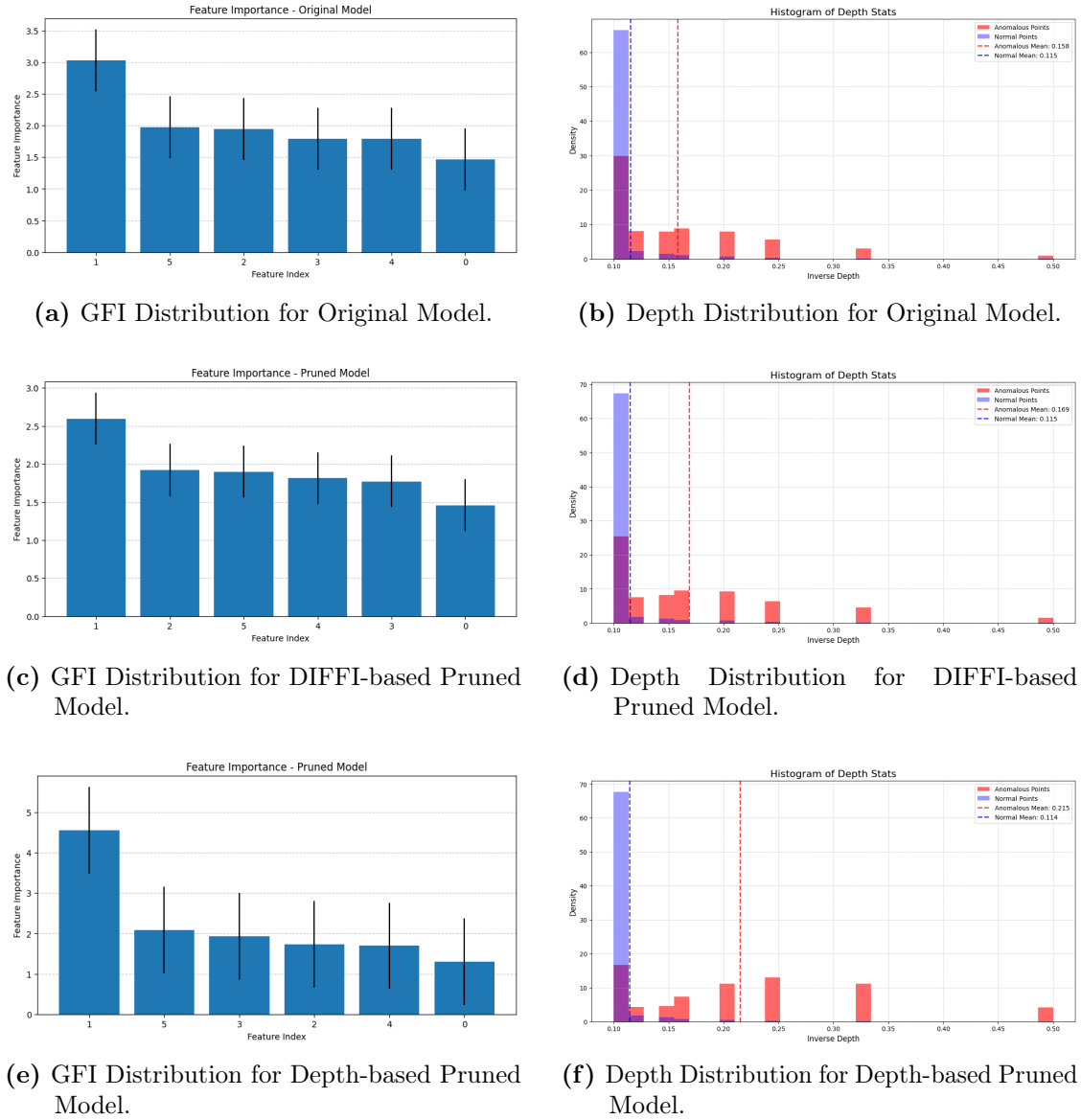


Figure 4.3: Comparison of GFI and Depth Distributions between Original, DIFFI-based Pruned, and Depth-based Pruned Isolation Forest models on the *thyroid* dataset, with `test_size` = 0.2 and `seed` = 0.

- with DIFFI strategy (Figure 4.3c), the pruned model shows a slight decrease in **feature 1**, with the highest GFI value, while the other features remain almost unchanged;
- with Depth strategy (Figure 4.3e), the pruned model exhibits a more pronounced focus on **feature 1**, with a significant increase in its GFI value, while the importance of other features, once again, remain almost unchanged.

Notice that **feature 1** is indeed the one selected by *Elbow* method (see Figure 3.4). This fact confirms that the relation between usage scores and anomaly detection capability of trees is not straightforward, while Depth-based metrics are more directly related.

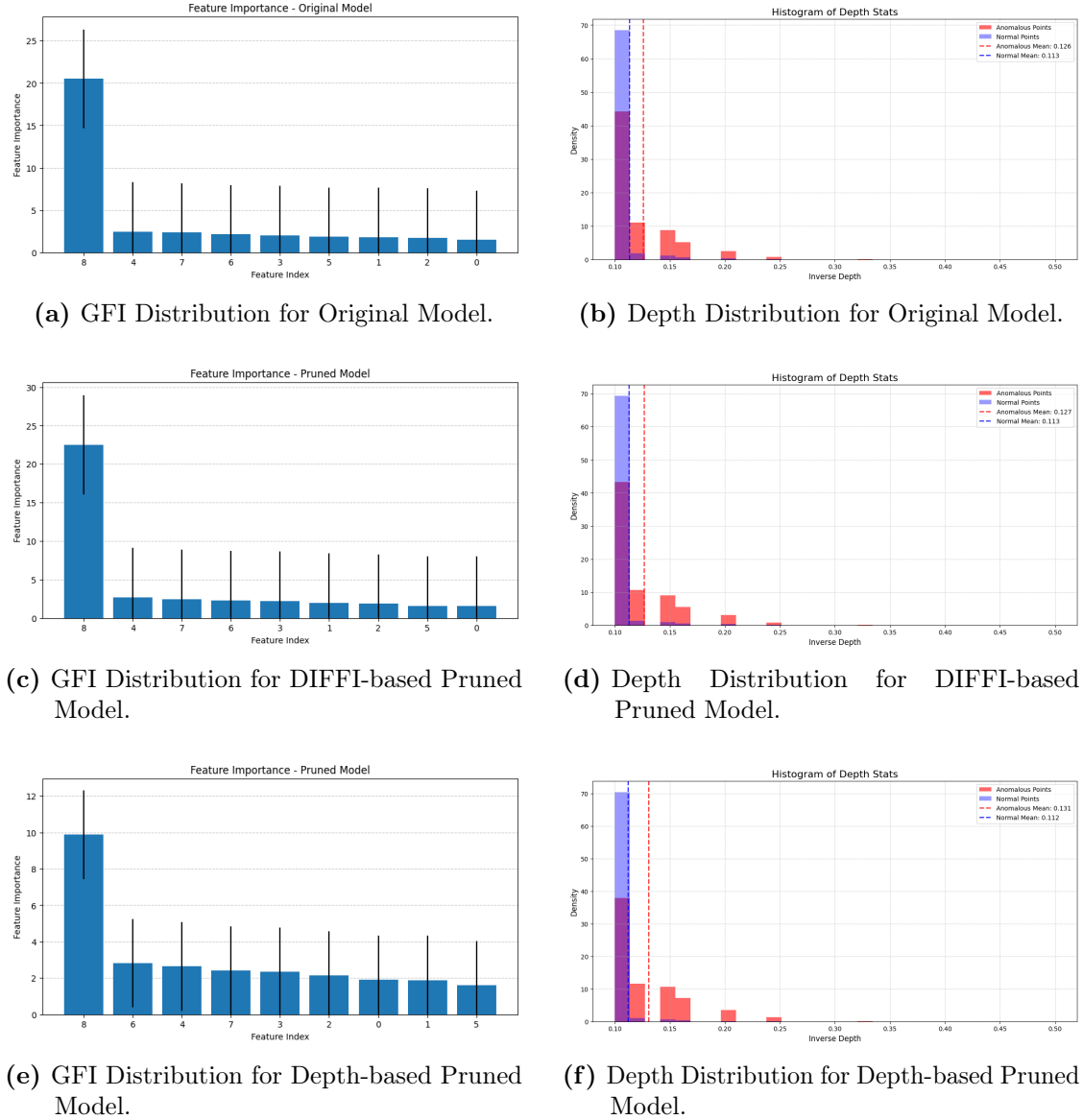


Figure 4.4: Comparison of GFI and Depth Distributions between Original, DIFFI-based Pruned, and Depth-based Pruned Isolation Forest models on the *breastw* dataset, with `test_size` = 0.2 and `seed` = 0.

While on *breastw* dataset:

- with DIFFI strategy (Figure 4.4c) there is a slight increase in GFI for **feature 8**, while the other features remain almost unchanged;
- with Depth-based strategy (Figure 4.4e), on the other hand, results in a significant decrease in GFI for **feature 8**, due to the fact that for this instance (`seed` = 0), we have the following results, using *80th Percentile*: Original F1-Score = 0.9010, pruned F1-Score = 0.8984, original AP = 0.9649, pruned AP = 0.9727. Therefore, it is consistent that the pruning process removes some trees that strongly depend on **feature 8**.

Moving to analyzing the depth distribution on *thyroid* dataset, with DIFFI strategy

(Figure 4.3d), we observe the following:

- The overlap between the depth distributions of normal and anomalous instances is slightly reduced after pruning, indicating a better separation between the two classes. For instance, from a density ~ 30 in the original model (Figure 4.3b), it reduces to ~ 25 in the pruned one.
- The mean depth distribution of *Anomalies* shifts from 0.158 to 0.169, while the mean depth distribution of *Normal Instances* remains unchanged (0.115). This proves that trees retained after pruning are more effective in isolating anomalies.

While with Depth-based strategy (Figure 4.3f), instead, we make the following considerations:

- A more pronounced reduction in the overlap between the depth distributions of normal and anomalous instances after pruning indicates an even better separation between the two classes. For instance, from a density ~ 30 in the original model (Figure 4.3b), it reduces to ~ 17 in the pruned one.
- The mean of the depth distribution of *Anomalies* shifts from 0.158 to 0.215, while the mean of depth distribution of the *Normal Instances* slightly decreases from 0.115 to 0.114. This further confirms that trees retained after pruning are significantly more effective in isolating anomalies.

Again, observing the depth distribution on *breastw* dataset, with DIFFI strategy (Figure 4.4d), we found the following:

- A slight increase in the overlap between the depth distributions of normal and anomalous instances after pruning indicates a slight deterioration in the separation between the two classes. For instance, from a density ~ 38 in the original model (Figure 4.4b), it increases to ~ 42 in the pruned model.
- The mean of *Anomalies* depth distribution shifts from 0.131 to 0.127, while the mean of *Normal Instances* depth distribution slightly increases from 0.112 to 0.113. This suggests that trees retained after pruning are marginally less effective in isolating anomalies.

While with Depth-based strategy, instead, (Figure 4.4f), it is not surprising that there is no change in depth distributions after pruning, as the performance degrades slightly.

In Figure 4.5 we provide a bar chart comparing the *Separation Ratio* metrics against the usage scores (i.e., the frequency of features usage) for every tree in a single Isolation Forest model. The selected model is trained on *thyroid* and *breastw* datasets, respectively.

From the plots, we first observe that the scores given to the trees by the two metrics are different, meaning that they do not rank the trees in the same order. Higher *Separation* scores mean that the tree is more effective at isolating anomalies, while a higher *Usage* score, even from previous analysis, does not necessarily imply that the tree is more effective. More in detail, further observations can be made:

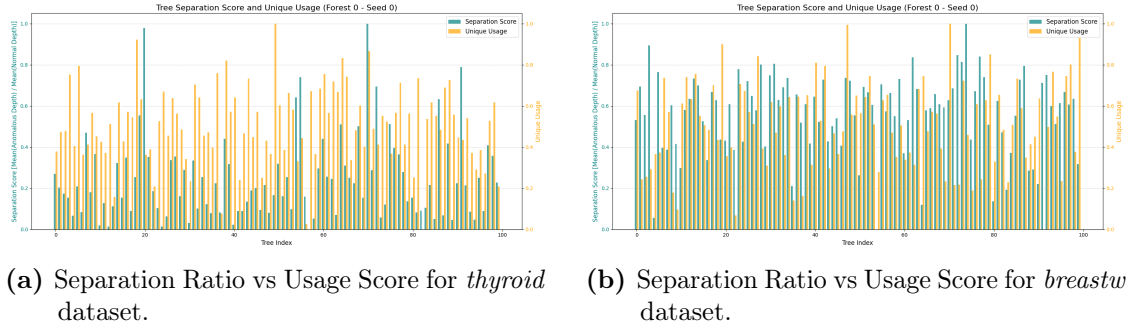


Figure 4.5: Comparison between Separation Ratio and Usage Score for each tree in single Isolation Forest model trained on *thyroid* and *breastw* datasets, *seed* = 0. Both metrics are normalized to the range $[0, 1]$ for better visualization.

- For *thyroid* dataset (Figure 4.5a), we note that trees with higher *Separation* scores (i.e., at least above 0.5) are few, proving that the ensemble size can be significantly reduced by up to 80% without losing detection capability, as mentioned in related works (Section 2.3). Moreover, the plot also confirms that a dataset with low initial performance, a more aggressive thresholding method (i.e., *80th Percentile*) is suitable, as there are many trees with low *Separation* scores that can be pruned (additional evidence is provided by Figure 3.8).
- For *breastw* dataset (Figure 4.5b), instead, is the opposite: Most trees have high *Separation* scores (i.e., at least above 0.5), indicating that the ensemble is already quite optimized and that only a few trees can be pruned without affecting performance. This observation supports the idea that for datasets with high initial performance, a less aggressive thresholding method (i.e., *50th Percentile* or *Mean*) is more appropriate to avoid removing trees that contribute positively to anomaly detection (additional evidence is provided by Figure 4.6).

4.4 Discussion about number of estimators

We discuss, in this section, the impact of the number of estimators (i.e., trees) in the Isolation Forest model on the effectiveness of the proposed pruning strategies. The original Isolation Forest algorithm suggests that a relatively small number of trees (e.g., 100) is sufficient for effective anomaly detection. However, given that our best pruning strategies proposed, i.e., *Separation Ratio*, tend to remove a significant portion of the trees, it is worth investigating whether starting with a larger number of trees could lead to better performance after pruning.

To further investigate this aspect, we conducted additional experiments on a subset of datasets from the *ODDS* collection by taking as representative those that showed the trends observed in previous sections: *thyroid* (significant performance increasing), *breastw* (moderate performance increasing) and *vertebral* (radical performance increasing). As a first experiment, we trained Isolation Forest models with varying numbers of estimators - 10, 20, 100, 200 and 400. For each configuration, we applied



Figure 4.6: Distribution of trees based on their Separation Ratio scores in a single Isolation Forest model. This example comes from *breastw* dataset, *seed* = 0.

the *Separation Ratio* pruning strategy, reducing the ensemble size by half, with *50th Percentile* thresholding method. The results are presented in Figures 4.7, 4.8, and 4.9. The idea behind this experiment is to identify a performance plateau, indicating that adding more trees to the original model does not yield significant benefits after pruning. This to confirm the thesis, mentioned in a related work (Section 2.3), in which adding more trees to the forest does not lead to a significant improvement in performance, but rather increases the computational complexity of the model (Bernard et al., 2009). Therefore, reducing the number of trees by pruning should ideally lead to a more efficient model without sacrificing detection capability.

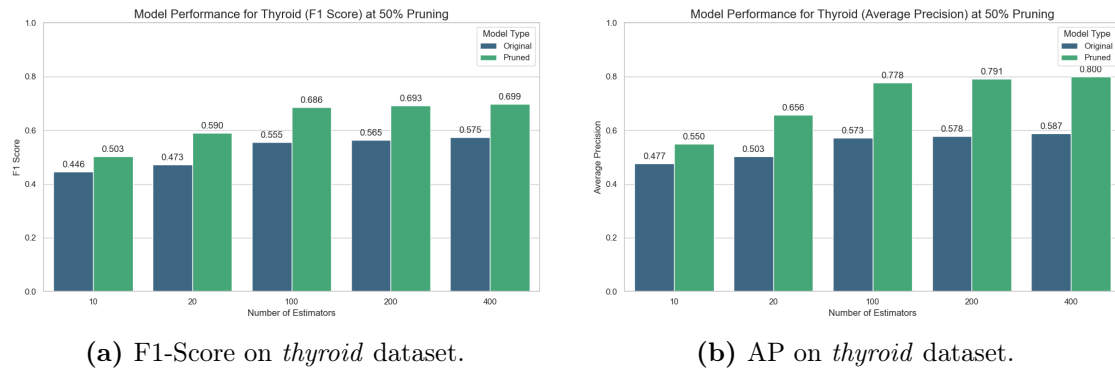


Figure 4.7: Performance of pruned Isolation Forest models on *thyroid* dataset with varying numbers of estimators using *Separation Ratio* pruning strategy and *50th Percentile* thresholding.

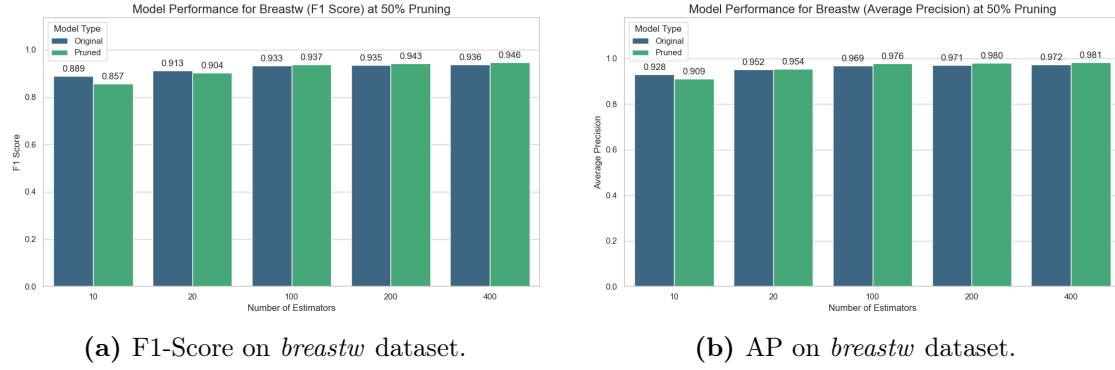


Figure 4.8: Performance of pruned Isolation Forest models on *breastw* dataset with varying numbers of estimators using *Separation Ratio* pruning strategy and *50th Percentile* thresholding.

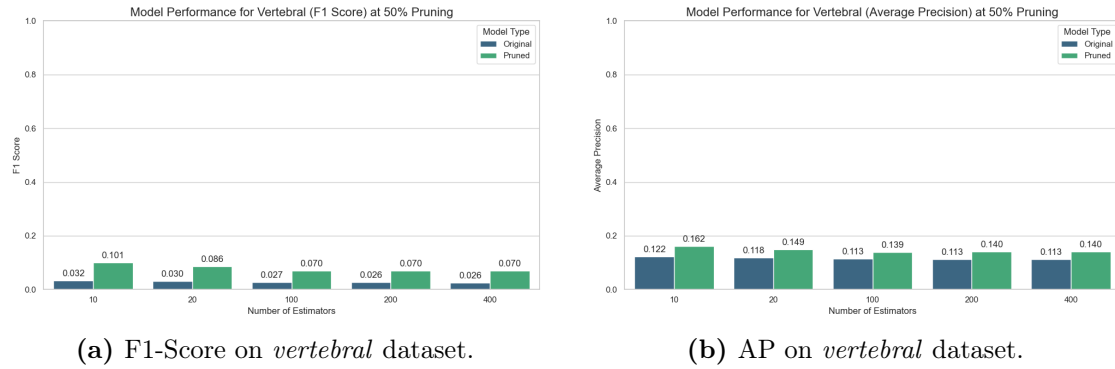


Figure 4.9: Performance of pruned Isolation Forest models on *vertebral* dataset with varying numbers of estimators using *Separation Ratio* pruning strategy and *50th Percentile* thresholding.

From the results, we can observe the following trends:

- **Thyroid Dataset:** As shown in Figure 4.7, there is a significant improvement in both F1-Score and AP as the number of estimators increases from 10 to 200. Beyond 200 estimators, performance improvements reach a plateau. From this we can deduce that, while pruning does indeed lead to a significant performance increase, this effect is more pronounced as the number of estimators in the original model increases, up to a certain point.
- **Vertebral Dataset:** In Figure 4.9, we see the opposite trend compared to *thyroid*: performance decreases as the number of estimators increases. This suggests that for datasets where the original model performs poorly and can significantly benefit from pruning, starting with a smaller number of trees may be more beneficial.
- **Breastw Dataset:** In Figure 4.8, the trend is different from the previous cases: we observe a mixed behavior where the performance decreases with 10 and 20 estimators, then increases with 100 and 200 estimators, and finally stabilizes with 400 estimators. This suggests that for datasets with good initial performance (i.e., > 0.80 F1/AP), a larger number of estimators may be necessary to maintain or improve performance after pruning. That is consistent with the observations made in previous sections: For this kind of datasets, a less aggressive pruning strategy is recommended; therefore, to benefit more from pruning, a larger pool of trees is needed.

In general, these findings show that with the increasing number of estimators in the original Isolation Forest model, the trend of performance after pruning is amplified, whether it is an increase or decrease. This indicates that the choice of the number of estimators should be made in conjunction with the pruning strategy to achieve optimal results.

Based on these insights, we conducted another experiment, only for *vertebral* dataset: the idea is to keep the same number of estimators after pruning, say 5 trees, since we know from previous results that the best performance is achieved with this number of trees. Therefore, we trained Isolation Forest models with varying the initial numbers of estimators - 100, 200. Then we applied the *Separation Ratio* pruning strategy, adjusting the threshold to keep exactly 5 trees in the pruned model. From the results presented in Figures 4.10 and 4.11, we observe that starting with a larger number of estimators and then pruning down to a fixed small number of trees, in order to achieve the best performance, produces better results than starting with a smaller number of estimators. This suggests that once the optimal number of trees in the pruned model has been identified, it is advantageous to start with a larger pool of trees in the original model to ensure that the best-performing trees are retained after pruning.

Unfortunately, due to computational constraints, we were unable to extend this experiment to the other two representative datasets to confirm the last observed trend. However, the insights gained from these experiments provide a valuable direction for future research on optimizing Isolation Forest models through pruning strategies.

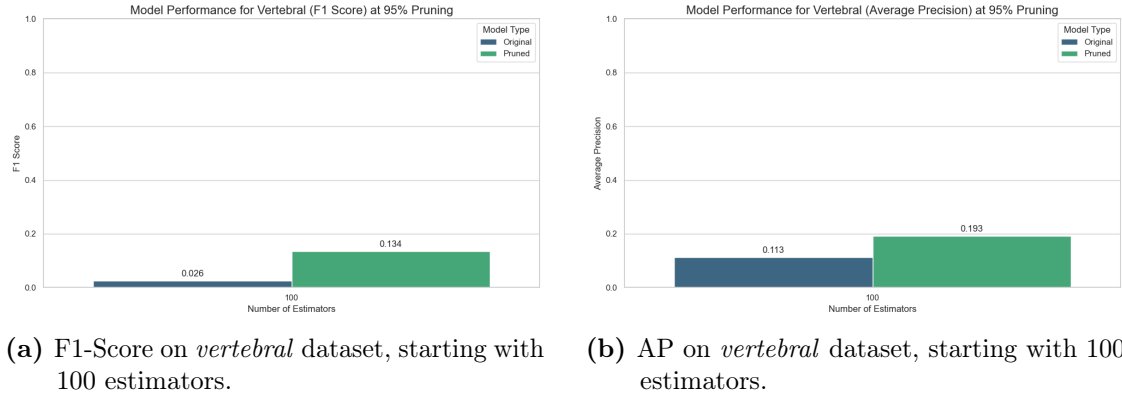


Figure 4.10: Performance of pruned Isolation Forest models on *vertebral* dataset, starting with 100 estimators, using *Separation Ratio* pruning strategy and adjusted thresholding to retain exactly 5 trees.

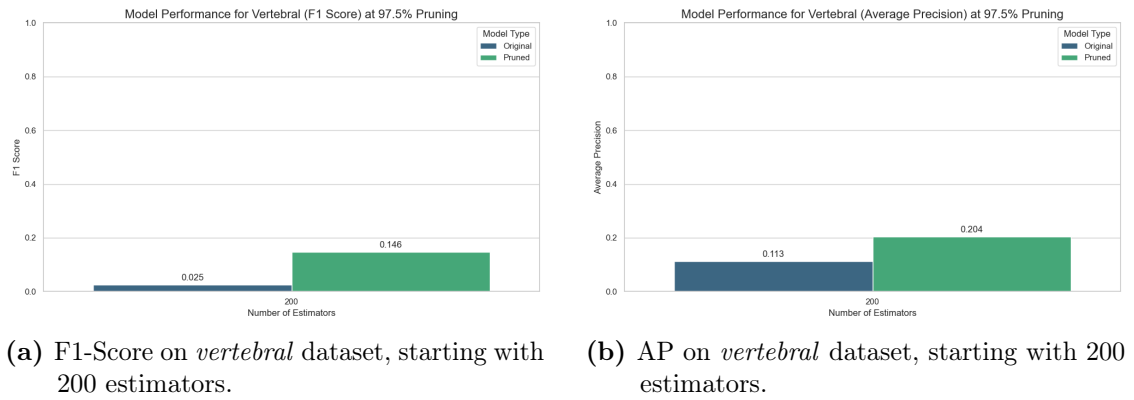


Figure 4.11: Performance of pruned Isolation Forest models on *vertebral* dataset, starting with 200 estimators, using *Separation Ratio* pruning strategy and adjusted thresholding to retain exactly 5 trees.

In conclusion, the number of estimators in the original Isolation Forest model plays a crucial role in determining the effectiveness of pruning strategies. Starting with a larger number of trees can provide a richer pool from which to select the most effective ones, leading to further improvements in the pruned model performance.

Chapter 5

Conclusion

This thesis addressed a fundamental challenge in applying Isolation Forests for Anomaly Detection: the presence of non-informative trees within the ensemble, which can degrade performance and increase computational costs. Motivated by the fact that in the literature there is a lack of unsupervised methods for pruning Isolation Forests, and considering the nature of anomaly detection tasks, we proposed and evaluated several strategies aimed at improving the effectiveness of Isolation Forests by removing the least useful trees from the ensemble in a more appropriated way.

To achieve this goal, we explored two distinct research directions. The first approach focused on exploiting DIFFI, an XAI technique, hypothesizing that the importance of the features used by each tree could serve as a proxy for its utility in anomaly detection. We developed pruning strategies based on **usage scores**, calculated from GFI values provided by DIFFI, which quantify the frequency of the most important features - those with high GFI scores - used in each tree's splits. The second approach centered on **depth-based metrics**, that rely on the fundamental principle of Isolation Forests: anomalies are isolated closer to the root of the tree, resulting in shorter paths. We proposed unsupervised pruning strategies that evaluate trees based on the average depth at which data points are isolated, with the intuition that the ability of trees to isolate not only anomalies in fewer splits but also normal instances at deeper levels could indicate their informational value.

Our empirical evaluations, conducted on heterogeneous datasets, from the synthetic one for initial validation to real-world ones from the *ODDS* collection, demonstrated the effectiveness of the proposed pruning strategies. In particular, the *Separation Ratio* one consistently outperformed other methods, achieving significant improvements in both performance metrics across multiple datasets. These results underscore the potential of pruning strategies to enhance anomaly detection performance while reducing ensemble size and computational overhead.

These empirical results confirm that the standard Isolation Forest model is often suboptimal, containing many non-informative or redundant trees. By carefully selecting and removing the least useful trees from the ensemble, we can achieve a more efficient and potentially more accurate anomaly detection system.

In addition, analyzing the impact of the number of estimators has highlighted that identifying the optimal starting point for the ensemble size and the number of trees

to retain after pruning is crucial to further maximizing performance.

5.1 Contributions

All the experiments conducted in this thesis produced several key discoveries:

- **Effectiveness of Pruning Strategies:** The proposed unsupervised pruning strategies, particularly those based on *Separation Ratio*, have demonstrated significant improvements in anomaly detection performance. This indicates that removing less informative trees can enhance the model’s ability to identify anomalies.
- **Reduction in Ensemble Size:** The strategies evaluated effectively reduce the size of Isolation Forest ensembles, diminishing the computational costs without sacrificing performance. This is especially important for real-world applications where efficiency is a crucial aspect.
- **Impact of the Number of Estimators:** Analysis of the number of estimators revealed that starting with a larger ensemble can be advantageous when applying pruning strategies. This finding suggests a trade-off between the initial ensemble size and the effectiveness of pruning, which deserves further investigation.
- **Complex Relationship with DIFFI:** The relationship between *usage scores* and pruning effectiveness is complex and not straightforward. Although DIFFI provides valuable insights on feature importance, its direct application to tree pruning requires further investigation to fully understand its implications.

5.2 Future works

However, although the results obtained in this thesis are promising, several limitations were identified during the research process:

- **Limited Dataset Diversity:** Despite the experiments included a variety of datasets, the scope could be expanded to include more diverse and larger datasets to validate the generalizability of the findings.
- **Computational Complexity Analysis:** An empirical evaluation of the computational complexity post-pruning was not performed, but only theoretically, so future work could focus on quantifying efficiency gains in terms of inference time and resource usage.
- **Deeper Exploration of DIFFI:** The complex relationship between usage scores and pruning effectiveness warrants further investigation to better understand how tree diversity influences ensemble performance.
- **Impact of Ensemble Size:** Further studies are needed, using a larger set of datasets, to explore the optimal initial set size and the number of trees to retain after pruning, particularly in relation to different dataset characteristics.

- **Majority vote Thresholds:** Future work could explore the impact of varying the majority vote threshold (e.g., > 0.5) on the performance of pruned ensembles, particularly in the context of Depth-based pruning strategies.
- **Analysis and Identification of Redundant Trees:** In this research, we did not specifically address the identification and eventual removal of redundant trees within the ensemble. Therefore, a comprehensive analysis of the presence of redundant trees and their impact on performance could be a valuable direction for future research.

Addressing these limitations in future research could further enhance the understanding and effectiveness of pruning strategies for Isolation Forests in anomaly detection tasks.

Regarding **usage scores**, a possible future direction could be to consider not only the frequency of use of high-GFI features, but also the depth at which they are used within each tree, since by simply counting their frequency, it is insufficient to capture their true importance. Therefore, a distinguishing factor could be introduced to weight the use of high-GFI features based on their depth of use in each tree. For example, high-GFI features used closer to the root of the tree could be assigned a higher weight than those used deeper in the tree, as they have a greater impact on the isolation process. This could lead to more accurate usage scores that better reflect the true importance of each tree in the ensemble.

Appendix A

Appendix A: Additional Experimental Results

This appendix provides supplementary experimental results that complement the findings presented in Section 4.3. Specifically, we include detailed tables that illustrate the performance of various pruning methods on different datasets with `test_size = 0`.

The following results show similar trends to those observed with `test_size = 0.2`, as discussed in the main body of the thesis. The additional data presented here serves to strengthen the conclusions drawn and offers a more comprehensive view of the effectiveness of the proposed pruning strategies.

Table A.1: Comparison of performance ($mean \pm std$) for Original vs. Pruned Models at different random pruning levels (50% to 80%). Results are obtained from the *Synthetic* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Pruning	Original	Pruned	Var. (%)
50%		0.4015 ± 0.0092	-4.95%
60%	0.4224	0.3903 ± 0.0080	-7.60%
70%	$\pm$ 0.0104	0.3744 ± 0.0065	-11.36%
80%		0.3497 ± 0.0055	-17.21%

(b) Average Precision

Pruning	Original	Pruned	Var. (%)
50%		0.3782 ± 0.0097	-7.03%
60%	0.4068	0.3633 ± 0.0085	-10.69%
70%	$\pm$ 0.0092	0.3456 ± 0.0070	-15.04%
80%		0.3163 ± 0.0074	-22.25%

Table A.2: Comparison of performance ($mean \pm std$) for Original and randomly Pruned Models. Pruning levels indicate the percentage of trees removed. Results are obtained from various *ODDS* datasets with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	80% Pruning		70% Pruning		60% Pruning		50% Pruning	
		Pruned	Var. (%)	Pruned	Var. (%)	Pruned	Var. (%)	Pruned	Var. (%)
anthroid	0.3253 $\pm$ 0.0022	0.3247 $\pm$ 0.0021	-0.18%	0.3245 $\pm$ 0.0014	-0.25%	0.3248 $\pm$ 0.0019	-0.15%	0.3256 $\pm$ 0.0017	+0.09%
breastw	0.9232 $\pm$ 0.0015	0.9060 $\pm$ 0.0005	-1.86%	0.9117 $\pm$ 0.0004	-1.25%	0.9154 $\pm$ 0.0004	-0.84%	0.9181 $\pm$ 0.0006	-0.55%
cardio	0.5105 $\pm$ 0.0110	0.4895 $\pm$ 0.0070	-4.11%	0.4978 $\pm$ 0.0071	-2.49%	0.5015 $\pm$ 0.0066	-1.76%	0.5062 $\pm$ 0.0083	-0.84%
ionosphere	0.6638 $\pm$ 0.0030	0.6701 $\pm$ 0.0026	+0.95%	0.6696 $\pm$ 0.0029	+0.87%	0.6684 $\pm$ 0.0029	+0.69%	0.6676 $\pm$ 0.0022	+0.57%
letter	0.0884 $\pm$ 0.0047	0.1082 $\pm$ 0.0041	+22.40%	0.1032 $\pm$ 0.0039	+16.74%	0.0980 $\pm$ 0.0039	+10.86%	0.0952 $\pm$ 0.0044	+7.69%
mnist	0.2887 $\pm$ 0.0067	0.2856 $\pm$ 0.0058	-1.07%	0.2875 $\pm$ 0.0066	-0.42%	0.2880 $\pm$ 0.0064	-0.24%	0.2893 $\pm$ 0.0060	+0.21%
opitdigits	0.0264 $\pm$ 0.0059	0.0533 $\pm$ 0.0069	+101.89%	0.0432 $\pm$ 0.0056	+63.64%	0.0375 $\pm$ 0.0057	+42.05%	0.0348 $\pm$ 0.0056	+31.82%
pendigits	0.3377 $\pm$ 0.0090	0.2957 $\pm$ 0.0084	-12.44%	0.3106 $\pm$ 0.0099	-8.02%	0.3159 $\pm$ 0.0094	-6.46%	0.3237 $\pm$ 0.0088	-4.15%
pima	0.5156 $\pm$ 0.0025	0.5039 $\pm$ 0.0025	-2.27%	0.5079 $\pm$ 0.0025	-1.49%	0.5110 $\pm$ 0.0032	-0.89%	0.5128 $\pm$ 0.0033	-0.54%
satellite	0.5716 $\pm$ 0.0071	0.5532 $\pm$ 0.0063	-3.22%	0.5595 $\pm$ 0.0054	-2.12%	0.5644 $\pm$ 0.0060	-1.26%	0.5670 $\pm$ 0.0060	-0.80%
satimage-2	0.8731 $\pm$ 0.0061	0.7935 $\pm$ 0.0088	-9.12%	0.8229 $\pm$ 0.0058	-5.75%	0.8394 $\pm$ 0.0054	-3.86%	0.8489 $\pm$ 0.0040	-2.77%
thyroid	0.5735 $\pm$ 0.0044	0.5319 $\pm$ 0.0077	-7.25%	0.5448 $\pm$ 0.0074	-5.00%	0.5525 $\pm$ 0.0075	-3.66%	0.5583 $\pm$ 0.0087	-2.65%
vertebral	0.0426 $\pm$ 0.0015	0.0406 $\pm$ 0.0020	-4.69%	0.0416 $\pm$ 0.0018	-2.35%	0.0423 $\pm$ 0.0018	-0.70%	0.0428 $\pm$ 0.0013	+0.47%
vowels	0.1872 $\pm$ 0.0178	0.1525 $\pm$ 0.0065	-18.54%	0.1596 $\pm$ 0.0077	-14.74%	0.1662 $\pm$ 0.0092	-11.22%	0.1723 $\pm$ 0.0100	-7.96%
wine	0.1680 $\pm$ 0.0084	0.2030 $\pm$ 0.0051	+20.83%	0.2044 $\pm$ 0.0059	+21.67%	0.1962 $\pm$ 0.0064	+16.79%	0.1978 $\pm$ 0.0060	+17.74%

(b) Average Precision

Dataset	Original	80% Pruning		70% Pruning		60% Pruning		50% Pruning	
		Pruned	Var. (%)	Pruned	Var. (%)	Pruned	Var. (%)	Pruned	Var. (%)
anthroid	0.3137 $\pm$ 0.0035	0.3028 $\pm$ 0.0047	-3.47%	0.3062 $\pm$ 0.0033	-2.39%	0.3078 $\pm$ 0.0039	-1.88%	0.3101 $\pm$ 0.0031	-1.15%
breastw	0.9674 $\pm$ 0.0008	0.9568 $\pm$ 0.0010	-1.10%	0.9613 $\pm$ 0.0008	-0.63%	0.9634 $\pm$ 0.0008	-0.41%	0.9649 $\pm$ 0.0008	-0.26%
cardio	0.5613 $\pm$ 0.0110	0.5072 $\pm$ 0.0094	-9.64%	0.5280 $\pm$ 0.0096	-5.93%	0.5371 $\pm$ 0.0095	-4.31%	0.5466 $\pm$ 0.0104	-2.62%
ionosphere	0.8080 $\pm$ 0.0023	0.7999 $\pm$ 0.0023	-1.00%	0.8034 $\pm$ 0.0025	-0.57%	0.8048 $\pm$ 0.0027	-0.40%	0.8060 $\pm$ 0.0024	-0.25%
letter	0.0893 $\pm$ 0.0020	0.0874 $\pm$ 0.0019	-2.13%	0.0881 $\pm$ 0.0020	-1.34%	0.0885 $\pm$ 0.0020	-0.90%	0.0888 $\pm$ 0.0022	-0.56%
mnist	0.2616 $\pm$ 0.0050	0.2449 $\pm$ 0.0050	-6.38%	0.2514 $\pm$ 0.0056	-3.90%	0.2556 $\pm$ 0.0050	-2.29%	0.2583 $\pm$ 0.0050	-1.26%
opitdigits	0.0548 $\pm$ 0.0041	0.0540 $\pm$ 0.0034	-1.46%	0.0542 $\pm$ 0.0031	-1.09%	0.0545 $\pm$ 0.0039	-0.55%	0.0545 $\pm$ 0.0039	-0.55%
pendigits	0.2609 $\pm$ 0.0069	0.2330 $\pm$ 0.0054	-10.69%	0.2452 $\pm$ 0.0071	-6.02%	0.2477 $\pm$ 0.0059	-5.06%	0.2519 $\pm$ 0.0059	-3.45%
pima	0.5011 $\pm$ 0.0027	0.4928 $\pm$ 0.0029	-1.66%	0.4962 $\pm$ 0.0029	-0.98%	0.4985 $\pm$ 0.0031	-0.52%	0.4994 $\pm$ 0.0029	-0.34%
satellite	0.6610 $\pm$ 0.0059	0.6407 $\pm$ 0.0068	-3.07%	0.6481 $\pm$ 0.0062	-1.95%	0.6532 $\pm$ 0.0063	-1.18%	0.6559 $\pm$ 0.0060	-0.77%
satimage-2	0.9254 $\pm$ 0.0016	0.8926 $\pm$ 0.0038	-3.54%	0.9069 $\pm$ 0.0026	-2.00%	0.9149 $\pm$ 0.0022	-1.13%	0.9187 $\pm$ 0.0017	-0.72%
thyroid	0.5487 $\pm$ 0.0074	0.5135 $\pm$ 0.0083	-6.42%	0.5256 $\pm$ 0.0075	-4.21%	0.5305 $\pm$ 0.0064	-3.32%	0.5369 $\pm$ 0.0086	-2.15%
vertebral	0.0943 $\pm$ 0.0004	0.0974 $\pm$ 0.0007	+3.29%	0.0964 $\pm$ 0.0006	+1.23%	0.0959 $\pm$ 0.0005	+1.70%	0.0953 $\pm$ 0.0004	+1.06%
vowels	0.1342 $\pm$ 0.0137	0.1118 $\pm$ 0.0064	-16.69%	0.1190 $\pm$ 0.0071	-11.33%	0.1238 $\pm$ 0.0086	-7.75%	0.1272 $\pm$ 0.0102	-5.22%
wine	0.2168 $\pm$ 0.0028	0.2073 $\pm$ 0.0031	-4.38%	0.2126 $\pm$ 0.0034	-1.94%	0.2157 $\pm$ 0.0033	-0.51%	0.2164 $\pm$ 0.0031	-0.18%

Table A.3: Comparison of performance (*mean $\pm$ std*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Elbow* method. Results are obtained from the *Synthetic* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Method	Original	Pruned	Variation (%)
Majority vote		0.4888 ± 0.0224	+15.72%
Mean		0.5051 ± 0.0183	+19.58%
Percentile 50%	0.4224	0.4998 ± 0.0141	+18.32%
Percentile 60%	± 0.0104	0.5138 ± 0.0208	+21.64%
Percentile 70%		0.5172 ± 0.0212	+22.44%
Percentile 80%		0.5232 ± 0.0185	+23.86%

(b) Average Precision

Method	Original	Pruned	Variation (%)
Majority vote		0.5042 ± 0.0292	+23.94%
Mean		0.5192 ± 0.0228	+27.63%
Percentile 50%	0.4068	0.5147 ± 0.0189	+26.52%
Percentile 60%	± 0.0092	0.5336 ± 0.0225	+31.17%
Percentile 70%		0.5446 ± 0.0250	+33.87%
Percentile 80%		0.5561 ± 0.0222	+36.70%

Table A.4: Comparison of performance (*mean*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Elbow* method. Results are obtained from various *ODDS* datasets with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	MV		Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3253	0.3650	+12.2%	0.3658	+12.4%	0.3636	+11.8%	0.3713	+14.1%	0.3778	+16.1%	0.3844	+18.2%
breastw	0.9232	0.9065	-1.8%	0.9065	-1.8%	0.9075	-1.7%	0.9037	-2.1%	0.8946	-3.1%	0.8848	-4.2%
cardio	0.5105	0.5035	-1.4%	0.5035	-1.4%	0.5102	-0.1%	0.4996	-2.1%	0.4905	-3.9%	0.4770	-6.6%
ionosphere	0.6638	0.6683	+0.7%	0.6707	+1.0%	0.6716	+1.2%	0.6735	+1.5%	0.6773	+2.0%	0.6822	+2.8%
letter	0.0884	0.1025	+16.0%	0.1015	+14.8%	0.1000	+13.1%	0.1026	+16.1%	0.0975	+10.3%	0.1084	+22.6%
mnist	0.2887	0.2924	+1.3%	0.2970	+2.9%	0.2976	+3.1%	0.2981	+3.3%	0.3005	+4.1%	0.3031	+5.0%
optdigits	0.0264	0.0285	+8.0%	0.0253	-4.2%	0.0273	+3.4%	0.0255	-3.4%	0.0220	-16.7%	0.0203	-23.1%
pendigits	0.3377	0.3230	-4.4%	0.3228	-4.4%	0.3231	-4.3%	0.3220	-4.7%	0.3082	-8.7%	0.2971	-12.0%
pima	0.5156	0.5130	-0.5%	0.5052	-2.0%	0.5065	-1.8%	0.5010	-2.8%	0.4915	-4.7%	0.4784	-7.2%
satellite	0.5716	0.5665	-0.9%	0.5686	-0.5%	0.5690	-0.5%	0.5669	-0.8%	0.5607	-1.9%	0.5602	-2.0%
satimage-2	0.8731	0.8343	-4.4%	0.8485	-2.8%	0.8482	-2.8%	0.8372	-4.1%	0.8235	-5.7%	0.7915	-9.3%
thyroid	0.5735	0.6264	+9.2%	0.6266	+9.3%	0.6227	+8.6%	0.6279	+9.5%	0.6342	+10.6%	0.6330	+10.4%
vertebral	0.0426	0.0359	-15.7%	0.0249	-41.5%	0.0300	-29.6%	0.0310	-27.2%	0.0328	-23.0%	0.0347	-18.5%
vowels	0.1872	0.1806	-3.5%	0.1760	-6.0%	0.1771	-5.4%	0.1739	-7.1%	0.1647	-12.0%	0.1652	-11.8%
wine	0.1680	0.1845	+9.8%	0.2119	+26.1%	0.2186	+30.1%	0.2140	+27.4%	0.2173	+29.3%	0.2098	+24.9%

(b) Average Precision

Dataset	Original	MV		Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3137	0.3753	+19.6%	0.3762	+19.9%	0.3731	+18.9%	0.3829	+22.1%	0.3920	+25.0%	0.3992	+27.3%
breastw	0.9674	0.9661	-0.1%	0.9661	-0.1%	0.9665	-0.1%	0.9659	-0.2%	0.9638	-0.4%	0.9603	-0.7%
cardio	0.5613	0.5566	-0.8%	0.5617	+0.1%	0.5661	+0.9%	0.5551	-1.1%	0.5476	-2.4%	0.5270	-6.1%
ionosphere	0.8080	0.8072	-0.1%	0.8097	+0.2%	0.8106	+0.3%	0.8103	+0.3%	0.8104	+0.3%	0.8102	+0.3%
letter	0.0893	0.0902	+1.0%	0.0896	+0.3%	0.0896	+0.3%	0.0889	-0.4%	0.0878	-1.7%	0.0904	+1.2%
mnist	0.2616	0.2641	+1.0%	0.2698	+3.1%	0.2694	+3.0%	0.2751	+5.2%	0.2824	+8.0%	0.2815	+7.6%
optdigits	0.0548	0.0493	-10.0%	0.0495	-9.7%	0.0500	-8.8%	0.0490	-10.6%	0.0469	-14.4%	0.0445	-18.8%
pendigits	0.2609	0.2487	-4.7%	0.2470	-5.3%	0.2503	-4.1%	0.2462	-5.6%	0.2344	-10.2%	0.2313	-11.3%
pima	0.5011	0.4974	-0.7%	0.4950	-1.2%	0.4951	-1.2%	0.4935	-1.5%	0.4890	-2.4%	0.4842	-3.4%
satellite	0.6610	0.6578	-0.5%	0.6585	-0.4%	0.6600	-0.1%	0.6575	-0.5%	0.6489	-1.8%	0.6468	-2.1%
satimage-2	0.9254	0.9152	-1.1%	0.9204	-0.5%	0.9216	-0.4%	0.9178	-0.8%	0.9106	-1.6%	0.8936	-3.4%
thyroid	0.5487	0.6497	+18.4%	0.6510	+18.6%	0.6447	+17.5%	0.6572	+19.8%	0.6646	+21.1%	0.6569	+19.7%
vertebral	0.0943	0.0941	-0.2%	0.0937	-0.6%	0.0936	-0.7%	0.0952	+1.0%	0.0958	+1.6%	0.0975	+3.4%
vowels	0.1342	0.1323	-1.4%	0.1305	-2.8%	0.1303	-2.9%	0.1280	-4.6%	0.1201	-10.5%	0.1200	-10.6%
wine	0.2168	0.2104	-3.0%	0.2181	+0.6%	0.2176	+0.4%	0.2185	+0.8%	0.2170	+0.1%	0.2137	-1.4%

Table A.5: Comparison of performance ($mean \pm std$) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Soft* method. Results are obtained from the *Synthetic* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Method	Original	Pruned	Variation (%)
Majority vote		0.4767 ± 0.0233	+12.86%
Mean		0.5154 ± 0.0150	+22.02%
Percentile 50%	0.4224	0.5067 ± 0.0133	+19.96%
Percentile 60%	± 0.0104	0.5283 ± 0.0170	+25.07%
Percentile 70%		0.5360 ± 0.0153	+26.89%
Percentile 80%		0.5307 ± 0.0110	+25.64%

(b) Average Precision

Method	Original	Pruned	Variation (%)
Majority vote		0.4882 ± 0.0277	+20.01%
Mean		0.5348 ± 0.0173	+31.47%
Percentile 50%	0.4068	0.5256 ± 0.0158	+29.20%
Percentile 60%	± 0.0092	0.5518 ± 0.0173	+35.64%
Percentile 70%		0.5655 ± 0.0169	+39.01%
Percentile 80%		0.5665 ± 0.0186	+39.26%

Table A.6: Comparison of performance (*mean*) for Original vs. Pruned Models with different *thresholding approach* on selected features by *Soft* method. Results are obtained from the *ODDS* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	MV		Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3253	0.3512	+8.0%	0.3557	+9.3%	0.3554	+9.2%	0.3642	+12.0%	0.3734	+14.8%	0.3788	+16.4%
breastw	0.9232	0.9065	-1.8%	0.9065	-1.8%	0.9075	-1.7%	0.9041	-2.1%	0.8965	-2.9%	0.8828	-4.4%
cardio	0.5105	0.5202	+1.9%	0.5127	+0.4%	0.5189	+1.6%	0.5117	+0.2%	0.5081	-0.5%	0.4864	-4.7%
ionosphere	0.6638	0.6706	+1.0%	0.6716	+1.2%	0.6725	+1.3%	0.6757	+1.8%	0.6770	+2.0%	0.6812	+2.6%
letter	0.0884	0.1097	+24.1%	0.0991	+12.1%	0.0986	+11.5%	0.0992	+12.2%	0.1007	+13.9%	0.1008	+14.0%
mnist	0.2887	0.2948	+2.1%	0.3055	+5.8%	0.3025	+4.8%	0.3068	+6.3%	0.3113	+7.8%	0.3126	+8.3%
optdigits	0.0264	0.0291	+10.2%	0.0183	-30.7%	0.0192	-27.3%	0.0171	-35.2%	0.0164	-37.9%	0.0180	-31.8%
pendigits	0.3377	0.3303	-2.2%	0.3342	-1.0%	0.3303	-2.2%	0.3314	-1.9%	0.3202	-5.2%	0.3069	-9.1%
pima	0.5156	0.5151	-0.1%	0.5063	-1.8%	0.5055	-2.0%	0.4988	-3.3%	0.4896	-5.0%	0.4763	-7.6%
satellite	0.5716	0.5602	-2.0%	0.5686	-0.5%	0.5685	-0.5%	0.5693	-0.4%	0.5710	-0.1%	0.5706	-0.2%
satimage-2	0.8731	0.8090	-7.3%	0.8440	-3.3%	0.8466	-3.0%	0.8357	-4.3%	0.8169	-6.4%	0.7959	-8.8%
thyroid	0.5735	0.5999	+4.6%	0.6230	+8.6%	0.6239	+8.8%	0.6233	+8.7%	0.6275	+9.4%	0.6253	+9.0%
vertebral	0.0426	0.0382	-10.3%	0.0291	-31.7%	0.0291	-31.7%	0.0215	-49.5%	0.0291	-31.7%	0.0349	-18.1%
vowels	0.1872	0.1690	-9.7%	0.1813	-3.1%	0.1838	-1.8%	0.1749	-6.6%	0.1673	-10.6%	0.1627	-13.1%
wine	0.1680	0.1699	+1.1%	0.2153	+28.1%	0.2173	+29.3%	0.1926	+14.6%	0.1874	+11.5%	0.2108	+25.5%

(b) Average Precision

Dataset	Original	MV		Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
annthyroid	0.3137	0.3468	+10.5%	0.3638	+16.0%	0.3626	+15.6%	0.3719	+18.5%	0.3815	+21.6%	0.3933	+25.4%
breastw	0.9674	0.9661	-0.1%	0.9664	-0.1%	0.9666	-0.1%	0.9656	-0.2%	0.9641	-0.3%	0.9612	-0.6%
cardio	0.5613	0.5716	+1.8%	0.5730	+2.1%	0.5784	+3.0%	0.5703	+1.6%	0.5633	+0.4%	0.5381	-4.1%
ionosphere	0.8080	0.8074	-0.1%	0.8100	+0.2%	0.8104	+0.3%	0.8098	+0.2%	0.8092	+0.1%	0.8067	-0.2%
letter	0.0893	0.0889	-0.4%	0.0895	+0.2%	0.0899	+0.7%	0.0898	+0.6%	0.0897	+0.4%	0.0898	+0.6%
mnist	0.2616	0.2650	+1.3%	0.2782	+6.3%	0.2754	+5.3%	0.2800	+7.0%	0.2864	+9.5%	0.2887	+10.4%
optdigits	0.0548	0.0481	-12.2%	0.0468	-14.6%	0.0489	-10.8%	0.0462	-15.7%	0.0447	-18.4%	0.0440	-19.7%
pendigits	0.2609	0.2560	-1.9%	0.2530	-3.0%	0.2522	-3.3%	0.2510	-3.8%	0.2435	-6.7%	0.2429	-6.9%
pima	0.5011	0.4989	-0.4%	0.4951	-1.2%	0.4947	-1.3%	0.4911	-2.0%	0.4889	-2.4%	0.4846	-3.3%
satellite	0.6610	0.6504	-1.6%	0.6576	-0.5%	0.6574	-0.5%	0.6596	-0.2%	0.6615	+0.1%	0.6578	-0.5%
satimage-2	0.9254	0.9027	-2.5%	0.9189	-0.7%	0.9197	-0.6%	0.9151	-1.1%	0.9092	-1.7%	0.8959	-3.2%
thyroid	0.5487	0.6057	+10.4%	0.6459	+17.7%	0.6436	+17.3%	0.6485	+18.2%	0.6525	+18.9%	0.6506	+18.6%
vertebral	0.0943	0.0946	+0.3%	0.0940	-0.3%	0.0945	+0.2%	0.0941	-0.2%	0.0953	+1.1%	0.0972	+3.1%
vowels	0.1342	0.1253	-6.6%	0.1324	-1.3%	0.1324	-1.3%	0.1318	-1.8%	0.1249	-6.9%	0.1222	-8.9%
wine	0.2168	0.2118	-2.3%	0.2220	+2.4%	0.2202	+1.6%	0.2248	+3.7%	0.2161	-0.3%	0.2230	+2.9%

Table A.7: Comparison of performance ($mean \pm std$) for Original vs. Depth-based Pruned Models, exploiting **depth ratio** between outliers and inliers, with different *thresholding approach*. Results are obtained from the *Synthetic* dataset with `test_size` = 0 by averaging the results over seeds.

(a) F1-Score

Method	Original	Pruned	Variation (%)
Mean		0.6350 ± 0.0234	+50.33%
Percentile 50%	0.4224 ± 0.0104	0.5850 ± 0.0162	+38.49%
Percentile 60%		0.6160 ± 0.0181	+45.83%
Percentile 70%		0.6408 ± 0.0215	+51.70%
Percentile 80%		0.6586 ± 0.0284	+55.92%

(b) Average Precision

Method	Original	Pruned	Variation (%)
Mean		0.7151 ± 0.0238	+75.79%
Percentile 50%	0.4068 ± 0.0092	0.6448 ± 0.0153	+58.51%
Percentile 60%		0.6922 ± 0.0224	+70.16%
Percentile 70%		0.7269 ± 0.0255	+78.69%
Percentile 80%		0.7491 ± 0.0268	+84.14%

Table A.8: Comparison of performance (*mean*) for Original vs. Depth-based Pruned Models, exploiting **depth ratio** between outliers and inliers, with different *thresholding approach*. Results are obtained from the *ODDS* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
anthyroid	0.3253	0.4033	+24.0%	0.3915	+20.3%	0.4060	+24.8%	0.4218	+29.7%	0.4362	+34.1%
breastw	0.9232	0.9352	+1.3%	0.9360	+1.4%	0.9339	+1.2%	0.9321	+1.0%	0.9267	+0.4%
cardio	0.5105	0.6438	+26.1%	0.6261	+22.6%	0.6447	+26.3%	0.6685	+31.0%	0.6690	+31.0%
ionosphere	0.6638	0.6976	+5.1%	0.6971	+5.0%	0.7013	+5.6%	0.7064	+6.4%	0.7093	+6.9%
letter	0.0884	0.1879	+112.6%	0.1773	+100.6%	0.1891	+113.9%	0.1969	+122.7%	0.2069	+134.0%
mnist	0.2887	0.4495	+55.7%	0.4303	+49.1%	0.4577	+58.5%	0.4775	+65.4%	0.4972	+72.2%
optdigits	0.0264	0.3389	+1183.7%	0.2698	+922.0%	0.3234	+1125.0%	0.3542	+1241.7%	0.3576	+1254.5%
pendigits	0.3377	0.5031	+49.0%	0.4824	+42.8%	0.4974	+47.3%	0.5036	+49.1%	0.5006	+48.2%
pima	0.5156	0.5468	+6.1%	0.5459	+5.9%	0.5472	+6.1%	0.5486	+6.4%	0.5448	+5.7%
satellite	0.5716	0.6159	+7.7%	0.6146	+7.5%	0.6196	+8.4%	0.6212	+8.7%	0.6217	+8.8%
satimage-2	0.8731	0.8733	+0.0%	0.8753	+0.2%	0.8724	-0.1%	0.8633	-1.1%	0.8412	-3.6%
thyroid	0.5735	0.7135	+24.4%	0.7051	+23.0%	0.7164	+24.9%	0.7269	+26.8%	0.7363	+28.4%
vertebral	0.0426	0.1066	+150.2%	0.0829	+94.6%	0.1018	+139.0%	0.1183	+177.7%	0.1414	+231.9%
vowels	0.1872	0.3222	+72.1%	0.3121	+66.7%	0.3209	+71.4%	0.3409	+82.1%	0.3453	+84.5%
wine	0.1680	0.5899	+251.1%	0.5332	+217.4%	0.5732	+241.2%	0.5965	+255.1%	0.6211	+269.7%

(b) Average Precision

Dataset	Original	Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
anthyroid	0.3137	0.4485	+43.0%	0.4350	+38.7%	0.4514	+43.9%	0.4688	+49.4%	0.4821	+53.7%
breastw	0.9674	0.9786	+1.2%	0.9789	+1.2%	0.9799	+1.3%	0.9807	+1.4%	0.9794	+1.2%
cardio	0.5613	0.7240	+29.0%	0.7050	+25.6%	0.7263	+29.4%	0.7522	+34.0%	0.7603	+35.5%
ionosphere	0.8080	0.8318	+2.9%	0.8318	+2.9%	0.8342	+3.2%	0.8375	+3.6%	0.8389	+3.8%
letter	0.0893	0.1342	+50.3%	0.1267	+41.9%	0.1360	+52.3%	0.1447	+62.0%	0.1569	+75.7%
mnist	0.2616	0.4464	+70.6%	0.4226	+61.5%	0.4550	+73.9%	0.4821	+84.3%	0.5087	+94.5%
optdigits	0.0548	0.2914	+431.7%	0.1978	+261.0%	0.2561	+367.3%	0.3281	+498.7%	0.4013	+632.3%
pendigits	0.2609	0.5193	+99.0%	0.4869	+86.6%	0.5140	+97.0%	0.5401	+107.0%	0.5550	+112.7%
pima	0.5011	0.5488	+9.5%	0.5478	+9.3%	0.5547	+10.7%	0.5617	+12.1%	0.5649	+12.7%
satellite	0.6610	0.7111	+7.6%	0.7104	+7.5%	0.7129	+7.8%	0.7141	+8.0%	0.7141	+8.0%
satimage-2	0.9254	0.9401	+1.6%	0.9391	+1.5%	0.9399	+1.6%	0.9398	+1.6%	0.9375	+1.3%
thyroid	0.5487	0.7999	+45.8%	0.7851	+43.1%	0.8006	+45.9%	0.8114	+47.9%	0.8178	+49.0%
vertebral	0.0943	0.1278	+35.5%	0.1149	+21.8%	0.1228	+30.2%	0.1326	+40.6%	0.1486	+57.6%
vowels	0.1342	0.3590	+167.5%	0.3294	+145.5%	0.3521	+162.4%	0.3760	+180.2%	0.3810	+183.9%
wine	0.2168	0.5848	+169.7%	0.4950	+128.3%	0.5583	+157.5%	0.6294	+190.3%	0.7011	+223.4%

Table A.9: Comparison of performance ($mean \pm std$) for Original vs. Depth-based Pruned Models, exploiting **depth of outliers**, with different *thresholding approach*. Results are obtained from the *Synthetic* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Method	Original	Pruned	Variation (%)
Mean		0.6241 ± 0.0214	+47.75%
Percentile 50%	0.4224 ± 0.0104	0.5763 ± 0.0129	+36.43%
Percentile 60%		0.6043 ± 0.0175	+43.06%
Percentile 70%		0.6328 ± 0.0209	+49.81%
Percentile 80%		0.6504 ± 0.0286	+53.98%

(b) Average Precision

Method	Original	Pruned	Variation (%)
Mean		0.7060 ± 0.0269	+73.55%
Percentile 50%	0.4068 ± 0.0092	0.6357 ± 0.0167	+56.27%
Percentile 60%		0.6825 ± 0.0188	+67.77%
Percentile 70%		0.7196 ± 0.0265	+76.89%
Percentile 80%		0.7442 ± 0.0303	+82.94%

Table A.10: Comparison of performance (*mean*) for Original vs. Depth-based Pruned Models, exploiting **depth of outliers**, with different *thresholding approach*. Results are obtained from the *ODDS* dataset with `test_size = 0` by averaging the results over seeds.

(a) F1-Score

Dataset	Original	Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
anthyroid	0.3253	0.3311	+1.8%	0.3313	+1.8%	0.3322	+2.1%	0.3333	+2.5%	0.3350	+3.0%
breastw	0.9232	0.9325	+1.0%	0.9326	+1.0%	0.9296	+0.7%	0.9266	+0.4%	0.9196	-0.4%
cardio	0.5105	0.5192	+1.7%	0.5165	+1.2%	0.5197	+1.8%	0.5266	+3.2%	0.5278	+3.4%
ionosphere	0.6638	0.6669	+0.5%	0.6677	+0.6%	0.6678	+0.6%	0.6712	+1.1%	0.6736	+1.5%
letter	0.0884	0.0787	-11.0%	0.0805	-8.9%	0.0785	-11.2%	0.0800	-9.5%	0.0820	-7.2%
mnist	0.2887	0.2865	-0.8%	0.2873	-0.5%	0.2855	-1.1%	0.2826	-2.1%	0.2841	-1.6%
optdigits	0.0264	0.0177	-33.0%	0.0187	-29.2%	0.0188	-28.8%	0.0158	-40.1%	0.0122	-53.8%
pendigits	0.3377	0.3879	+14.9%	0.3844	+13.8%	0.3881	+14.9%	0.3843	+13.8%	0.3829	+13.4%
pima	0.5156	0.5172	+0.3%	0.5151	-0.1%	0.5123	-0.6%	0.5103	-1.0%	0.5043	-2.2%
satellite	0.5716	0.5752	+0.6%	0.5757	+0.7%	0.5743	+0.5%	0.5721	+0.1%	0.5655	-1.1%
satimage-2	0.8731	0.8643	-1.0%	0.8661	-0.8%	0.8596	-1.5%	0.8512	-2.5%	0.8224	-5.8%
thyroid	0.5735	0.6321	+10.2%	0.6251	+9.0%	0.6316	+10.1%	0.6323	+10.3%	0.6348	+10.7%
vertebral	0.0426	0.0133	-68.8%	0.0139	-67.4%	0.0075	-82.4%	0.0036	-91.5%	0.0082	-80.8%
vowels	0.1872	0.1766	-5.7%	0.1765	-5.7%	0.1778	-5.0%	0.1725	-7.9%	0.1703	-9.0%
wine	0.1680	0.2025	+20.5%	0.1976	+17.6%	0.2152	+28.1%	0.1974	+17.5%	0.2183	+29.9%

(b) Average Precision

Dataset	Original	Mean		P50		P60		P70		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.	Pruned	Var.
anthyroid	0.3137	0.3282	+4.6%	0.3302	+5.3%	0.3311	+5.5%	0.3325	+6.0%	0.3299	+5.2%
breastw	0.9674	0.9758	+0.9%	0.9757	+0.9%	0.9758	+0.9%	0.9748	+0.8%	0.9735	+0.6%
cardio	0.5613	0.5792	+3.2%	0.5754	+2.5%	0.5799	+3.3%	0.5832	+3.9%	0.5781	+3.0%
ionosphere	0.8080	0.8081	+0.0%	0.8082	+0.0%	0.8075	-0.1%	0.8064	-0.2%	0.8041	-0.5%
letter	0.0893	0.0841	-5.8%	0.0849	-4.9%	0.0838	-6.2%	0.0828	-7.3%	0.0803	-10.1%
mnist	0.2616	0.2449	-6.4%	0.2461	-5.9%	0.2433	-7.0%	0.2396	-8.4%	0.2373	-9.3%
optdigits	0.0548	0.0464	-15.3%	0.0482	-12.0%	0.0467	-14.8%	0.0446	-18.6%	0.0412	-24.8%
pendigits	0.2609	0.3127	+19.9%	0.3068	+17.6%	0.3120	+19.6%	0.3130	+20.0%	0.3214	+23.2%
pima	0.5011	0.5068	+1.1%	0.5058	+0.9%	0.5057	+0.9%	0.5057	+0.9%	0.5016	+0.1%
satellite	0.5716	0.6685	+1.1%	0.6688	+1.2%	0.6680	+1.1%	0.6668	+0.9%	0.6605	-0.1%
satimage-2	0.9254	0.9375	+1.3%	0.9358	+1.1%	0.9368	+1.2%	0.9363	+1.2%	0.9333	+0.8%
thyroid	0.5487	0.6851	+24.9%	0.6721	+22.5%	0.6857	+25.0%	0.6953	+26.7%	0.6993	+27.4%
vertebral	0.0943	0.0915	-3.0%	0.0919	-2.5%	0.0912	-3.3%	0.0904	-4.1%	0.0912	-3.3%
vowels	0.1342	0.1232	-8.2%	0.1248	-7.0%	0.1223	-8.9%	0.1244	-7.3%	0.1229	-8.4%
wine	0.2168	0.2524	+16.4%	0.2514	+16.0%	0.2517	+16.1%	0.2564	+18.3%	0.2537	+17.0%

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